A.B.V.M. Agrawal Jatiya Kosh's
Ghanshyamdas Jalan College of Science & Commerce
Malad (East), Mumbai
(Affiliated to University of Mumbai)

DEPARTMENT OF INFORMATION TECHNOLOGY

Data Visualization Practical

B.Sc. Information Technology

Submitted by

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Under the Guidance of

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Academic Year 2026–27

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Ghanshyamdas Jalan College of Science & Commerce

Malad (East), Mumbai

(Affiliated to University of Mumbai)

Certificate by the Institute

Department of Information Technology

I hereby certify that Ms. **Ayesha Sarguru**, Seat Number: 502520019, a student of Ghanshyamdas Jalan College of Science & Commerce, Malad (East), studying in **B.Sc. Information Technology**, has successfully completed the **Data Visualization Practical** for the academic year 2026–27, under the guidance of Prof. Raviprakash Jaiswal. To the best of my knowledge, the work of the student is original and the information included in the practical is correct.

Internal Guide	**External Examiner**	**Head of the Department**

Date: _____ **College Seal:**

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Practical 1: Getting Started with Tableau and Building Basic Visualizations

Aim: To learn the Tableau interface, connect to a CSV dataset, identify measures and dimensions, create basic visualizations, and build an interactive dashboard.

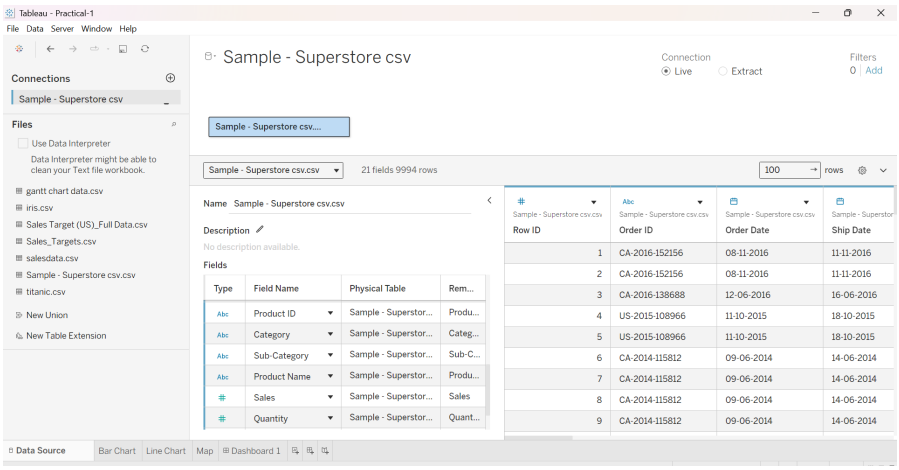
Software Required

- Tableau Desktop / Tableau Public

Procedure

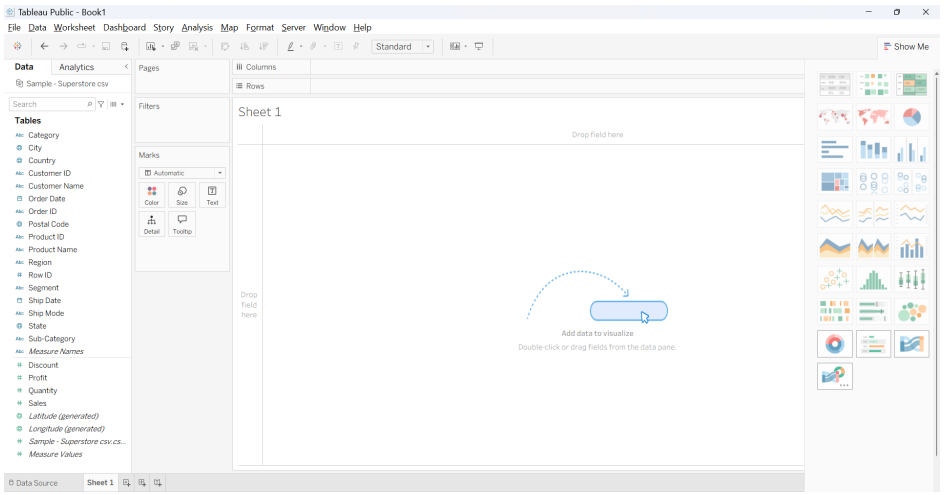
a) Connect to a CSV Dataset and Explore the Tableau Interface

1. Open Tableau Public/Desktop.
2. Click Connect → Text File.
3. Select Sample Superstore.csv.
4. Tableau displays the Data Source page.
5. Verify all fields are imported correctly.



Data Source page with Sample - Superstore.csv connected (21 fields, 9,994 rows).

1. Click Sheet 1.



Sheet 1 showing the Data pane, Columns and Rows shelves, Marks card and Show Me panel.

Observation

The interface contains:

- Data Pane
- Columns Shelf
- Rows Shelf
- Marks Card

- Filters Shelf
- Pages Shelf
- Show Me Panel

b) Identify and Differentiate Measures and Dimensions

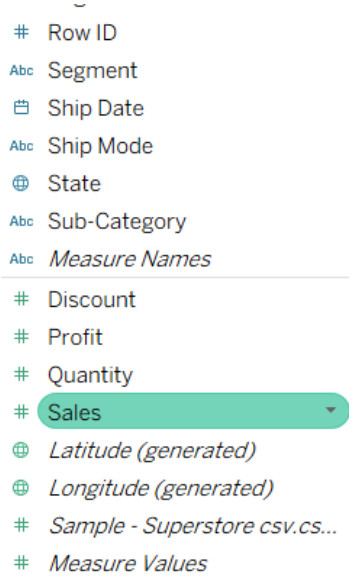
Observe the left Data Pane.

Dimensions

- Order Date
- Customer Name
- Region
- State
- City
- Category
- Segment

Measures

- Sales
- Profit
- Quantity
- Discount

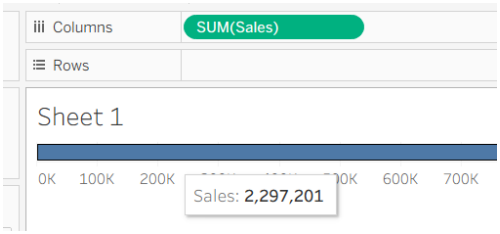


Data pane fields with their data type icons (zoomed view).

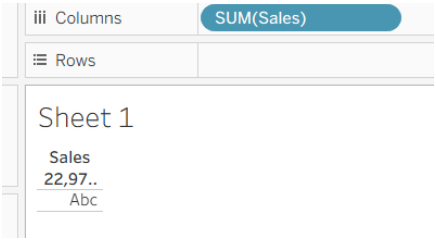
Dimensions	Measures
Qualitative data	Quantitative data
Used for grouping	Used for calculations
Usually text/date	Usually numeric

c) Convert Fields Between Discrete and Continuous

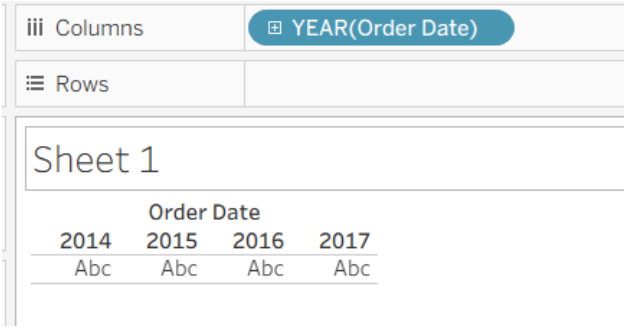
1. Drag Sales to Columns.
2. Click the Sales pill and select Continuous (green).
3. Select Discrete (blue) and observe the chart change.
4. Similarly convert Order Date and Profit.



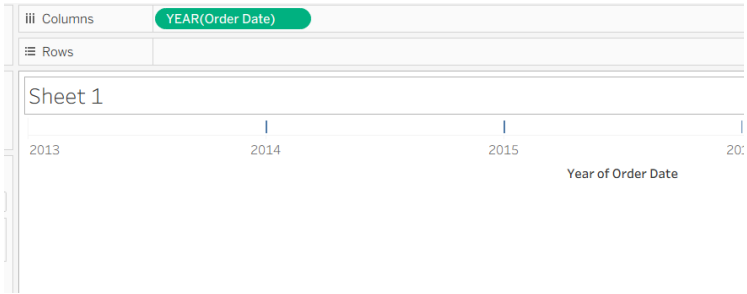
Sales as a Continuous measure (axis).



Sales as a Discrete measure (header).



Order Date as Discrete (year headers).



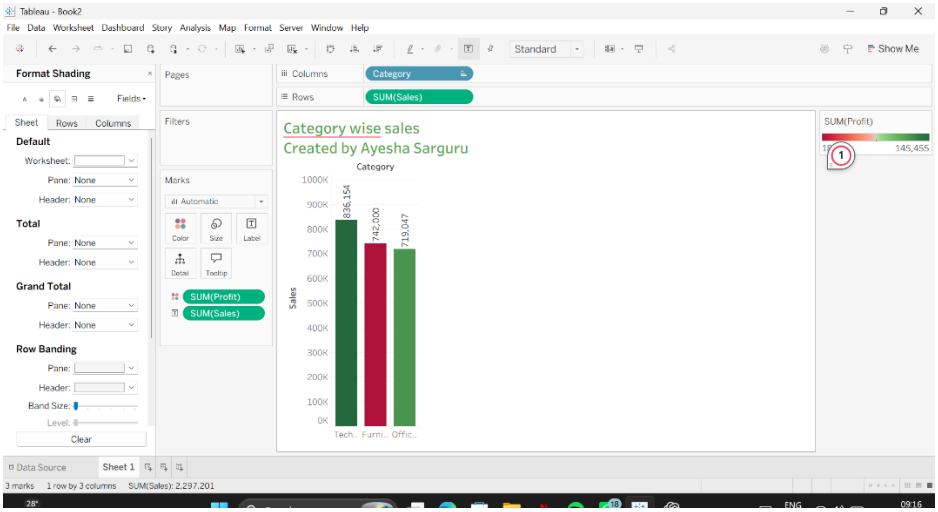
Order Date as Continuous (date axis).

Observation

- Continuous creates an axis.
- Discrete creates headers.

d) Create and Customize a Basic Bar Chart

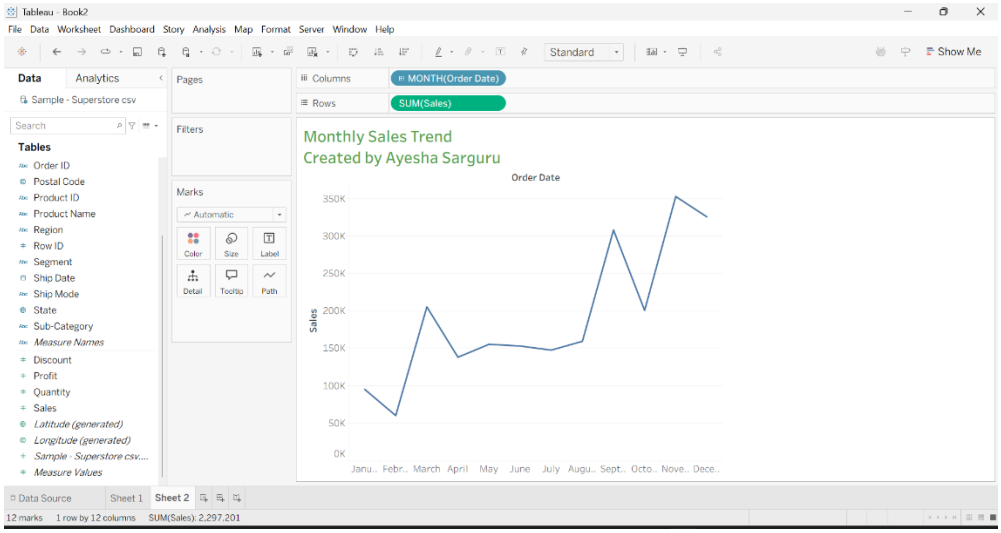
1. Drag Category to Rows.
2. Drag Sales to Columns.
3. Choose Bar Chart from Show Me.
4. Sort Ascending.
5. Drag Profit to Color.
6. Drag Sales to Label.
7. Format colors and titles.



Bar chart: Sales by Category.

e) Build a Line Chart for Time-Series Analysis

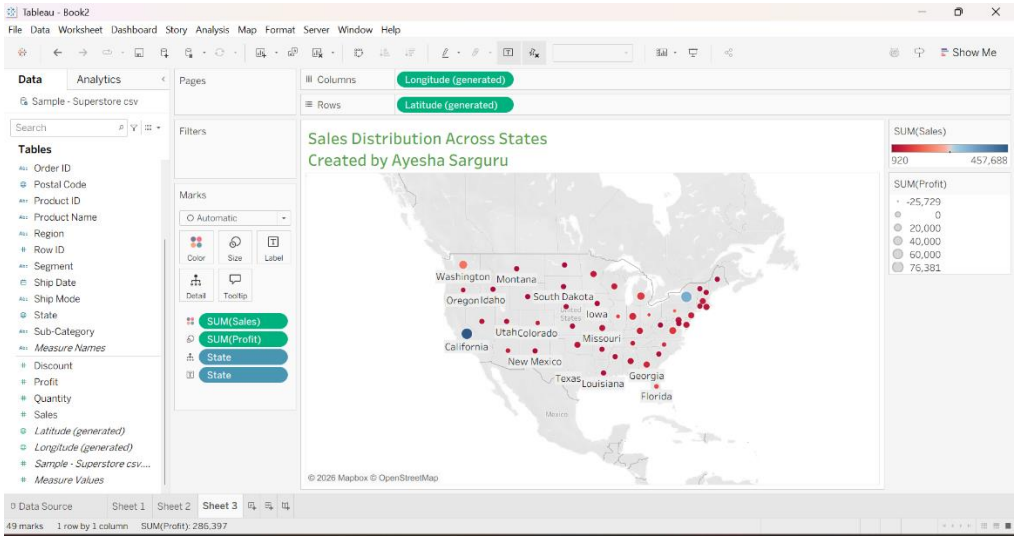
1. Open a new worksheet.
2. Drag Order Date to Columns and select Month.
3. Drag Sales to Rows.
4. Choose Line Chart.
5. Format axis and title.



Line chart: Monthly Sales Trend.

f) Create a Geographic Map Visualization

1. Open another worksheet.
2. Drag State to the view; Tableau automatically generates Latitude and Longitude.
3. Drag Sales to Color.
4. Drag Profit to Size.
5. Drag State to Label.

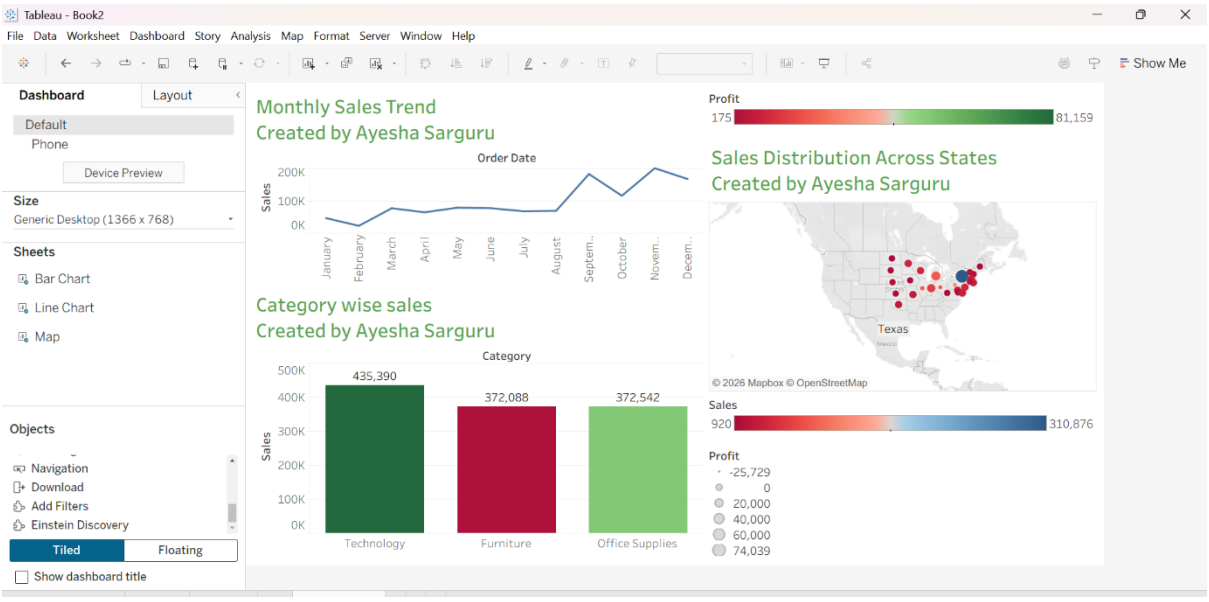


Map: Sales distribution across states.

g) Combine Multiple Charts into an Interactive Dashboard

1. Click New Dashboard.
2. Drag the Bar Chart, Line Chart and Map onto the dashboard.
3. Add a Region filter.
4. Right-click the filter and select Apply to Worksheets → All Using This Data Source.
5. Enable Use as Filter on the Bar Chart.

The dashboard contains Sales by Category, Monthly Sales Trend, Sales by State and a Region filter.



Interactive dashboard combining the Bar Chart, Line Chart and Map sheets.

Result

Successfully connected a CSV dataset in Tableau, explored the interface, differentiated Measures and Dimensions, created bar, line, and map visualizations, and built an interactive dashboard.

Practical 2: Working with Data Sources and Data Preparation in Tableau

Aim: To learn how to connect Tableau to different data sources, prepare data using joins and blending, manage metadata, apply filters, and optimize workbook performance.

Software Required

- Tableau Desktop / Tableau Public
- Microsoft Excel
- CSV Dataset (Sample Superstore.csv)
- Database (MySQL/SQL Server)

Dataset Used

Sample Superstore dataset (9,994 records). For joins and blending, two additional tables are used:

Orders.csv

Order ID	Customer Name	Category	Sales	Profit
CA-2018-100001	John Smith	Furniture	261.96	41.91
CA-2018-100002	David Jones	Technology	731.94	219.58

Returns.csv

Order ID	Returned
CA-2018-100001	Yes
CA-2018-100005	Yes

Procedure

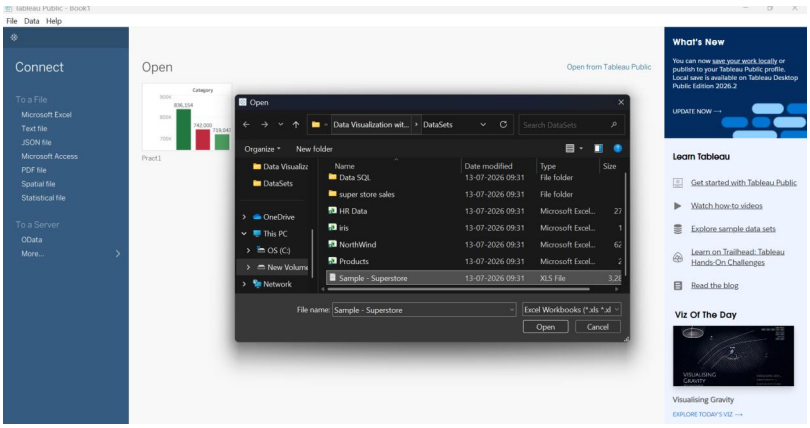
a) Connect to Different Data Sources (Excel, CSV, Database)

CSV File

1. Open Tableau.
2. Click Connect → Text File.
3. Select Sample Superstore.csv.
4. Verify the data preview.

Excel File

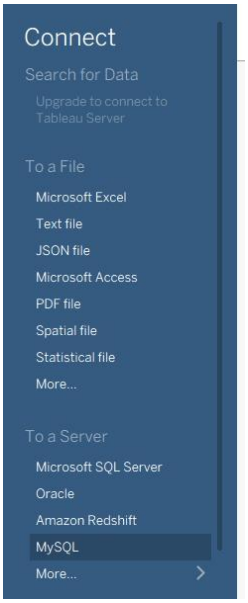
1. Click Connect → Microsoft Excel.
2. Select Sample Superstore.xlsx.
3. Drag the required sheet to the canvas.



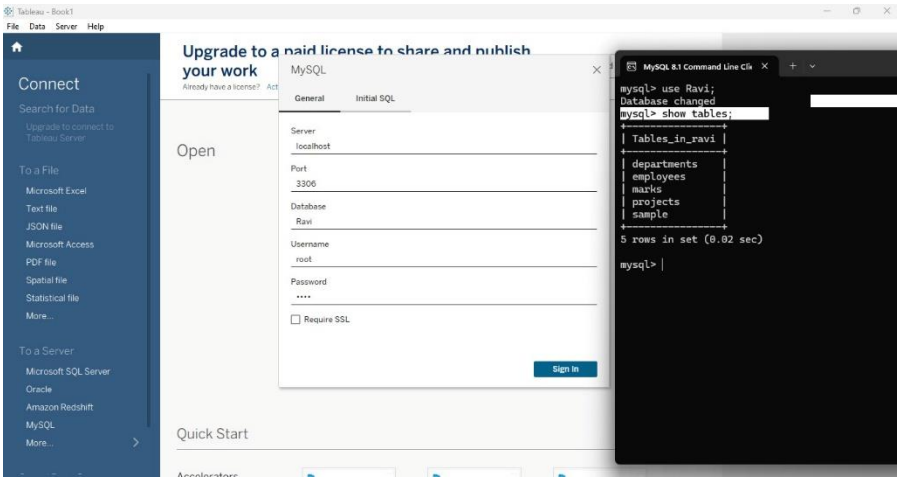
Selecting the Excel file in the Open dialog.

Database Connection

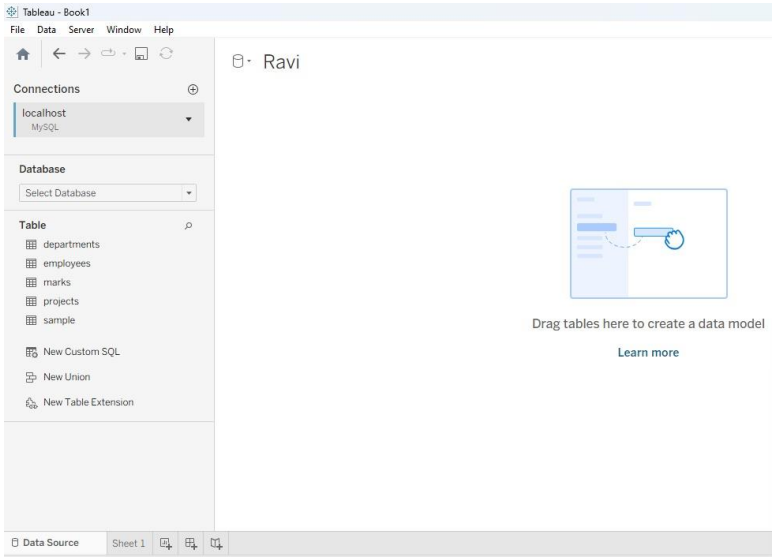
1. Click Connect → MySQL (or SQL Server).
2. Enter the Server Name, Database Name, Username and Password.
3. Click Sign In.
4. Select the required table.



Connect pane listing the data source types.



MySQL connection dialog (server, port, database, username, password).



MySQL database tables listed on the Data Source page.

Observation

Tableau supports Excel, CSV, SQL Server, MySQL, Oracle, PostgreSQL, Google Sheets and cloud databases.

b) Create and Use Data Extracts vs Live Connections

Live Connection

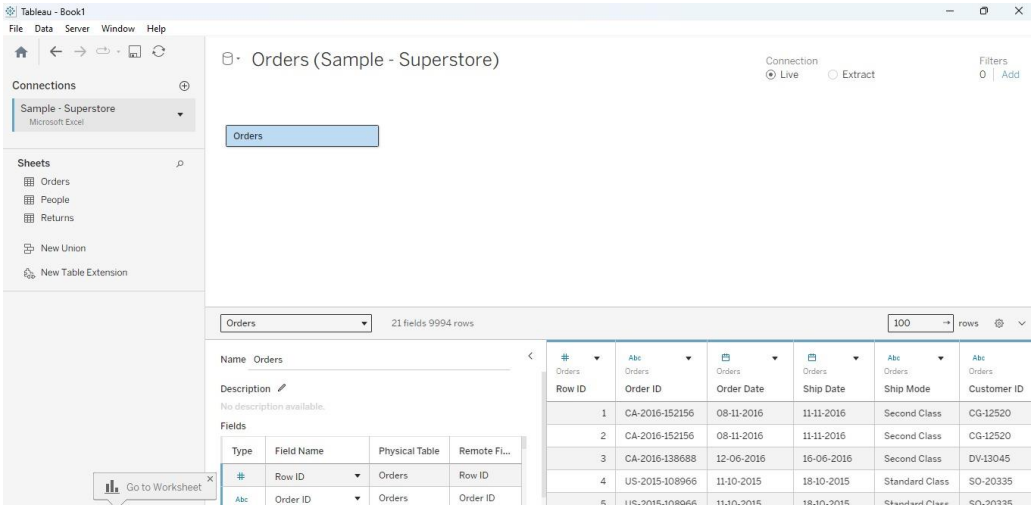
Tableau reads data directly from the source, so updates in the source are reflected immediately.

1. Select the data source.
2. Choose Live.

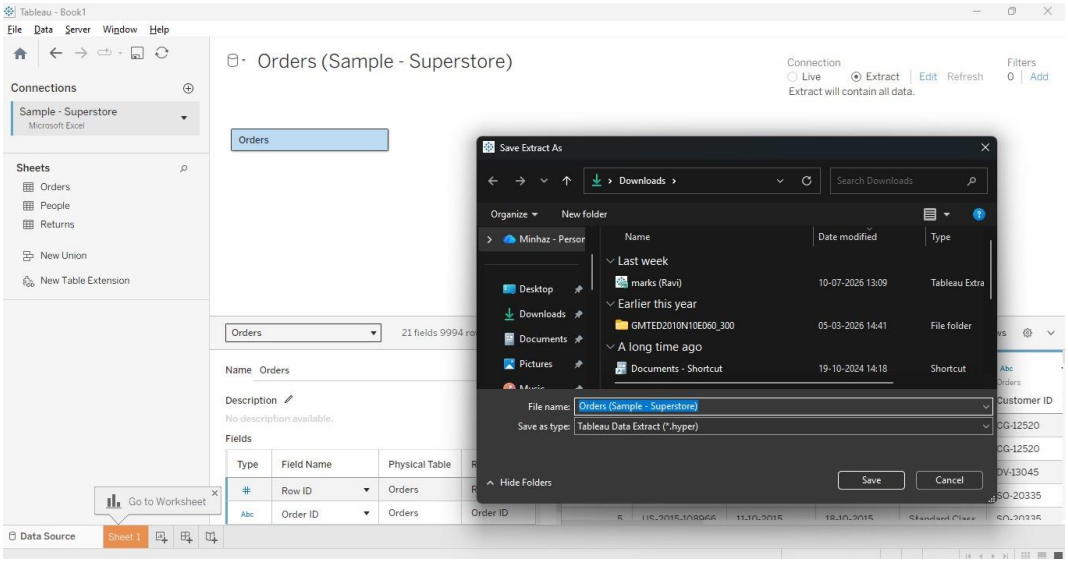
Extract Connection

Tableau creates a local snapshot of the data, which improves performance.

1. Select Extract.
2. Click Create Extract.
3. Save the extract file.



Data Source page showing the Live and Extract options.



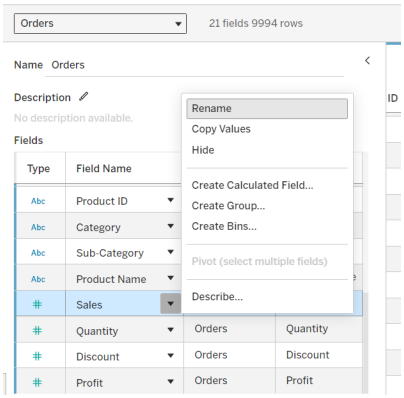
Save Extract As dialog.

Live Connection	Extract
Reads real-time data	Stores a local copy of data
Slower for large datasets	Faster performance
Always up to date	Requires refresh
Depends on database connection	Works offline

c) Manage Metadata

Rename Fields

1. Right-click a field.
2. Select Rename.
3. Enter a new name (example: Sales → Total Sales).



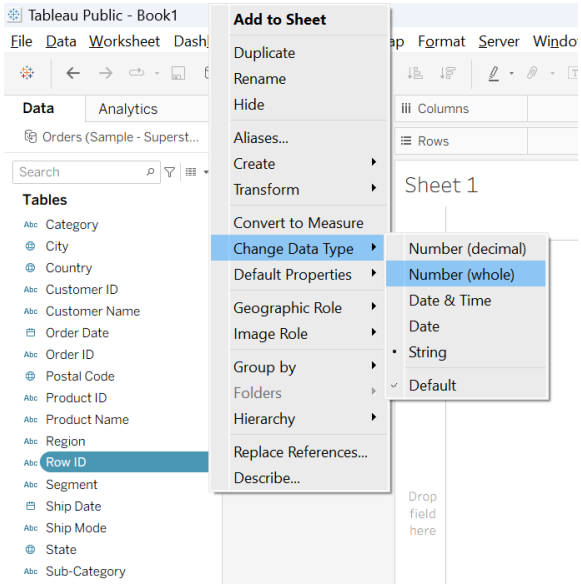
Rename option on a field.

Hide Fields

1. Right-click a field.
2. Select Hide.

Change Data Type

Right-click the field → Change Data Type. Available types: String, Number (Whole), Number (Decimal), Date, Date & Time, Boolean, Geographic Role.



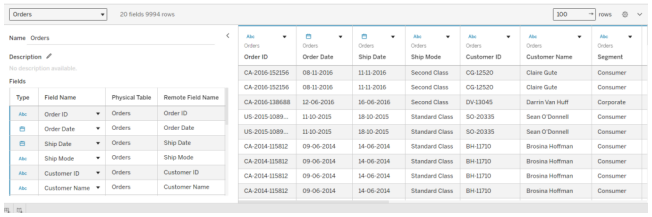
Change Data Type options.

d) Perform Joins (Inner, Left, Right)

1. Open the Data Source page.
2. Drag the Orders table.
3. Drag the Returns table.
4. Select the join icon.
5. Choose Inner, Left, Right or Full Outer.

Orders Table

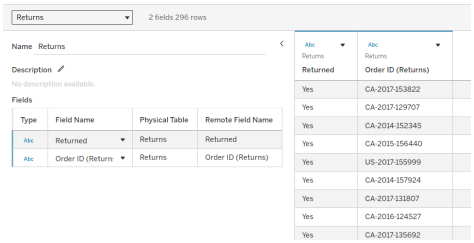




Orders table.

Returns Table

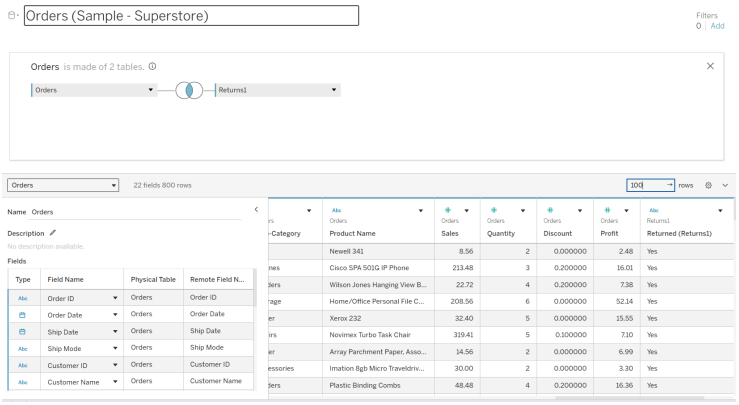




Returns table.

Inner Join

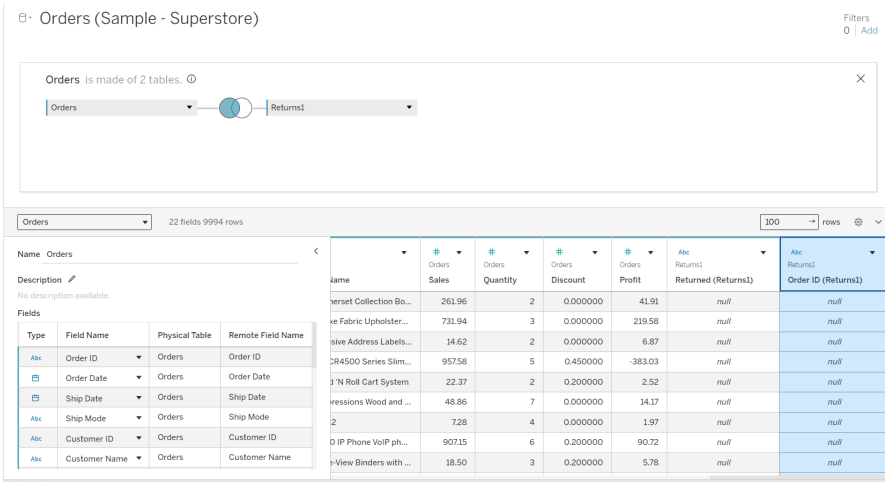
Returns only matching records.



Inner join result.

Left Join

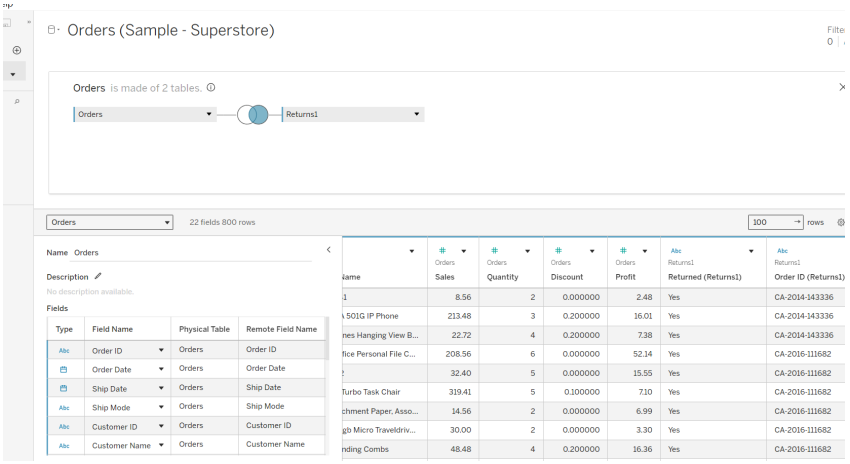
Returns all records from the left table.



Left join result.

Right Join

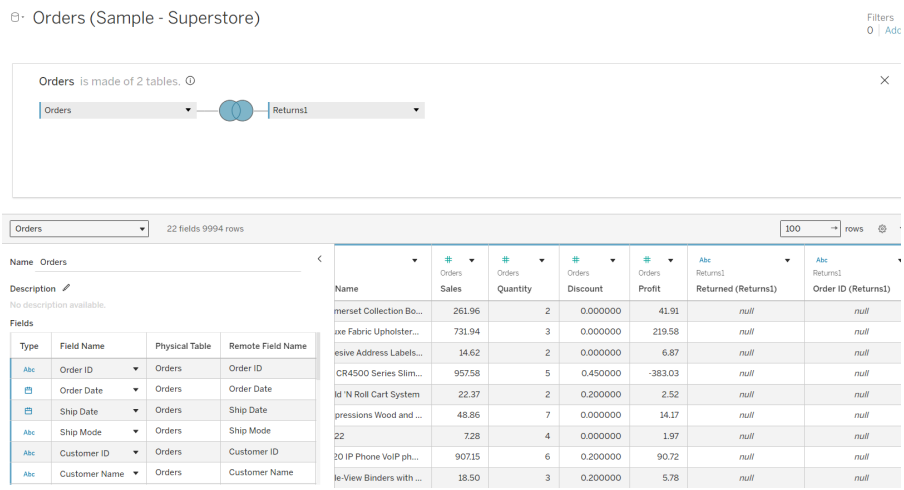
Returns all records from the right table.



Right join result.

Full Outer Join

Returns all records from both tables.

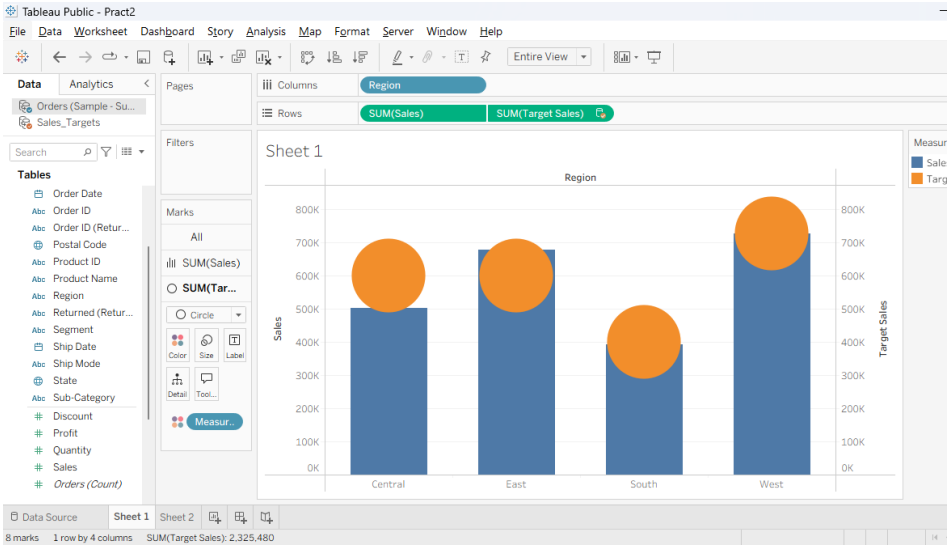


Full outer join result.

e) Apply Data Blending Between Two Sources

Primary source: Orders. Secondary source: Sales_Targets. Common field: Region.

1. Connect Orders and connect Sales_Targets.
2. Open a worksheet.
3. Select Orders as Primary.
4. Select Targets as Secondary.
5. Link using Region.



Blended chart: Sales (bars) and Sales Target (circles) by Region.

Observation

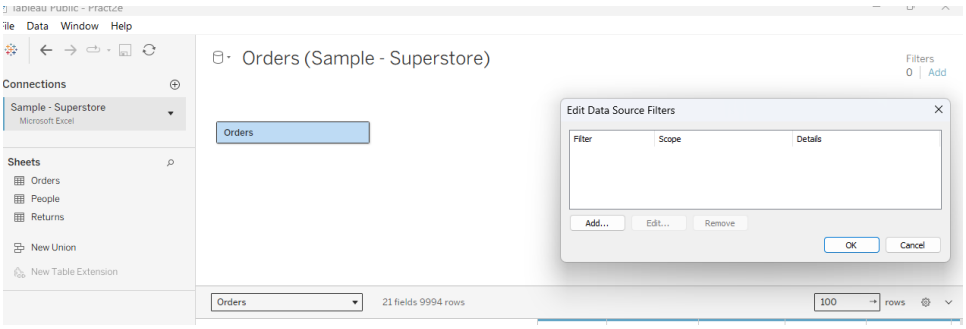
Data blending combines data from multiple sources at the visualization level without physically joining the tables.

f) Implement Different Types of Filters

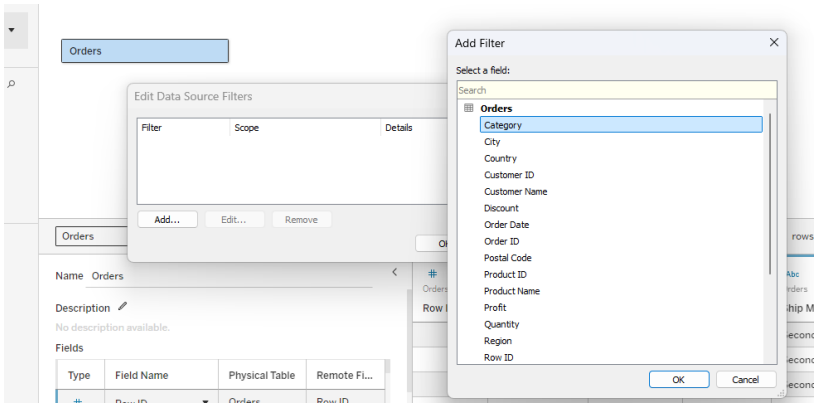
Filter types: Dimension, Measure, Date, Table Calculation, Context, Data Source and Extract filters.

Data Source Filter

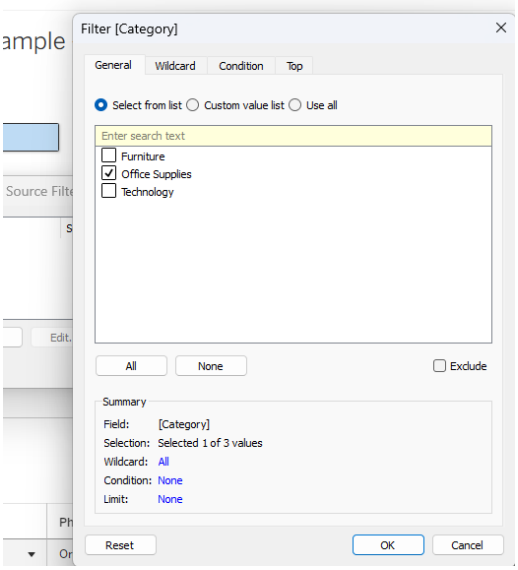
1. Drag Category to Filters.
2. Select Office Supplies.



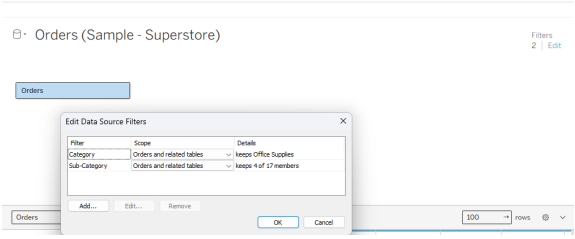
Edit Data Source Filters dialog.



Add Filter dialog.

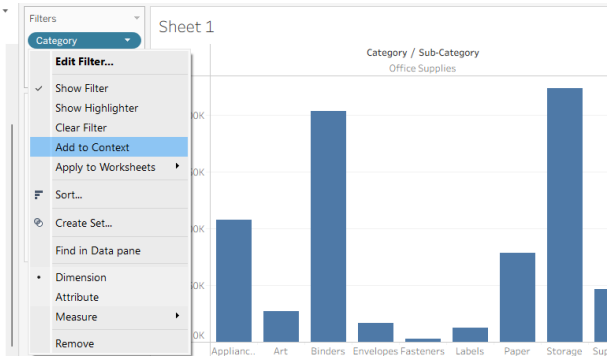


Filter [Category] with Office Supplies selected.

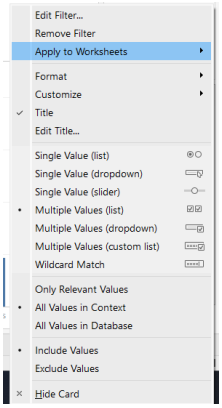


Data source filters listed in the dialog.

Context Filter



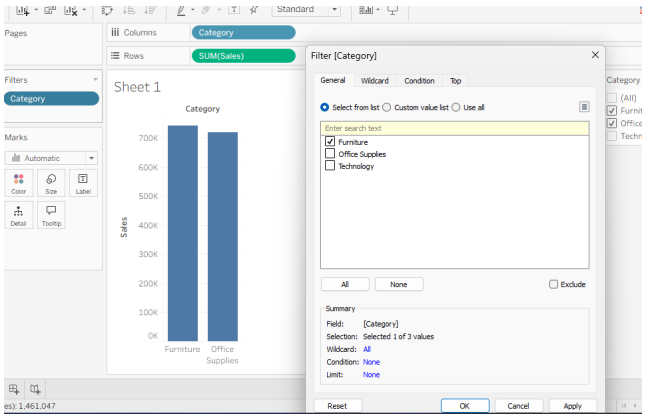
Add to Context option on the filter menu.



Apply to Worksheets option on the filter menu.

Dimension Filter (Category = Furniture)

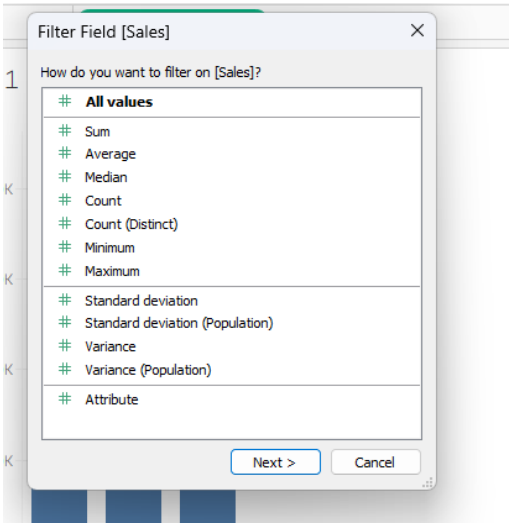
1. Drag Category to Filters.
2. Select Furniture.



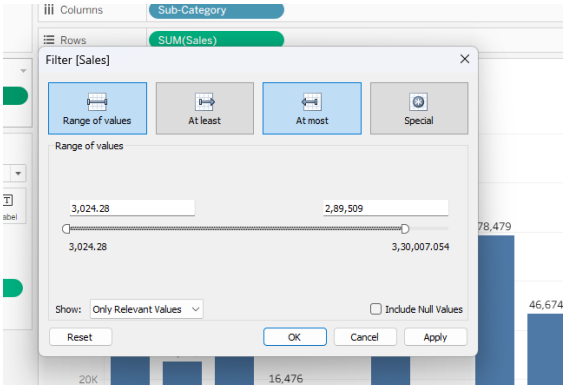
Dimension filter: Filter [Category] dialog.

Measure Filter (Sales > 500)

1. Drag Sales to Filters.
2. Choose SUM.
3. Set the condition.



Filter Field [Sales]: aggregation choices.



Filter [Sales]: range of values.

Table Calculation Filter

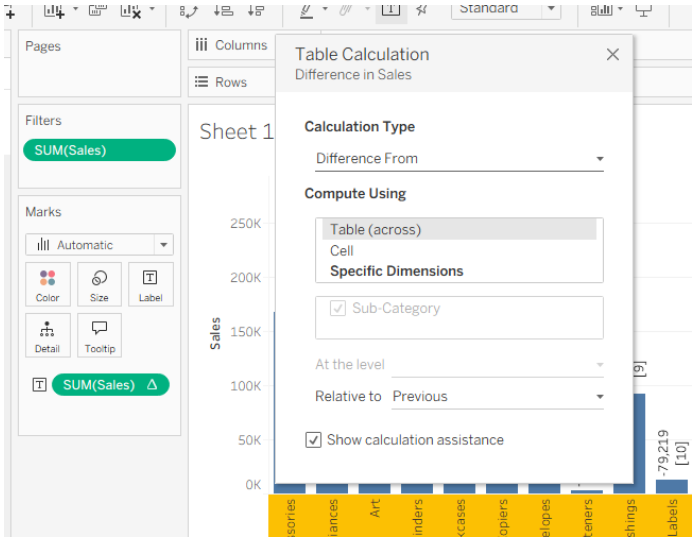
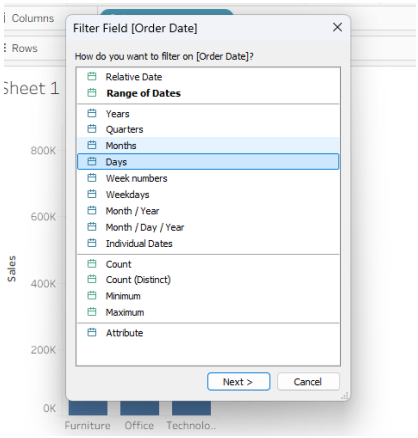


Table Calculation dialog used as a filter.

Date Filter (Year = 2023)

1. Drag Order Date to Filters.
2. Select Year.
3. Choose the required year.



Filter Field [Order Date] options.

g) Optimize Performance Using Extracts and Filtering

- Use an Extract instead of a Live Connection.
- Hide unused fields.
- Remove unnecessary worksheets.
- Apply filters before calculations.
- Limit the number of marks displayed.
- Reduce the number of joins.
- Use context filters when appropriate.
- Aggregate data where possible.

Result

Successfully connected Tableau to Excel, CSV, and a database, created live and extract connections, managed metadata, performed joins and data blending, applied different types of filters, and optimized workbook performance using extracts and filtering.

Practical 3: Creating Advanced Visualizations in Tableau

Aim: To create advanced visualizations in Tableau, including comparison charts, distribution charts, relationship charts, and time-based visualizations for effective data analysis.

Software Required

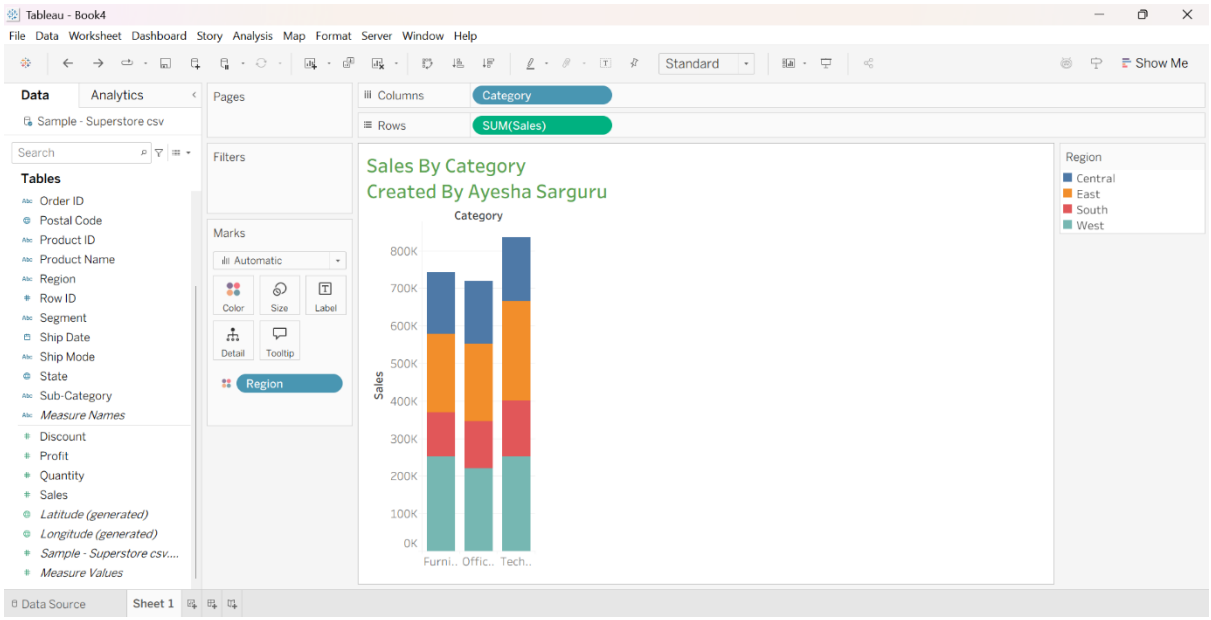
- Tableau Desktop / Tableau Public
- Sample Superstore.csv dataset (9,994 records)

Procedure

a) Create Stacked and Grouped Bar Charts for Comparison

1. Stacked Bar Chart

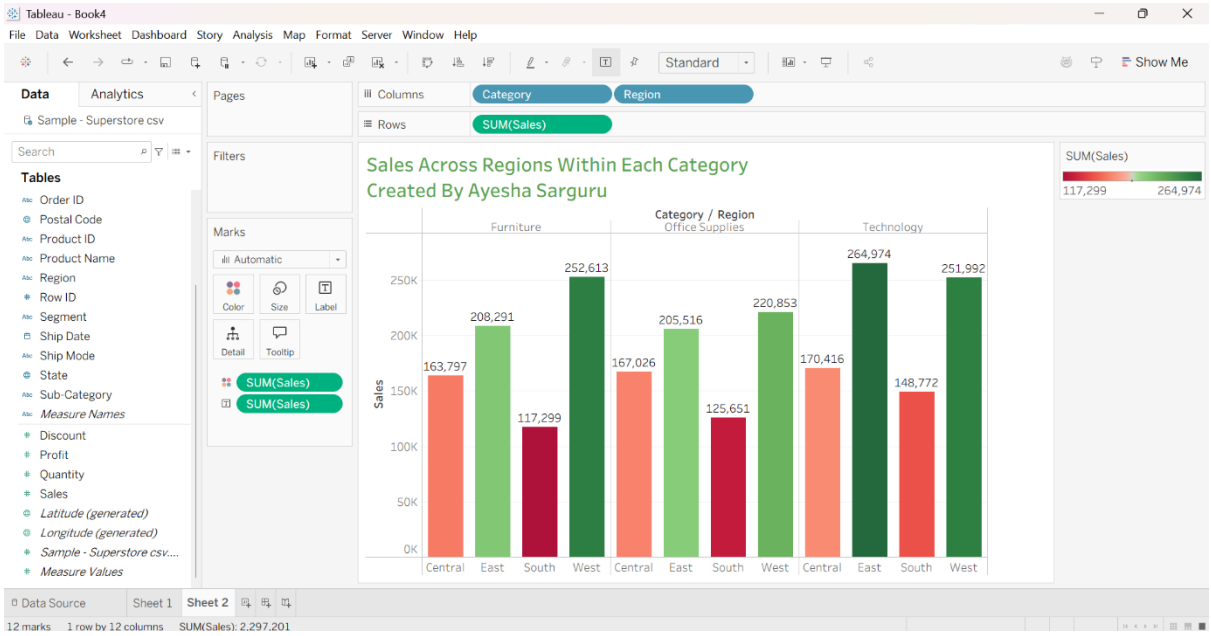
1. Open a new worksheet.
2. Drag Category to Columns.
3. Drag Sales to Rows.
4. Drag Region to Color on the Marks card.
5. Select Bar as the chart type.
6. Ensure Analysis → Stack Marks → On.



Stacked bar chart: Sales by Category, segmented by Region.

2. Grouped (Side-by-Side) Bar Chart

1. Drag Category to Columns.
2. Drag Region next to Category on Columns.
3. Drag Sales to Rows.
4. Choose Bar chart.

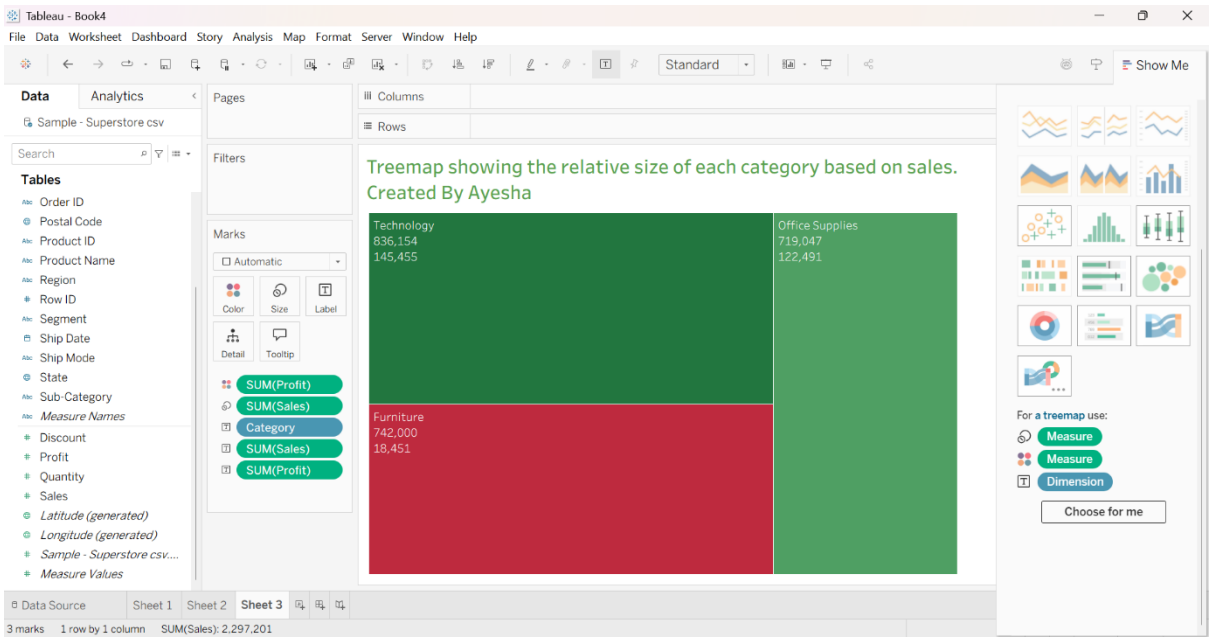


Grouped bars comparing Sales across Regions within each Category.

b) Build Treemaps and Pie Charts for Part-to-Whole Analysis

1. Treemap

1. Open a new worksheet.
2. Click Show Me → Treemap.
3. Drag Category to Label.
4. Drag Sales to Size.
5. Drag Profit to Color.

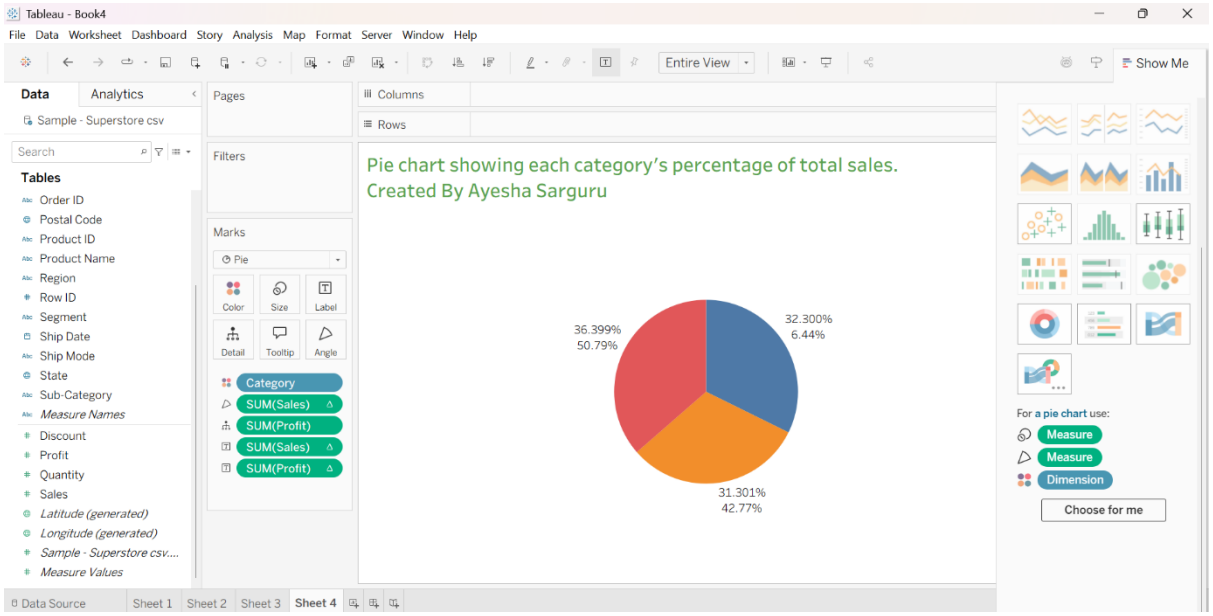


Treemap showing the relative size of each category based on sales.

2. Pie Chart

1. Open a new worksheet.
2. Select Pie from the Marks card.
3. Drag Category to Color.
4. Drag Sales to Angle.
5. Drag Sales to Label.

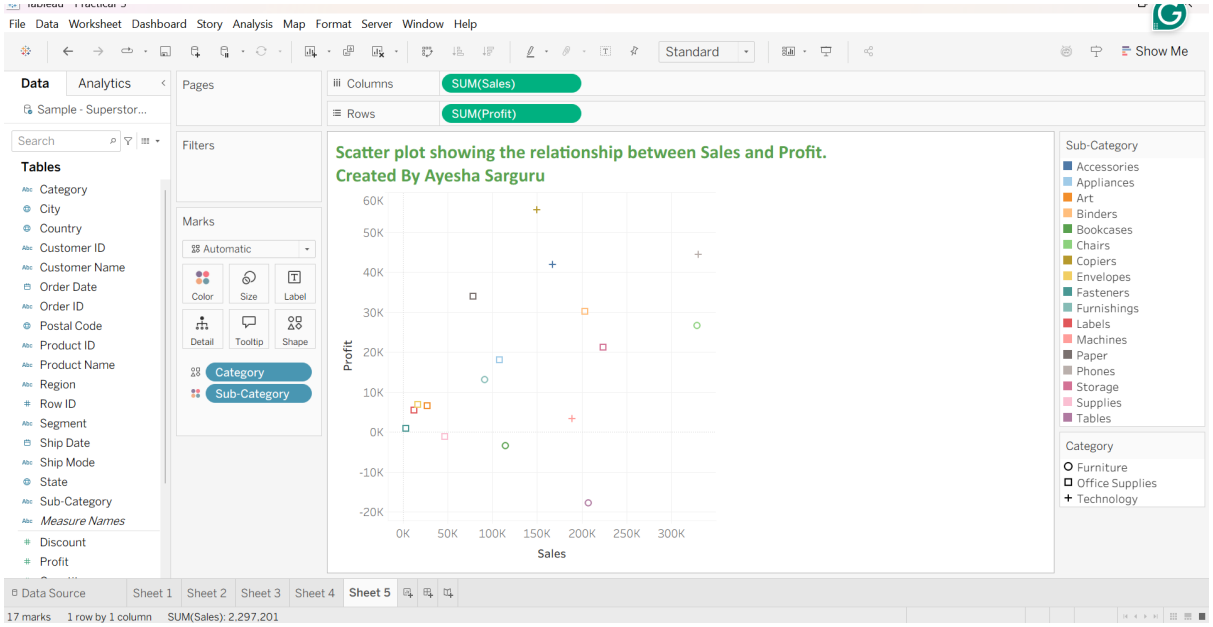
6. Click Quick Table Calculation → Percent of Total.



Pie chart showing each category's percentage of total sales.

c) Design Scatter Plots to Identify Relationships

1. Open a new worksheet.
2. Drag Sales to Columns.
3. Drag Profit to Rows.
4. Drag Category to Color.
5. Drag Sub-Category to Detail.
6. Select Scatter Plot from Show Me.



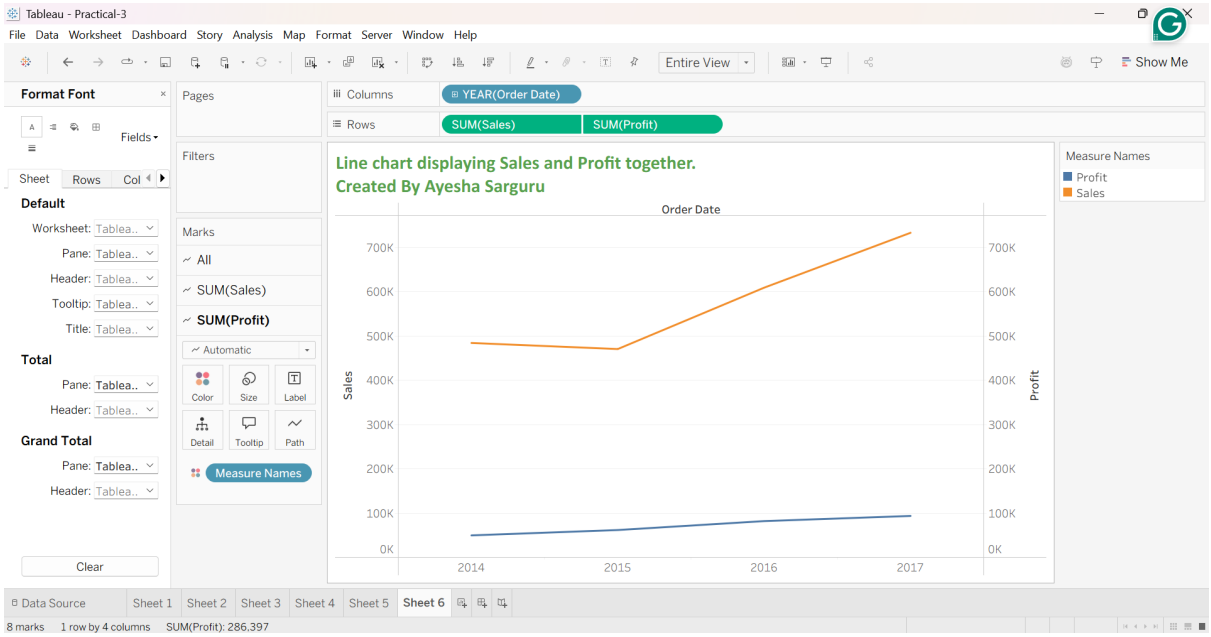
Scatter plot showing the relationship between Sales and Profit.

d) Create Dual-Axis and Combination Charts

1. Dual-Axis Chart

1. Drag Order Date to Columns.
2. Drag Sales to Rows.
3. Drag Profit to the Rows shelf beside Sales.
4. Right-click the second axis and select Dual Axis.

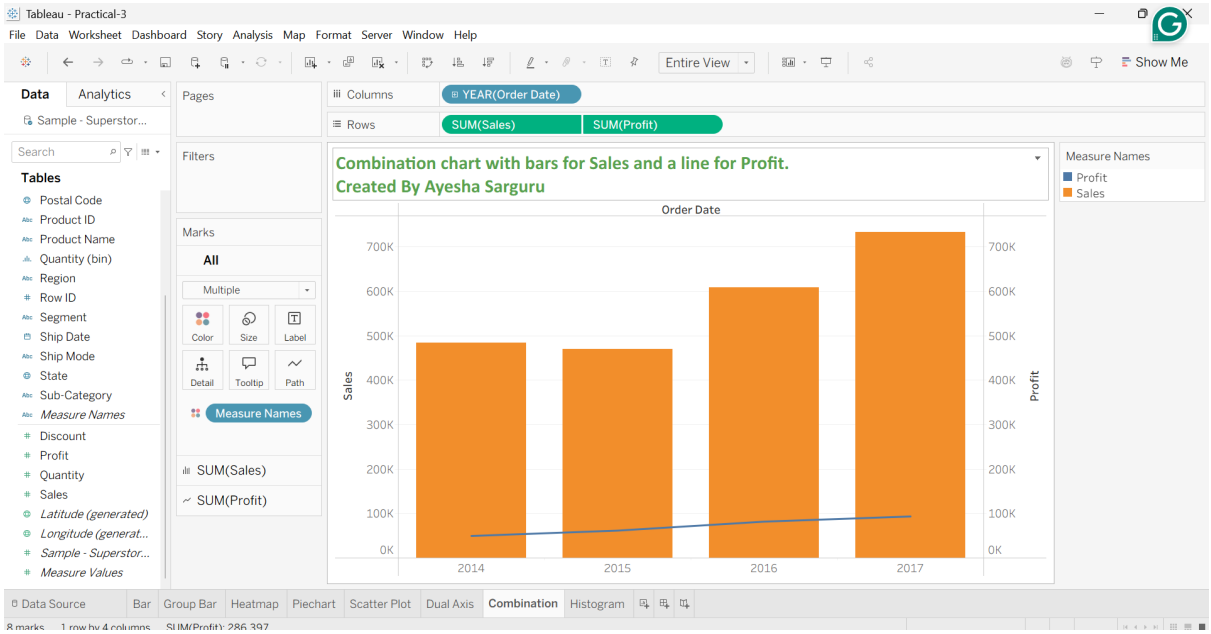
5. Synchronize the axes if required.



Line chart displaying Sales and Profit together.

2. Combination Chart

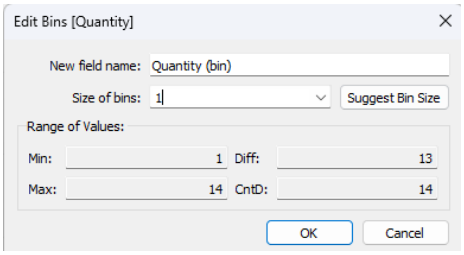
1. Create a dual-axis chart.
2. Set Sales as Bar.
3. Set Profit as Line.



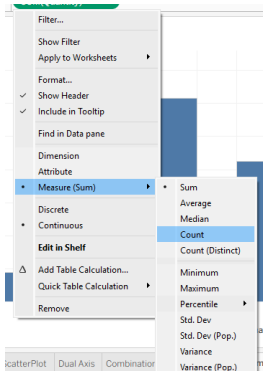
Combination chart with bars for Sales and a line for Profit.

e) Develop Histograms for Distribution Analysis

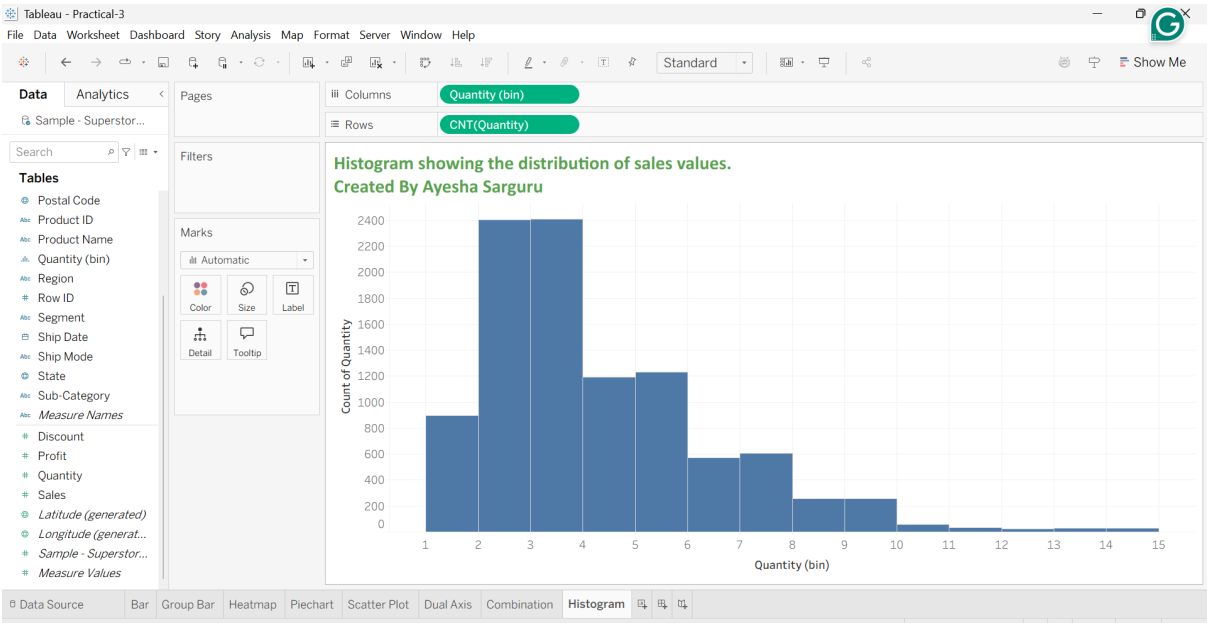
1. Open a new worksheet.
2. Right-click Quantity → Create → Bins.
3. Change the default bin size to 1.
4. Drag Quantity (bin) to Columns.
5. Drag Sum(Quantity) to Rows and change the aggregation to Count.
6. Select Histogram from Show Me.



Create Bins dialog for Quantity.



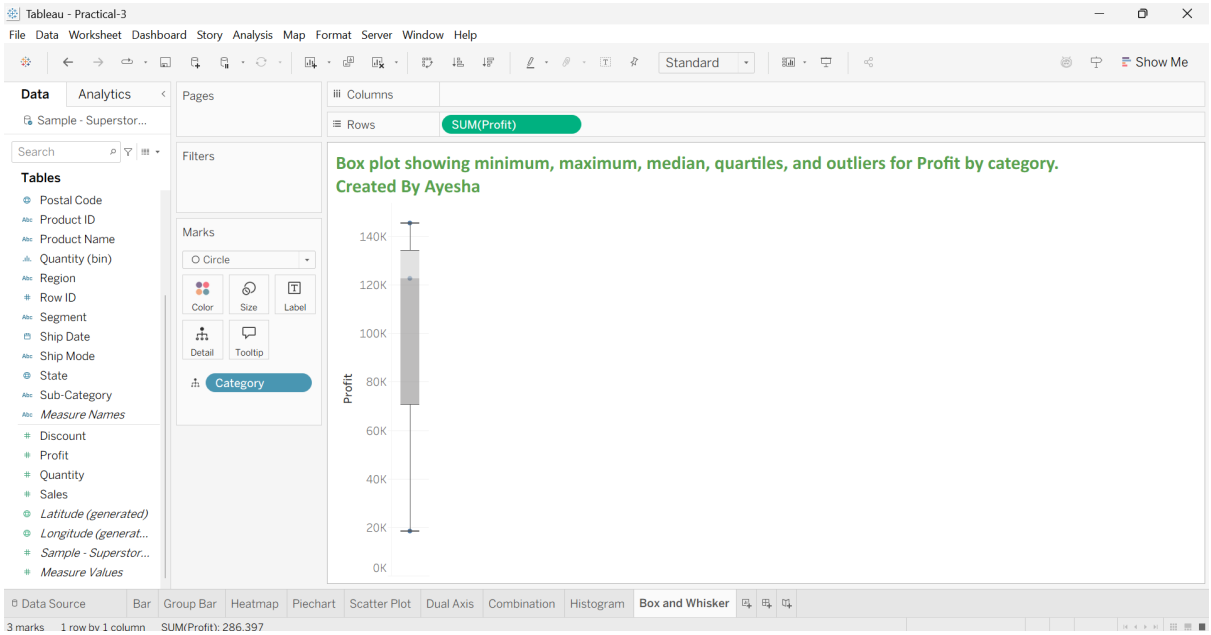
Quantity (bin) and Count of Quantity placed on the shelves.



Histogram showing the distribution of Quantity.

f) Build Box-and-Whisker Plots

1. Open a new worksheet.
2. Drag Category to Columns.
3. Drag Profit to Rows.
4. Click Show Me → Box-and-Whisker Plot.



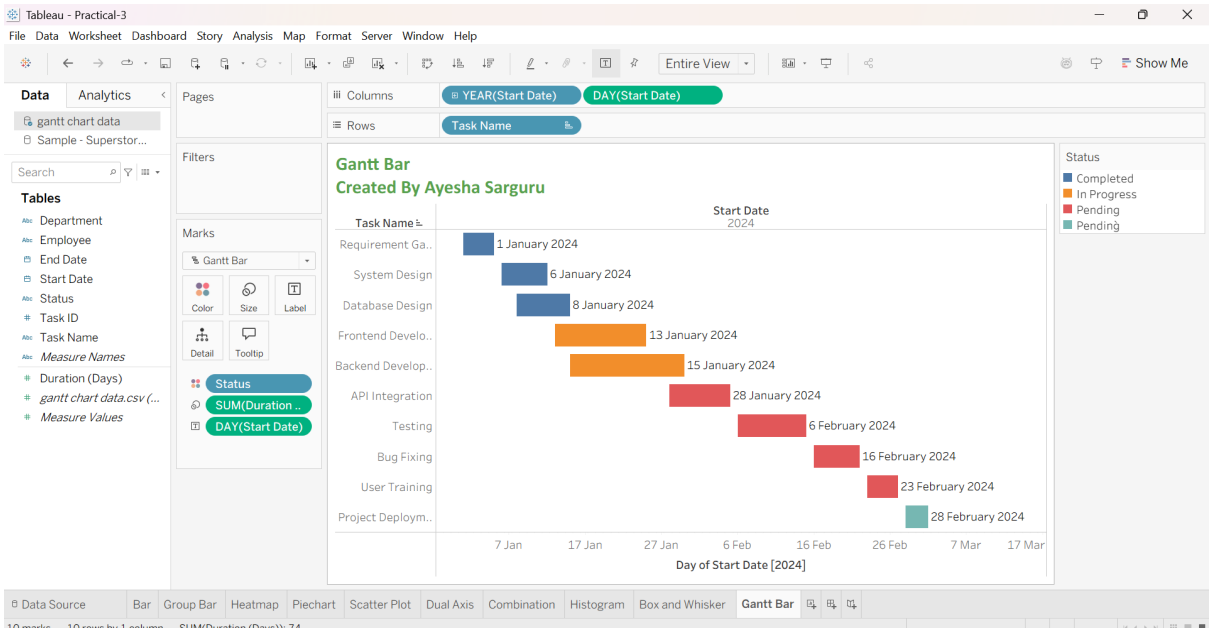
Box plot showing minimum, maximum, median, quartiles and outliers for Profit by category.

g) Use Gantt Charts and Advanced Date Visualizations

1. Gantt Chart

1. Open a new worksheet.
2. Connect to Project_Tasks.csv.
3. Drag Task Name to Rows.
4. Drag Start Date to Columns.
5. Change the Marks type to Gantt Bar.
6. Drag Duration (Days) to Size. If only Start Date and End Date are available, create a calculated field Duration = DATEDIFF('day', [Start Date], [End Date]).
7. Drag Status to Color.
8. Drag Employee to Label (optional).
9. Format colors, labels and title.

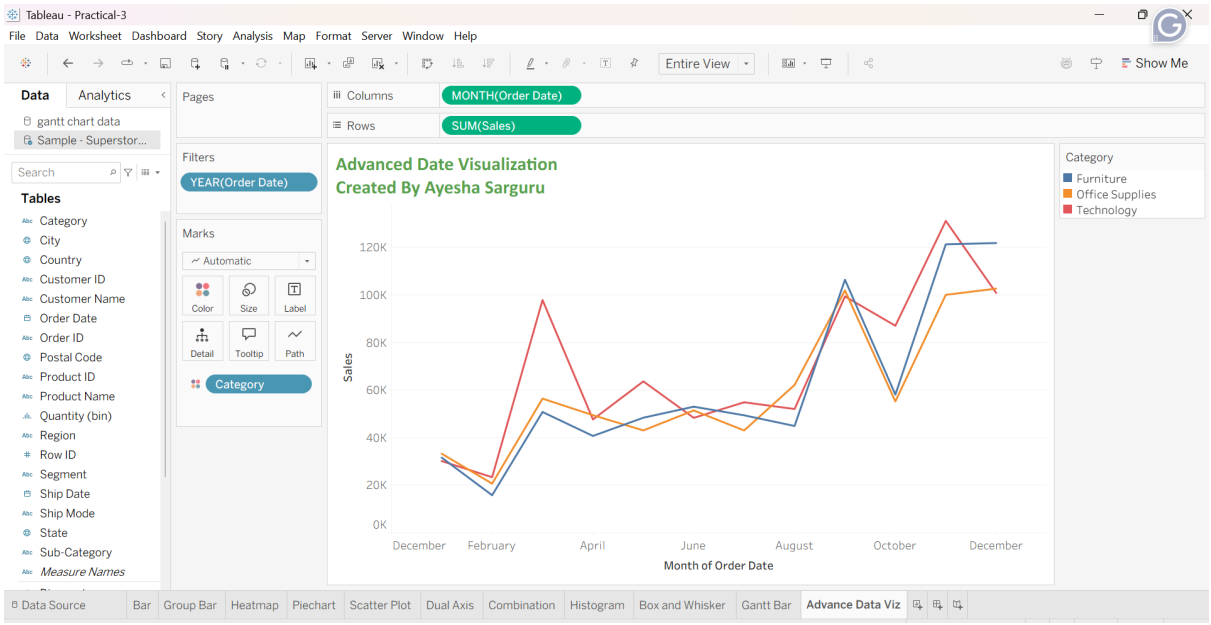
Each bar represents a task: its position shows the Start Date, its length the Duration (Days), and its color the Status (Completed, In Progress, Pending).



Gantt chart: tasks by start date, colored by Status.

2. Advanced Date Visualization

1. Drag Order Date to Columns.
2. Select Month (Continuous).
3. Drag Sales to Rows.
4. Drag Category to Color.
5. Add Year as a filter if required.



Monthly sales trend by category.

Result

Successfully created advanced Tableau visualizations including stacked and grouped bar charts, treemaps, pie charts, scatter plots, dual-axis and combination charts, histograms, box-and-whisker plots, Gantt charts, and advanced date visualizations using the Sample Superstore dataset.

Practical 4: Performing Data Analysis using Calculations in Tableau

Aim: To perform data analysis in Tableau using calculated fields, row-level calculations, aggregate calculations, Level of Detail (LOD) expressions, parameters, and data transformation techniques.

Software Required

- Tableau Desktop / Tableau Public
- Sample Superstore.csv dataset (9,994 records)

Procedure

a) Create Basic Calculated Fields Using Formulas

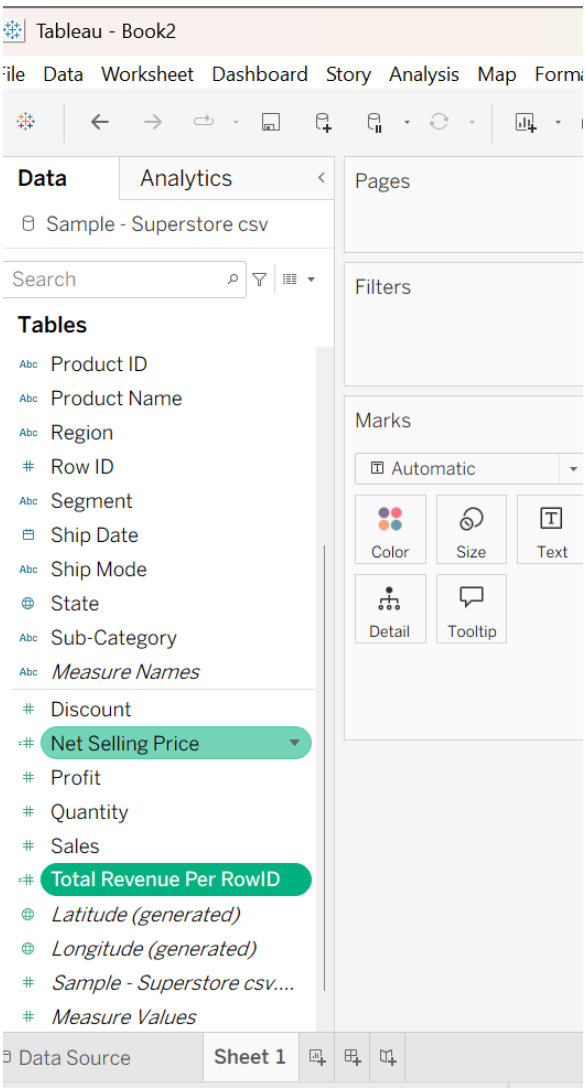
1. Open Tableau.
2. Right-click in the Data Pane.
3. Select Create → Calculated Field.
4. Enter the field name.
5. Write the formula.
6. Click OK.

Example 1: Net Selling Price

```
[Sales]-[Sales]*[Discount]
```

Example 2: Total Revenue Per RowID

```
[Net Selling Price]*[Quantity]
```



Calculated fields added to the Data pane.

b) Perform Row-Level Calculations

Row-level calculations are performed individually for each record before aggregation.

Example 1: Profit Percentage

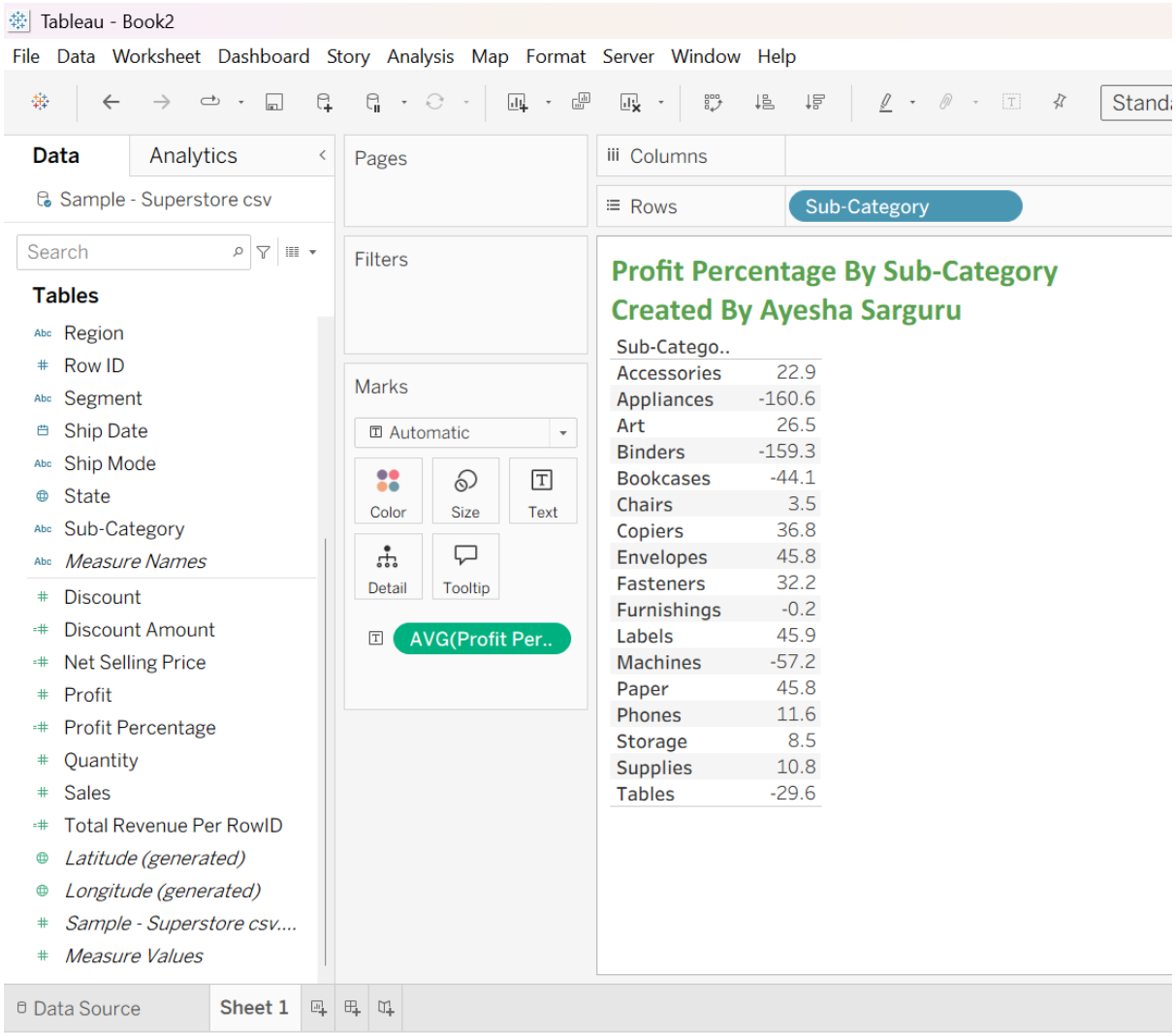
```
([Profit] / [Net Selling Price]) * 100
```

Example 2: Discount Amount

```
[Sales] * [Discount]
```

Example 3: Shipping Days Duration

```
DATEDIFF('day',[Order Date],[Ship Date])
```



Profit Percentage by Sub-Category.

c) Perform Aggregate-Level Calculations

Aggregate calculations operate on summarized data.

Example 1: Total Sales

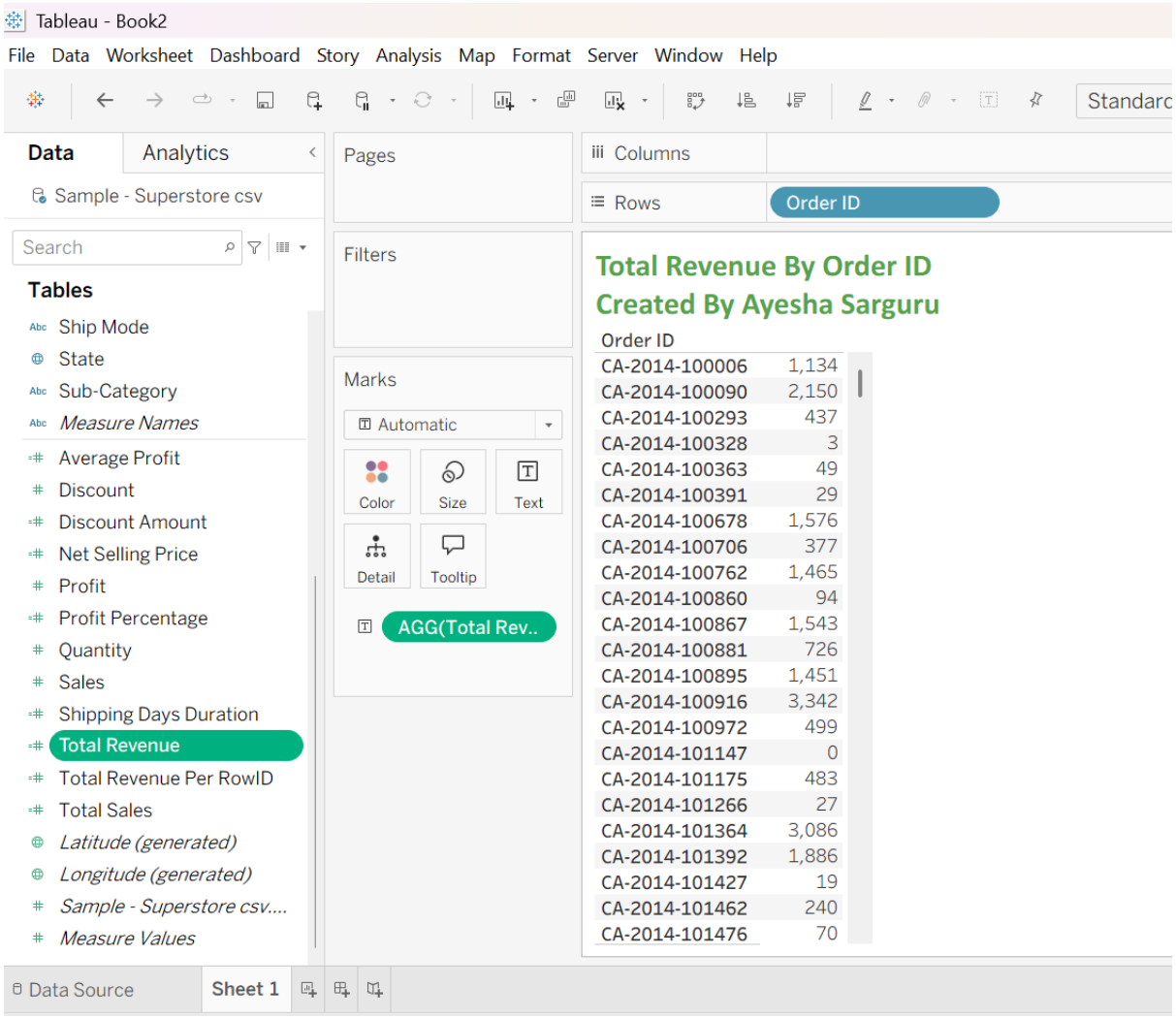
```
SUM([Sales])
```

Example 2: Average Profit

```
AVG([Profit])
```

Example 3: Total Revenue

```
SUM([Total Revenue Per RowID])
```



Total Revenue by Order ID.

Output

Values are calculated after grouping the data.

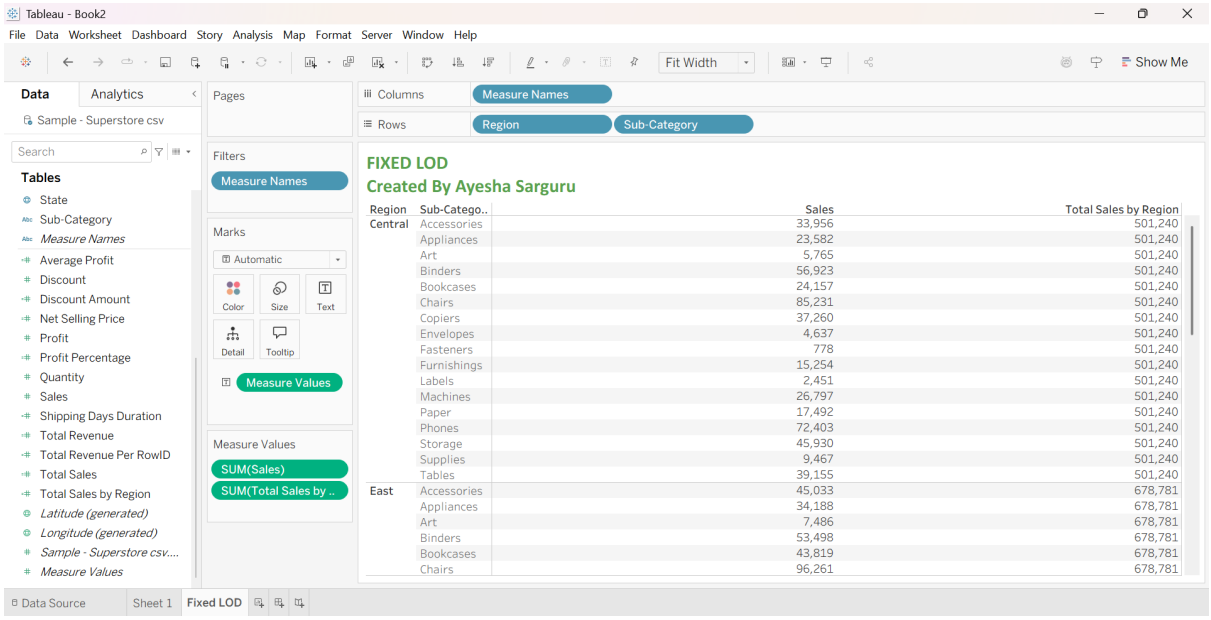
d) Implement FIXED Level of Detail (LOD) Calculations

FIXED LOD calculates values at a specified level of detail, regardless of the view.

Example 1: Total Sales by Region

```
{ FIXED [Region] : SUM([Sales]) }
```

1. Connect to the Sample - Superstore dataset.
2. Right-click in the Data pane → Create Calculated Field and name it Total Sales by Region.
3. Enter the formula above and click OK.
4. Drag Region and Sub-Category to Rows.
5. Drag Measure Names to Columns and Measure Values to Text on the Marks card.
6. In Measure Values keep only SUM(Sales) and SUM(Total Sales by Region).
7. Drag Measure Names to Filters and select only Sales and Total Sales by Region.



FIXED LOD: Total Sales by Region repeated for every Sub-Category.

Example 2: Average Profit by Category

```
{ FIXED [Category] : AVG([Profit]) }
```

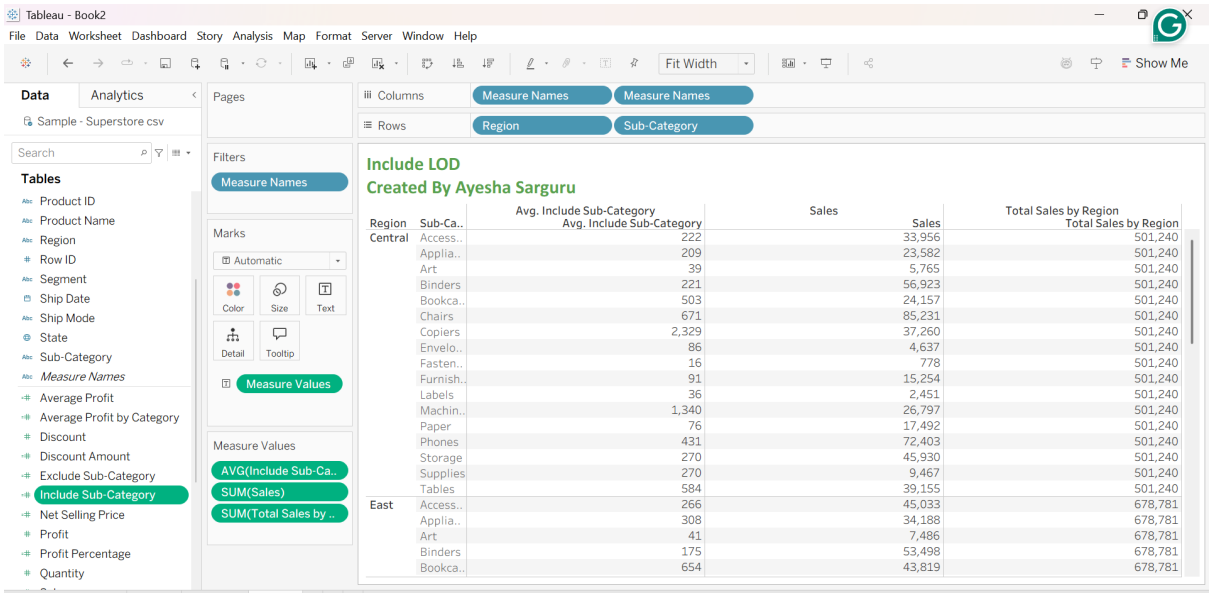
Each Region or Category displays the same fixed value, even if additional dimensions are added.

e) Use INCLUDE and EXCLUDE LOD Expressions

INCLUDE (adds a dimension to the calculation level)

```
{ INCLUDE [Sub-Category] : SUM([Sales]) }
```

1. Create a calculated field named Include Sub-Category with the formula above.
2. Drag Region and Sub-Category to Rows.
3. Drag Measure Names to Columns and Measure Values to Text.
4. Keep SUM(Sales), SUM(Total Sales by Region) and the Include field in Measure Values; change the Include field aggregation to AVG.

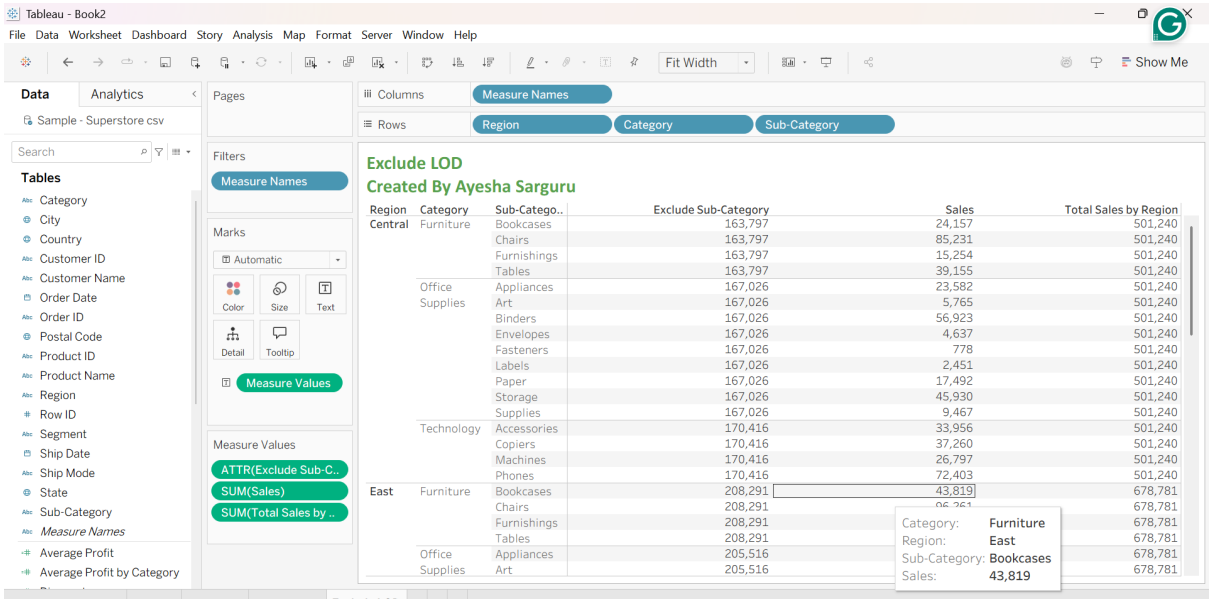


INCLUDE LOD result.

EXCLUDE (removes a dimension from the calculation)

```
{ EXCLUDE [Sub-Category] : SUM([Sales]) }
```

1. Create a calculated field named Exclude Sub-Category with the formula above.
2. Drag Region, Category and Sub-Category to Rows.
3. Drag Measure Names to Columns and Measure Values to Text.
4. Keep SUM(Sales), SUM(Total Sales by Region) and the Exclude field in Measure Values.



EXCLUDE LOD result.

FIXED	INCLUDE	EXCLUDE
Ignores view level	Adds dimensions	Removes dimensions
Independent of visualization	Depends on view	Depends on view
Most commonly used	More detailed calculations	Higher-level summaries

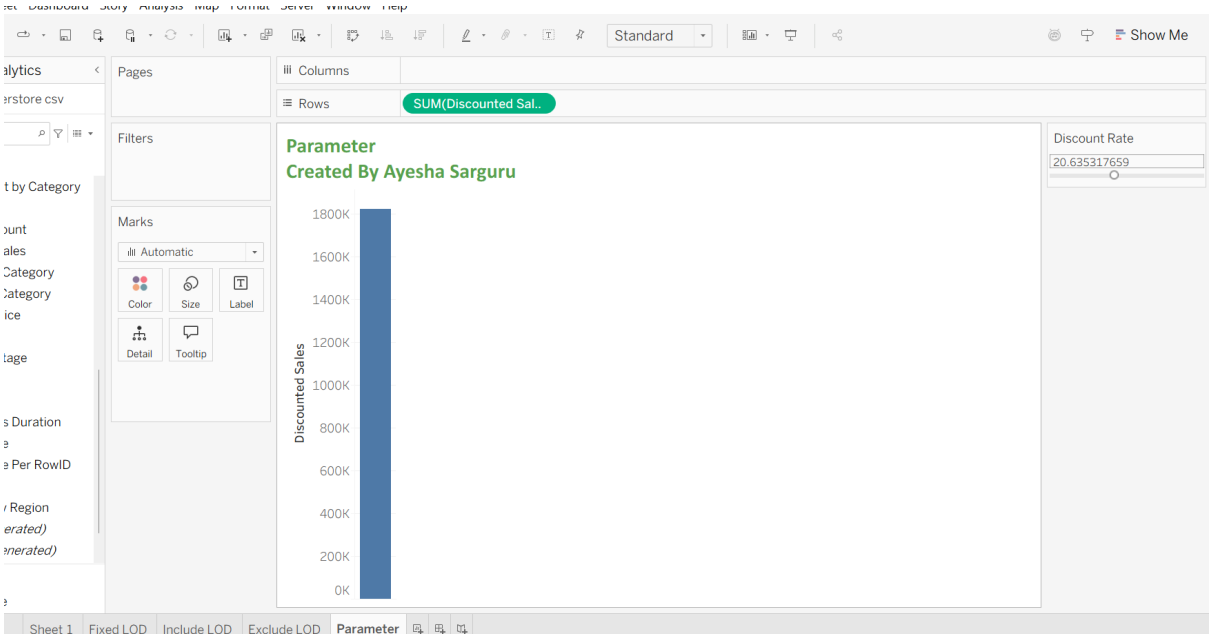
f) Create and Use Parameters for Dynamic Analysis

Parameters allow users to interactively change values used in calculations. Example: Discount Percentage.

1. Right-click the Data Pane.
2. Select Create Parameter.
3. Configure: Name: Discount Rate; Data Type: Float; Current Value: 10; Range: 0 to 50.
4. Click OK.
5. Right-click the parameter.
6. Select Show Parameter.

Create Calculated Field

```
[Sales] - ([Sales] * [Discount Rate] / 100)
```



Discount Rate parameter control with the Discounted Sales bar.

Changing the parameter dynamically updates the calculated sales values.

g) Apply Calculations to Fix Data Issues and Enhance Insights

Example 1: Replace Null Values

```
IFNULL([Profit],0)
```

Example 2: Handle Missing Sales

```
ZN([Sales])
```

Example 3: Profit Category

```
IF [Profit] > 0 THEN
"Profit"
ELSE
"Loss"
END
```

Example 4: Sales Classification

```
IF [Sales] >= 1000 THEN
"High"
ELSEIF [Sales] >= 500 THEN
"Medium"
ELSE
"Low"
END
```

Example 5: Shipping Status

```
IF DATEDIFF('day',[Order Date],[Ship Date]) <= 3 THEN
"Fast Delivery"
ELSE
"Delayed"
```

```
END
```

These calculations improve data quality, classify records, and provide better analytical insights.

Result

Successfully created calculated fields, performed row-level and aggregate calculations, implemented FIXED, INCLUDE, and EXCLUDE LOD expressions, created parameters for dynamic analysis, and applied calculations to clean data and generate meaningful business insights in Tableau.

Practical 5: Analyzing Data using Table Calculations in Tableau

Aim: To perform advanced data analysis using Table Calculations in Tableau, including running totals, moving averages, year-over-year growth, ranking, window functions, addressing and partitioning, and dynamic titles.

Software Required

1. Tableau Desktop / Tableau Public
2. Sample Superstore.csv dataset (9,994 records)

Dataset Used

Sample Superstore (9,994 records)

Important Fields

1. Order Date
2. Sales
3. Profit
4. Quantity
5. Category
6. Sub-Category
7. Region
8. Customer Name
9. State

Procedure

a) Apply Quick Table Calculations (Running Total, Percent of Total)

1. Running Total

Purpose

Displays a cumulative total over time.

Steps

1. Open a new worksheet.
2. Drag Order Date (Month) to Columns.
3. Drag Sales to Rows.
4. Right-click SUM(Sales).
5. Select Quick Table Calculation → Running Total.

Output

The chart displays cumulative sales month by month.

2. Percent of Total

Purpose

Shows each category's contribution to the overall total.

Steps

1. Drag Category to Rows.
2. Drag Sales to Columns.
3. Right-click SUM(Sales).
4. Select Quick Table Calculation → Percent of Total.

Output

Each category displays its percentage contribution to total sales.

b) Create Moving Averages and Cumulative Metrics

Moving Average

Purpose

Smooths fluctuations in data over time.

Steps

1. Create a line chart using Order Date and Sales.
2. Right-click SUM(Sales).
3. Select Quick Table Calculation → Moving Average.

Output

Displays the average sales across a moving time window.

Cumulative Metric

Formula

```
RUNNING_SUM(SUM([Sales]))
```

Output

Shows cumulative sales over time.

c) Perform Year-over-Year (YoY) Growth Analysis

Purpose

Compares performance with the previous year.

Steps

1. Drag Year(Order Date) to Columns.
2. Drag Sales to Rows.
3. Right-click SUM(Sales).
4. Select Quick Table Calculation → Year over Year Growth (or use Percent Difference From).

Alternative Formula

```
(SUM([Sales]) - LOOKUP(SUM([Sales]),-1))
/
ABS(LOOKUP(SUM([Sales]),-1))
```

Output

Displays yearly growth percentage.

d) Use Ranking Functions (Top N, Bottom N)

Rank Function

Formula

```
RANK(SUM([Sales]))
```

Top 10 Products

Formula

```
RANK(SUM([Sales])) <= 10
```

Bottom 10 Products

Formula

```
RANK(SUM([Sales]),'asc') <= 10
```

Output

Displays products ranked by sales.

e) Implement Window Functions (WINDOW_SUM, WINDOW_AVG)

WINDOW_SUM

Purpose

Calculates the total across a specified window.

Formula

```
WINDOW_SUM(SUM([Sales]))
```

WINDOW_AVG

Purpose

Calculates the average over a specified window.

Formula

```
WINDOW_AVG(SUM([Sales]))
```

WINDOW_MAX

```
WINDOW_MAX(SUM([Sales]))
```

WINDOW_MIN

```
WINDOW_MIN(SUM([Sales]))
```

Output

Displays aggregated values within the selected partition.

f) Customize Addressing and Partitioning

Purpose

Control how table calculations are computed.

Steps

1. Right-click the Table Calculation.
2. Select Edit Table Calculation.
3. Choose:
    a. Table Across
    b. Table Down
    c. Pane Across
    d. Pane Down

e. Specific Dimensions

Example

Running Total

Addressing:

1. Month

Partitioning:

1. Region

Output

Running totals are calculated separately for each region.

g) Create Dynamic Titles and Labels using Table Calculations

Dynamic Title

Create a Calculated Field:

```
"Total Sales : " + STR(WINDOW_SUM(SUM([Sales])))
```

Insert the calculated field into the worksheet title.

Dynamic Label

Formula

```
"Sales = " + STR(SUM([Sales]))
```

Place the calculated field on Label in the Marks card.

Dynamic Rank Label

```
"Rank : " + STR(RANK(SUM([Sales])))
```

Output

Titles and labels update automatically as filters or data change.

Common Table Calculation Functions

Function	Purpose
RUNNING_SUM()	Running total
RUNNING_AVG()	Running average
WINDOW_SUM()	Sum over a window
WINDOW_AVG()	Average over a window
WINDOW_MAX()	Maximum value in a window
WINDOW_MIN()	Minimum value in a window
LOOKUP()	Access previous or next values
RANK()	Ranking values
INDEX()	Row number within a partition
FIRST()	Offset from the first row
LAST()	Offset from the last row

Result

Successfully applied quick table calculations, created moving averages and cumulative metrics, performed year-over-year growth analysis, implemented ranking and window functions, customized addressing and partitioning, and created dynamic titles and labels using table calculations in Tableau.

Practical 5: Visualization Outputs

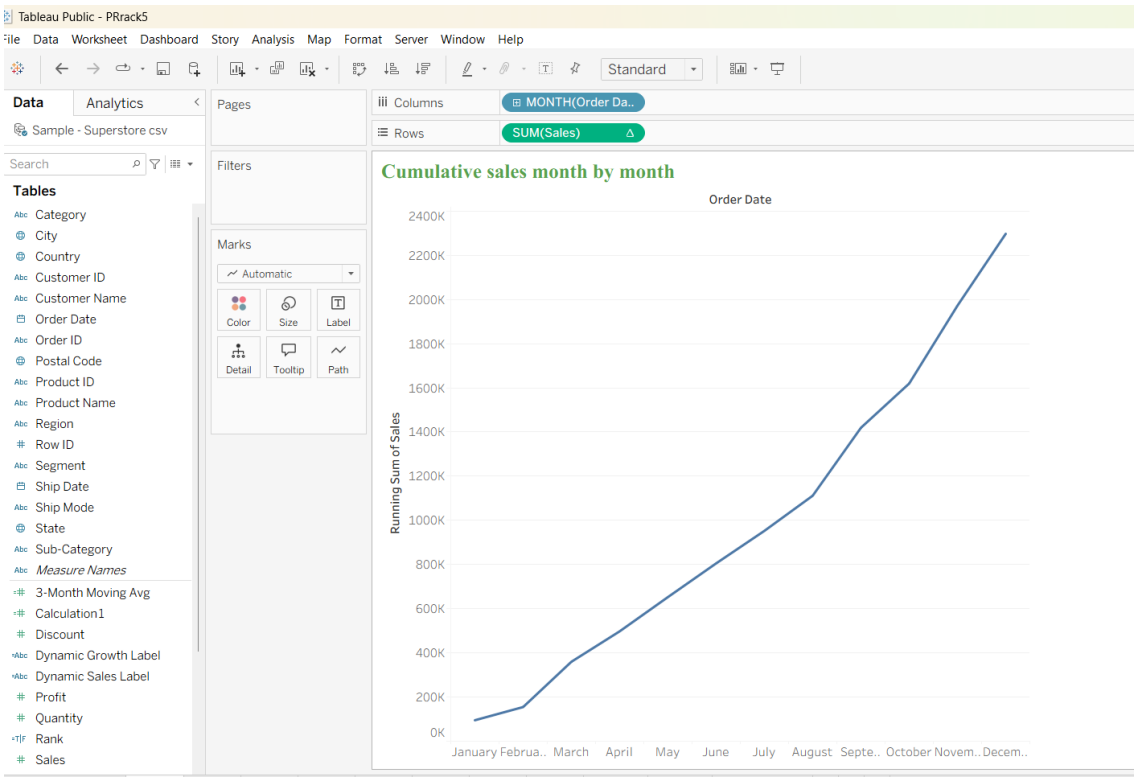


Figure 26: Running Total: cumulative sales month by month.

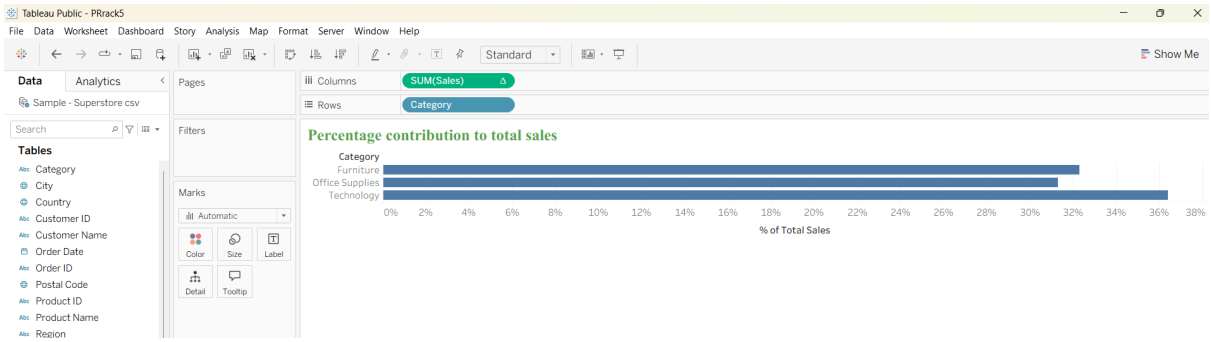


Figure 27: Percent of Total: each Category's contribution to total sales.

B.)

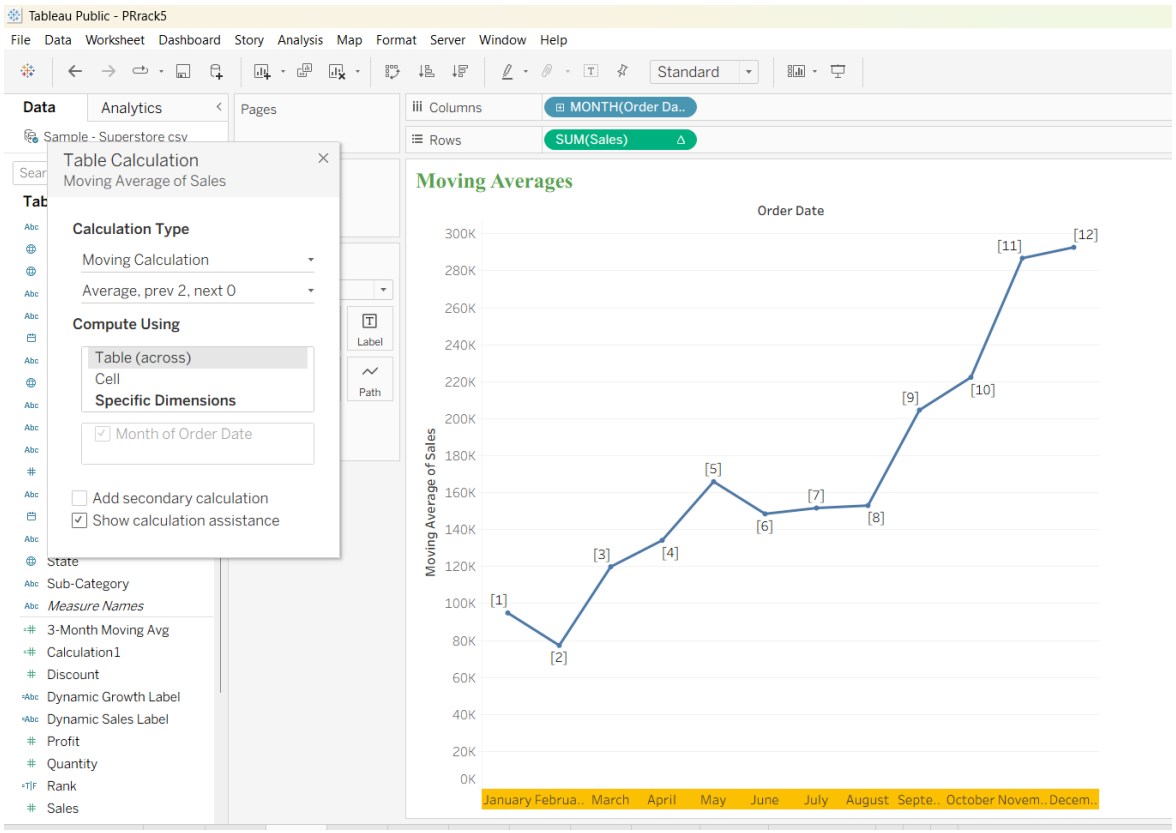


Figure 28: Moving Average of Sales with the Table Calculation dialog.

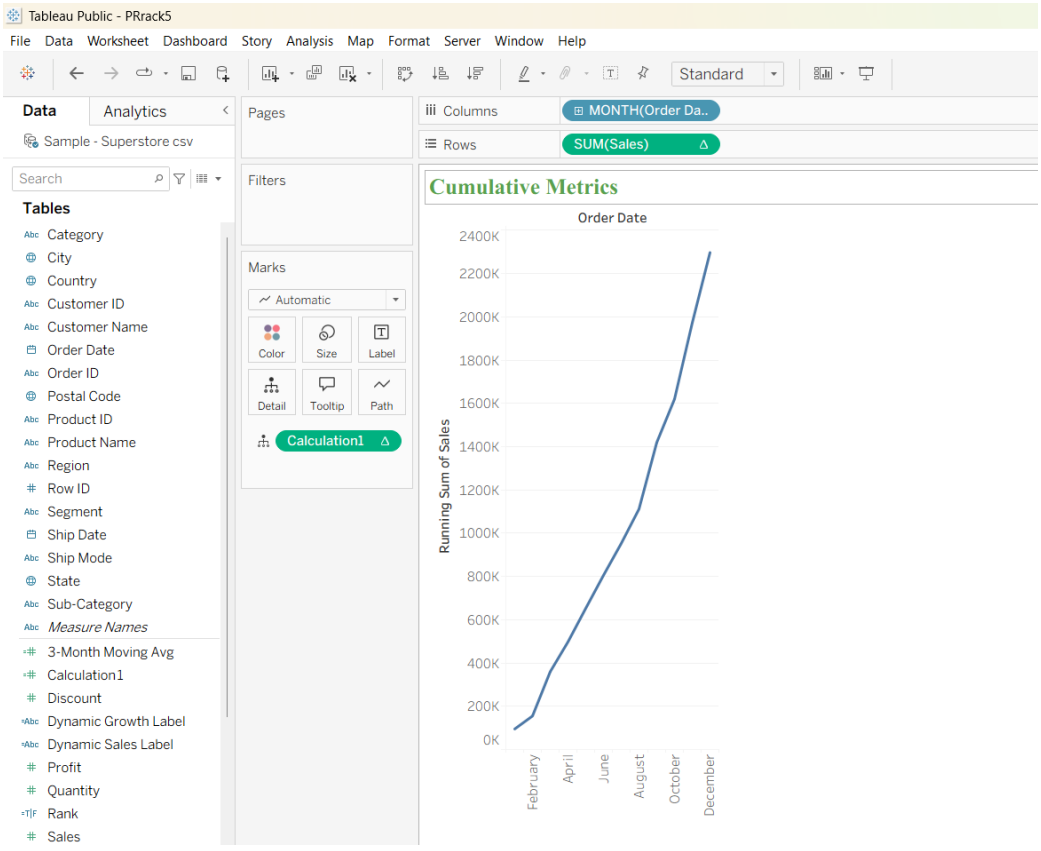


Figure 29: Cumulative Metrics: running sum of Sales.

c.)

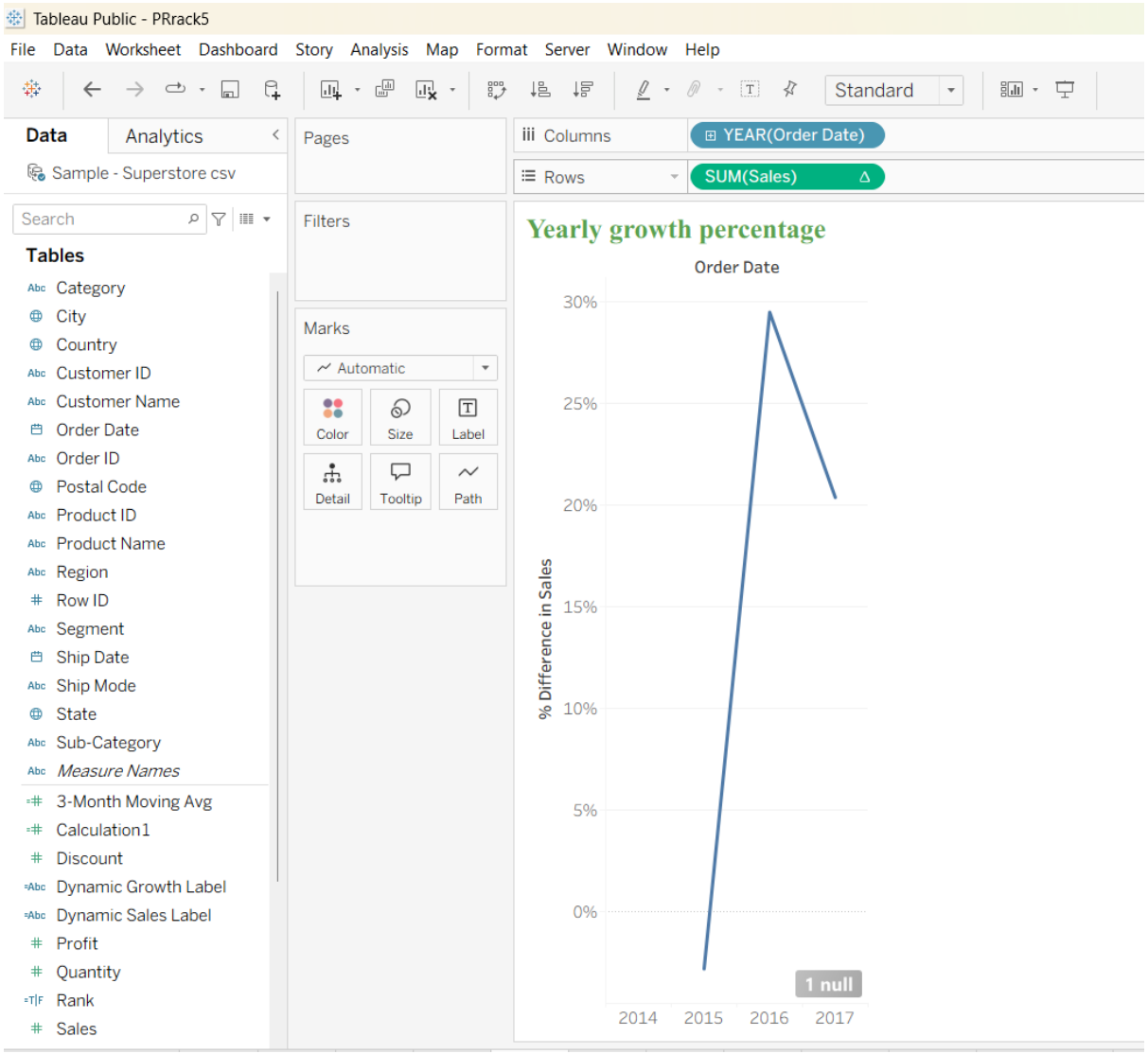


Figure 30: Year-over-year growth percentage of Sales.

D.)

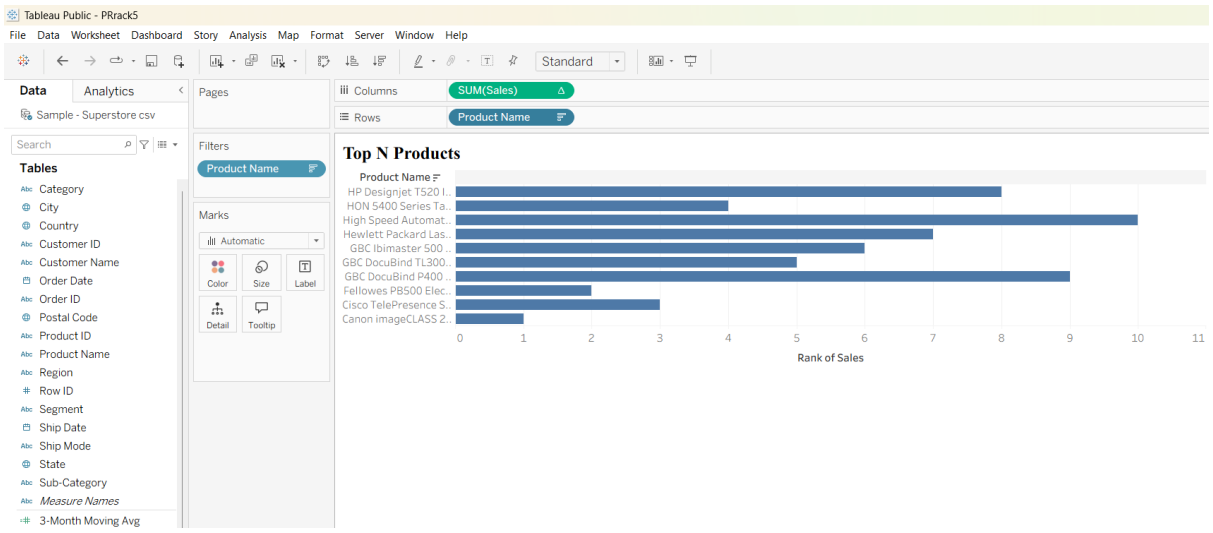


Figure 31: Top N Products by rank of Sales.

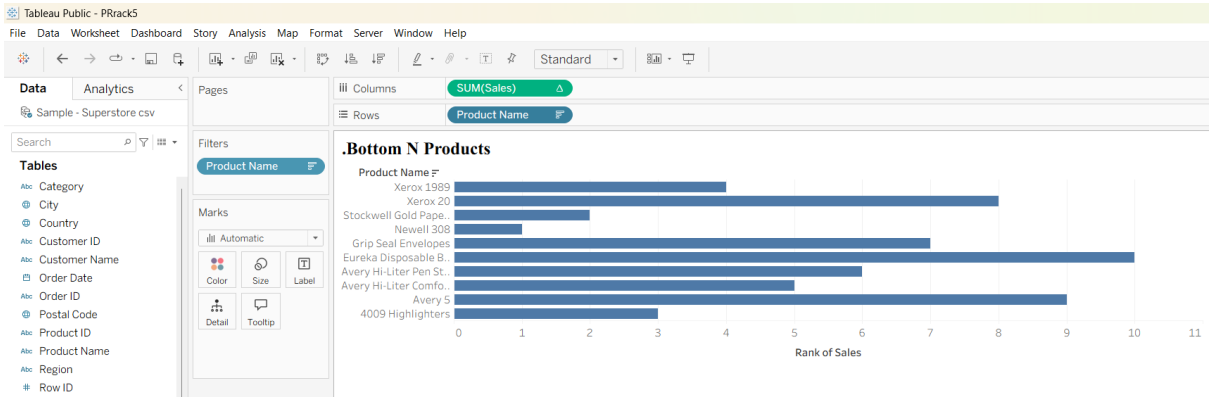


Figure 32: Bottom N Products by rank of Sales.

e.)

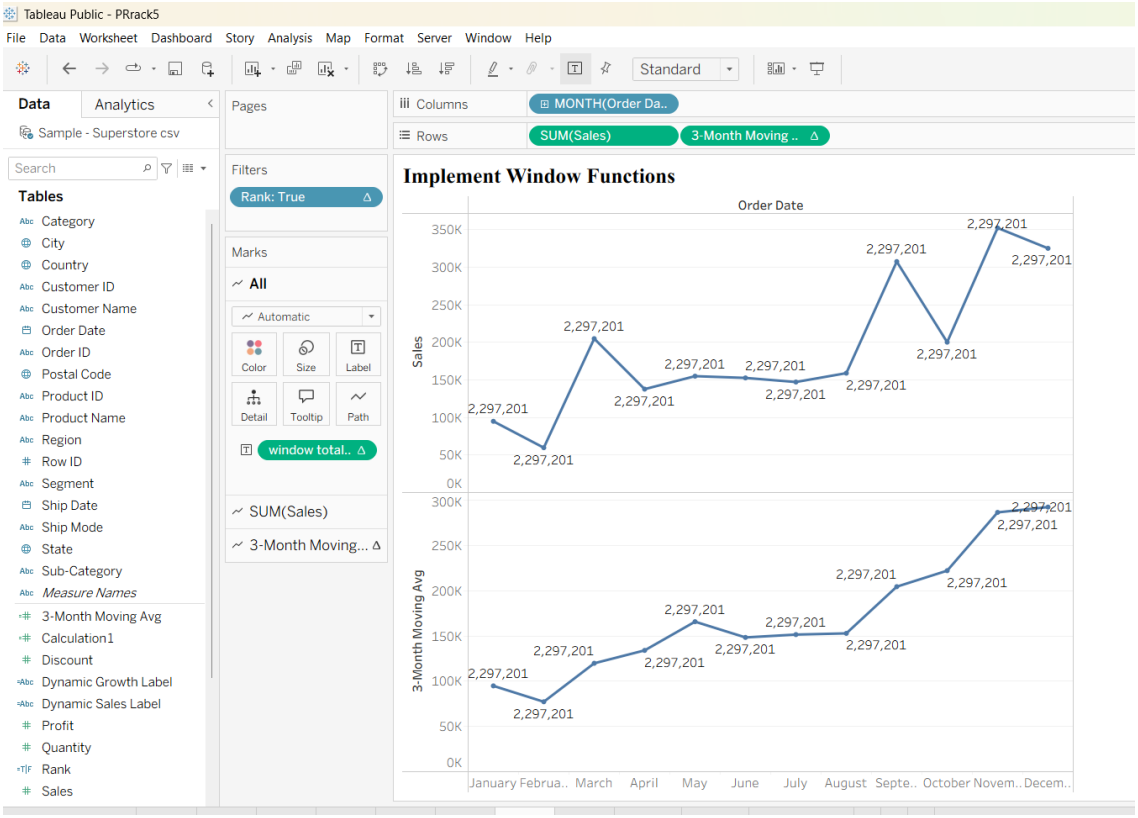


Figure 33: Window functions applied to monthly Sales and the 3-Month Moving Average.

f)

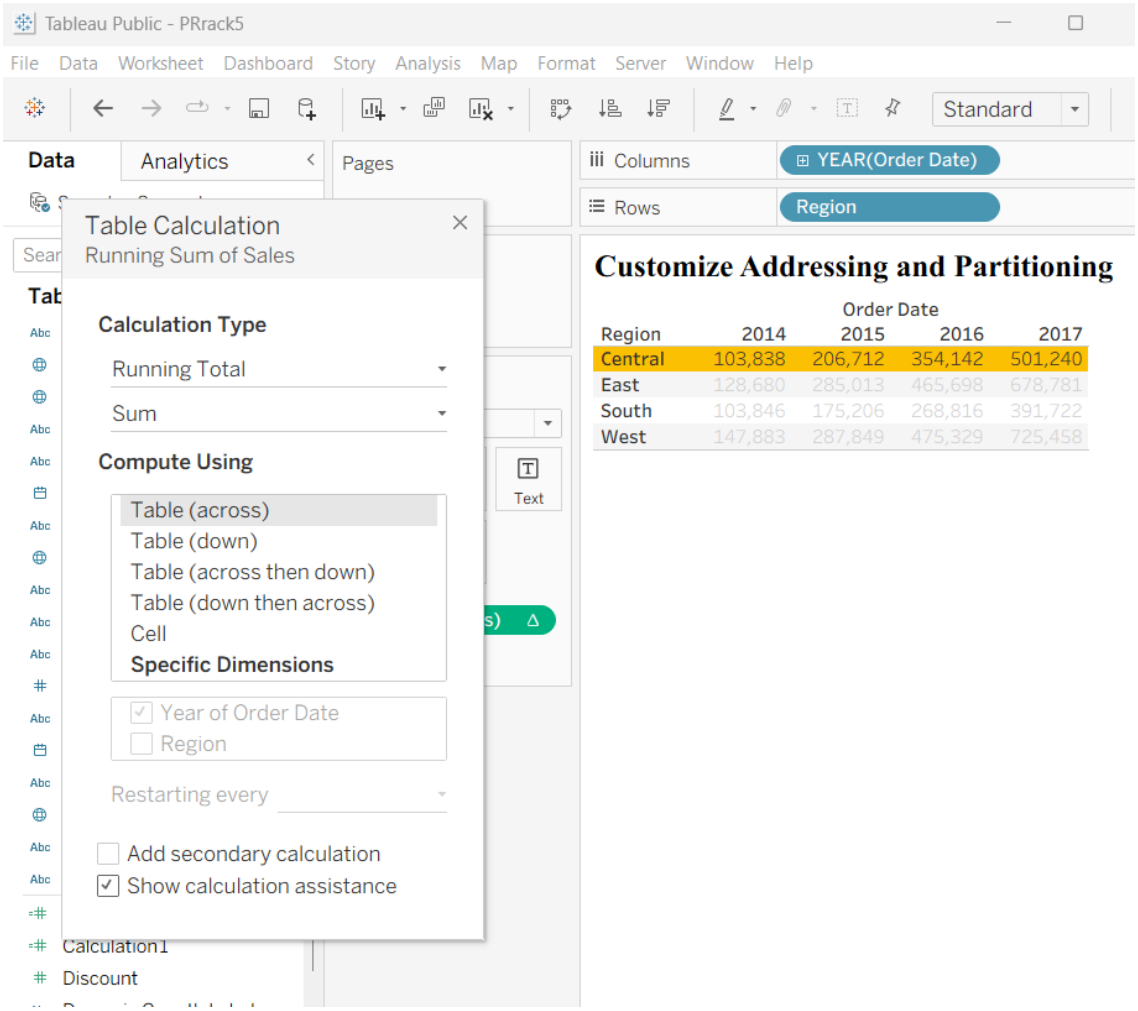


Figure 34: Addressing and partitioning: Running Total of Sales by Region and Year.

G.)

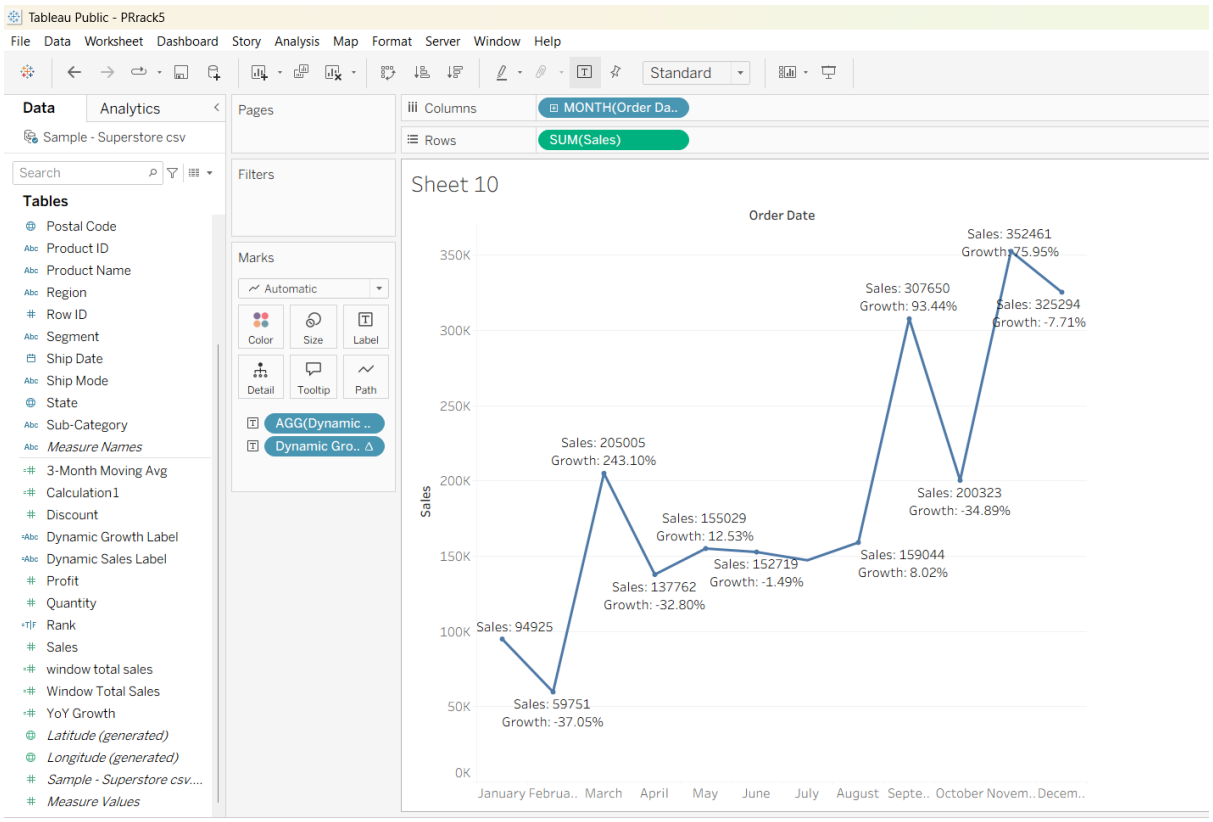


Figure 35: Dynamic labels showing Sales and Growth on each month.

Steps (Dashboard)

1. Create a new dashboard.
2. Drag the sheets created in this practical onto the canvas.
3. Arrange them in Tiled layout and title the dashboard.

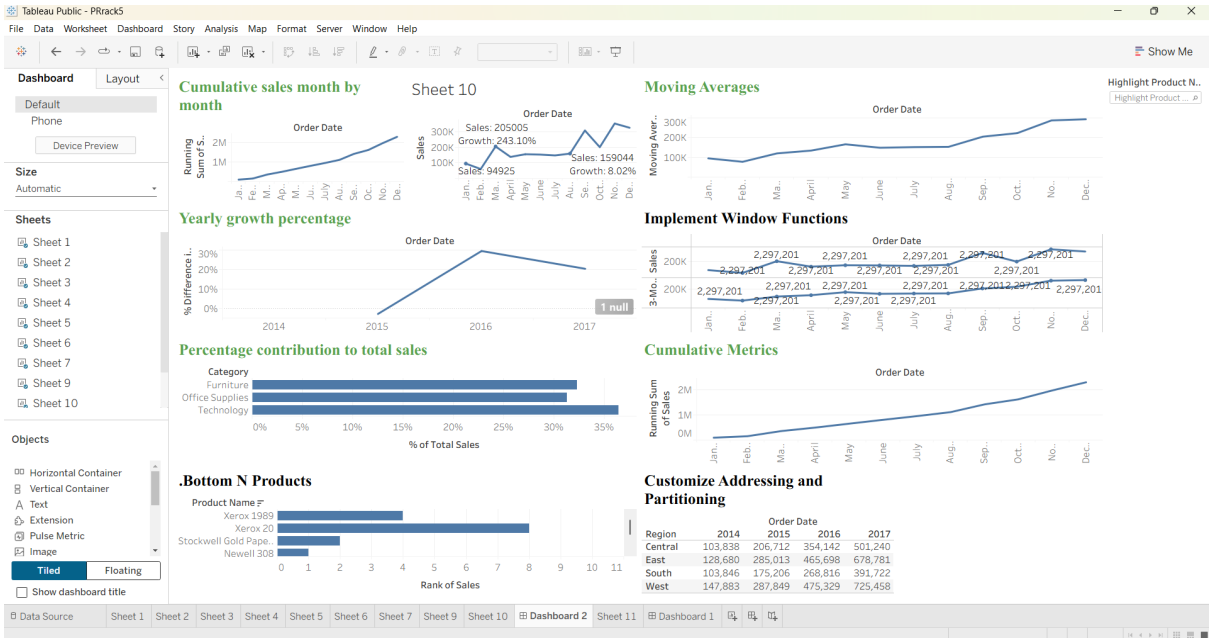


Figure 36: Dashboard combining all Practical 5 sheets.

Practical 6: Designing Effective and Visually Appealing Dashboards

Aim: To design an effective, interactive and visually appealing Tableau dashboard by applying formatting, color schemes, layouts, tooltips, Viz in Tooltip and readability principles.

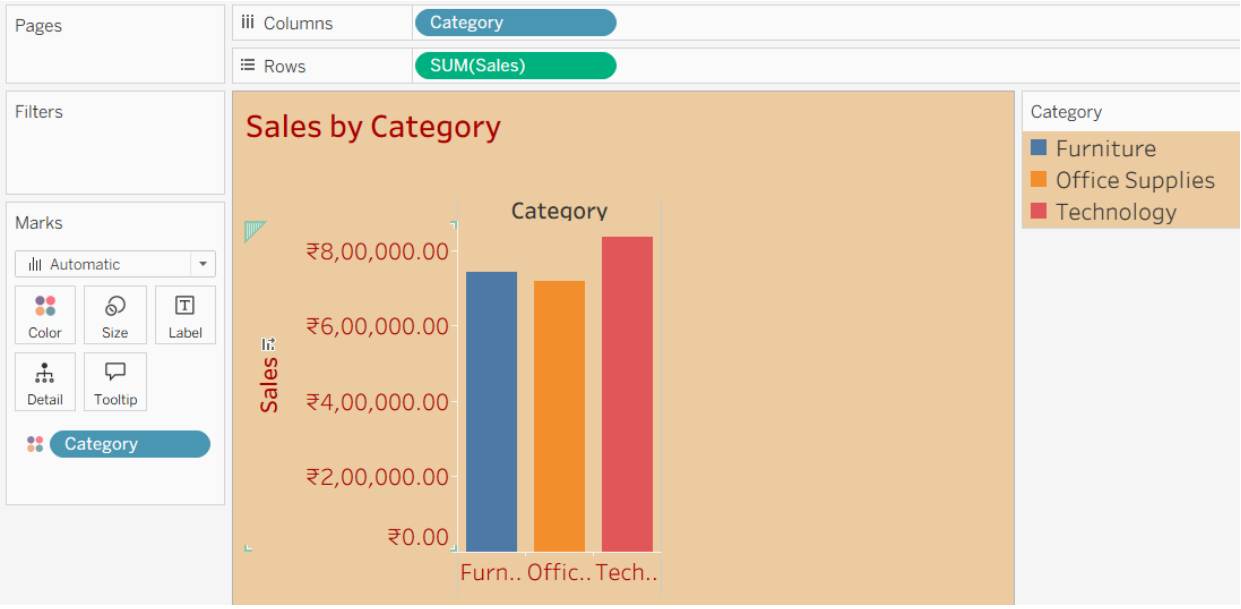
Procedure

a) Create the Worksheets

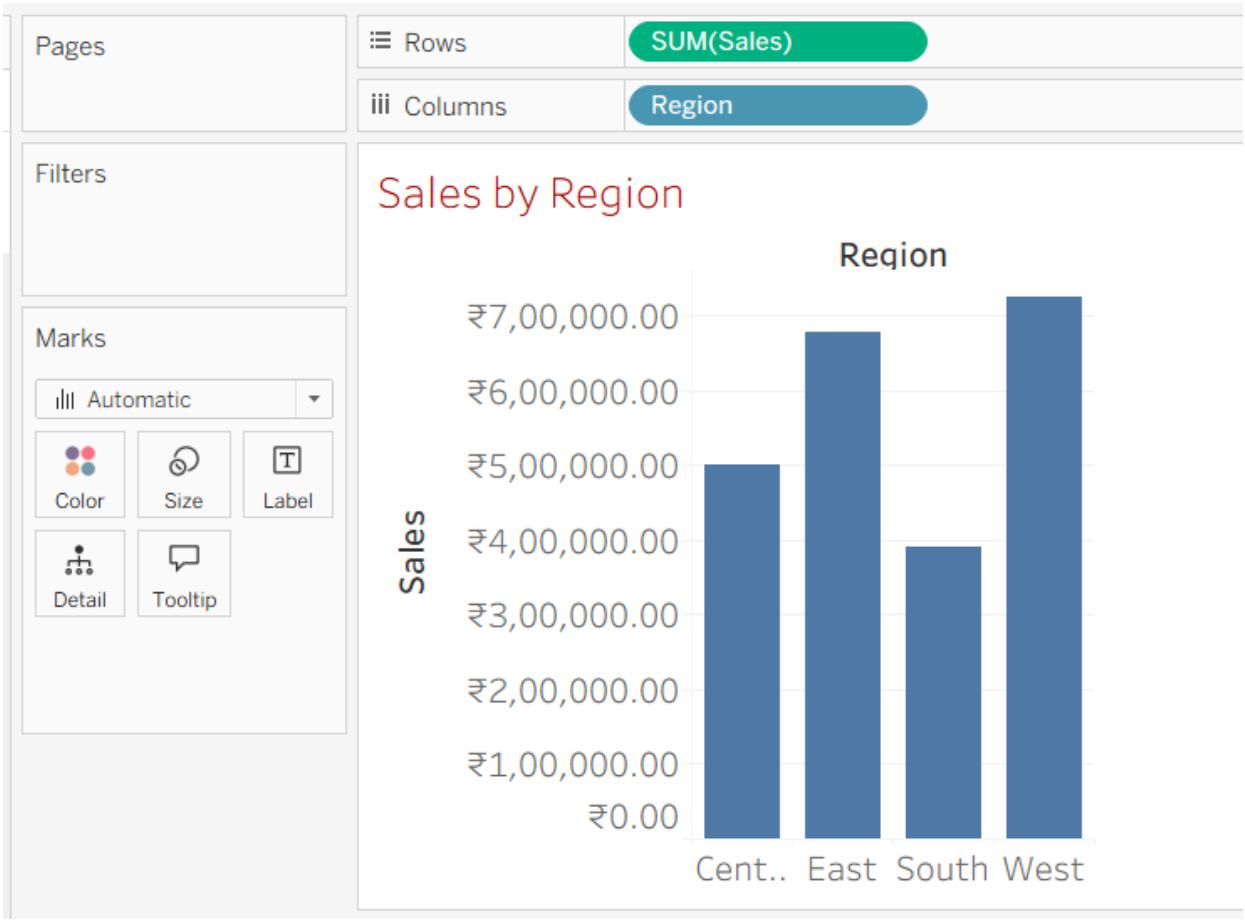
1. Sales by Category: drag Category to Rows and Sales to Columns, select Bar Chart and rename the sheet.
2. Sales by Region: drag Region to Rows and Sales to Columns, select Bar Chart and rename the sheet.
3. Monthly Sales Trend: drag Order Date (Month) to Columns and Sales to Rows, select Line Chart and rename the sheet.
4. Profit by Category: drag Category to Rows and Profit to Columns, select Bar Chart and rename the sheet.

b) Apply Worksheet-Level and Workbook-Level Formatting

1. Open the worksheet and select Format → Font to change the font type and size.
2. Format the worksheet title, borders, gridlines and shading.
3. Use the same fonts, colors, titles, number formats and backgrounds across all worksheets for a consistent workbook.



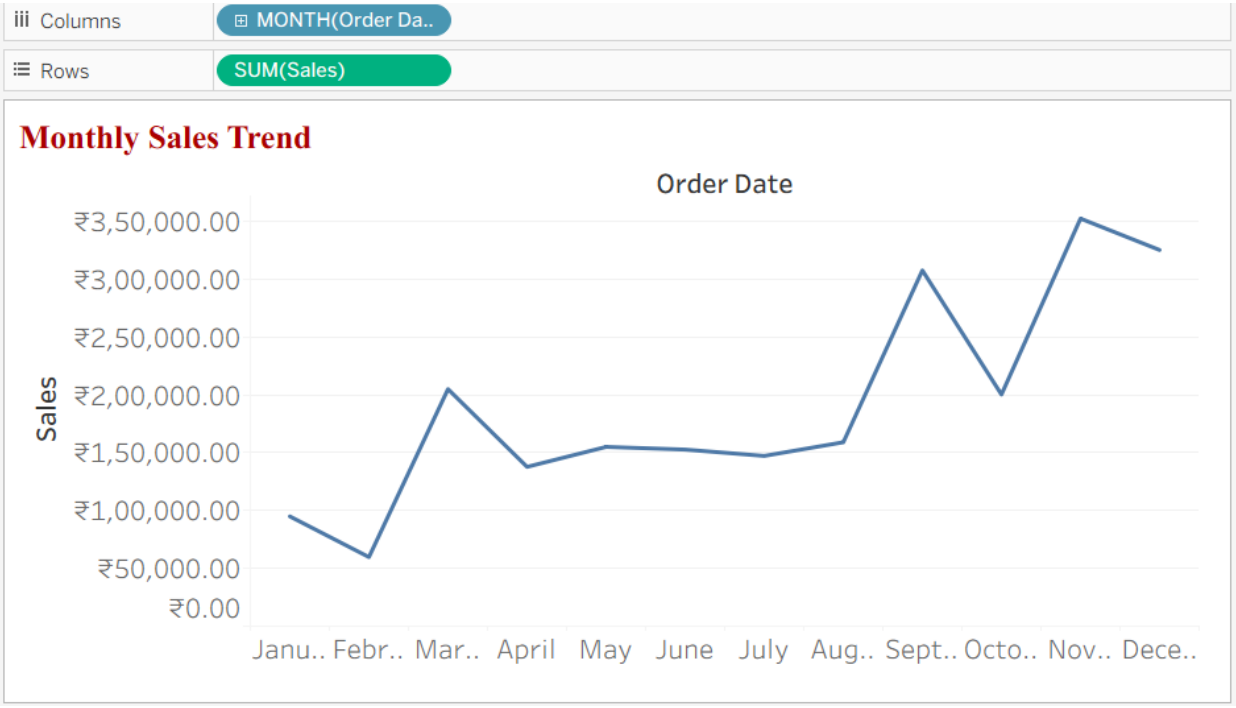
Sales by Category with formatted title, background and axis.



Sales by Region with formatted title and axis.

c) Customize Number, Date and Null Value Formats

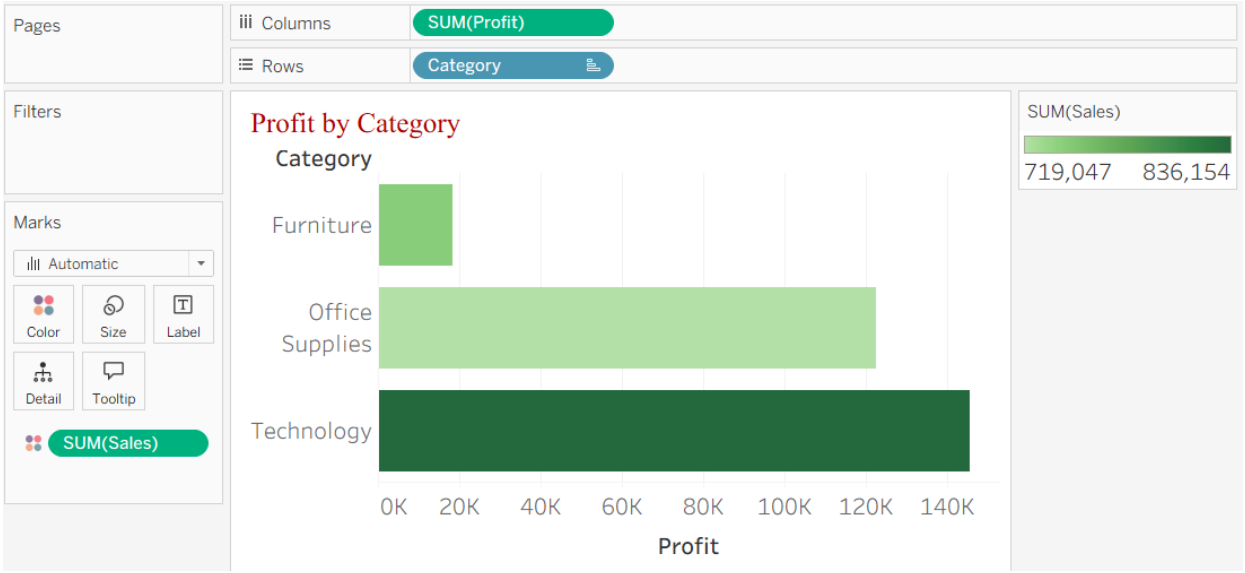
1. Number: right-click Sales → Default Properties → Number Format → Currency (Custom) and set the decimal places.
2. Percentage: right-click Discount → Default Properties → Number Format → Percentage.
3. Date: right-click Order Date → Default Properties → Date Format and select the required format.
4. Null values: right-click the field → Format → Special Values, or use IFNULL([Profit], 0).



Monthly Sales Trend with formatted title and currency axis.

d) Design Effective Color Schemes

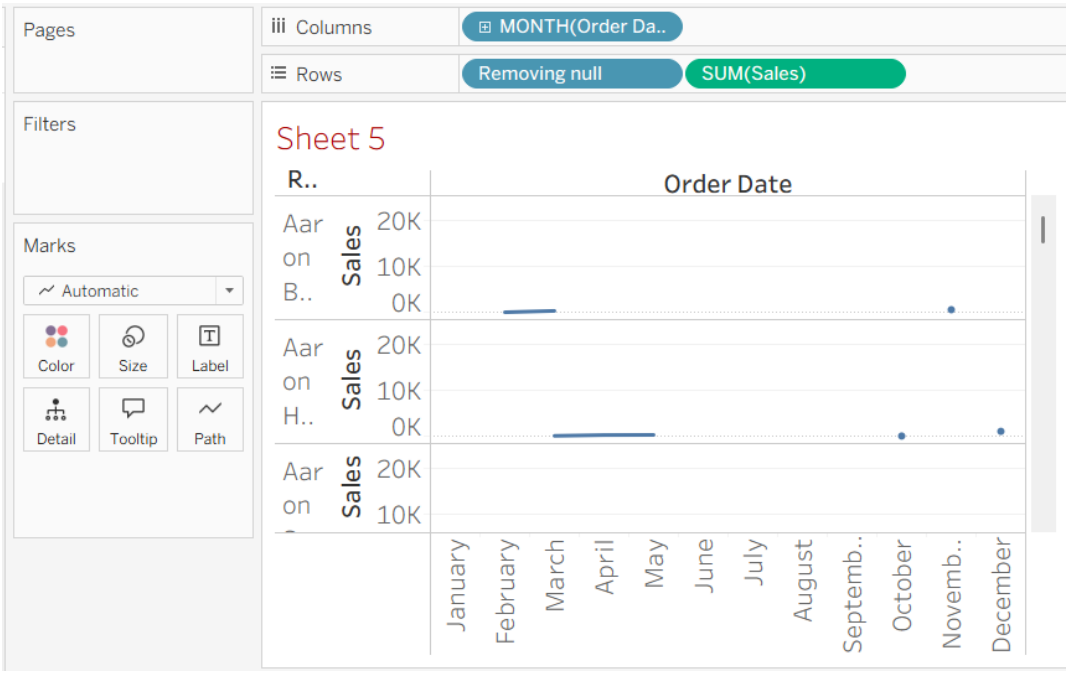
1. Select the worksheet.
2. Drag Profit (or Sales) to Color.
3. Click Color → Edit Colors and select an appropriate palette.
4. Enable Use Full Color Range and click Apply.



Profit by Category with SUM(Sales) on Color.

e) Create Informative Tooltips

1. Open the worksheet and select Tooltip on the Marks card.
2. Edit the text, for example Category: <Category>, Sales: <SUM(Sales)>, Profit: <SUM(Profit)>.
3. Optionally add a calculated field such as Profit Margin = SUM([Profit]) / SUM([Sales]) formatted as a percentage.
4. Click OK.



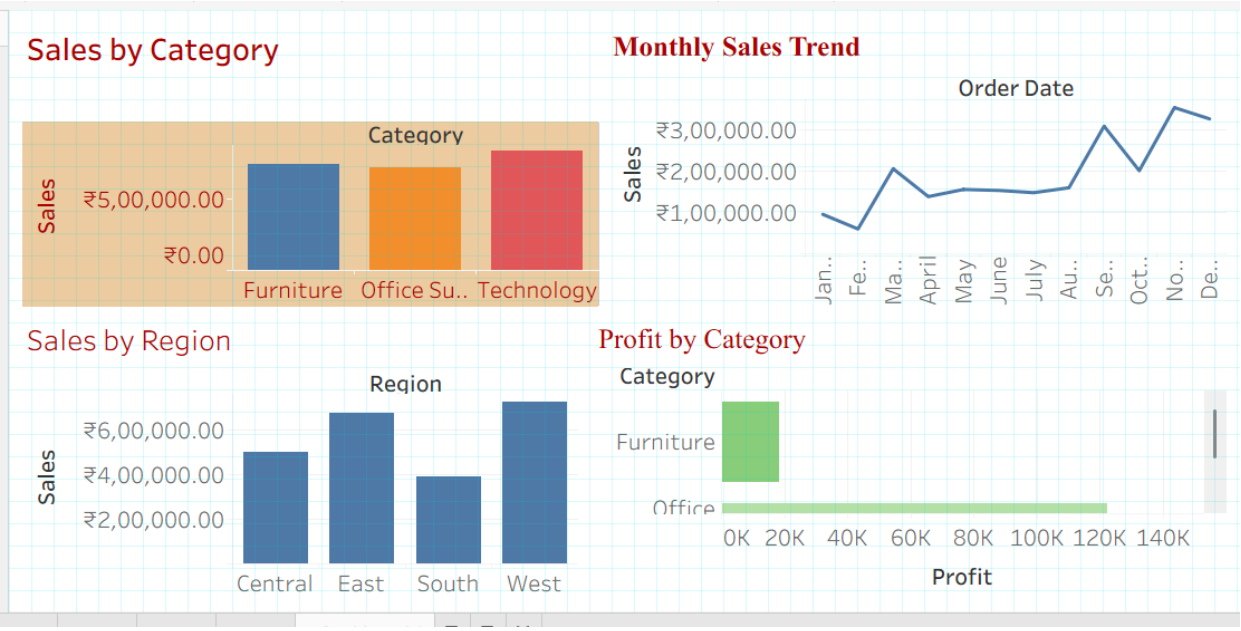
Sheet 5: monthly Sales with the Removing null field on Rows.

f) Implement Viz in Tooltip

1. Create the main worksheet Sales by Region (Region on Rows, Sales on Columns).
2. Create a second worksheet Category Sales Detail (Category on Rows, Sales on Columns).
3. In Sales by Region click Tooltip on the Marks card → Insert → Sheets and select Category Sales Detail.
4. Make sure Region is passed as a filter so the tooltip shows the selected region.

g) Build the Dashboard

1. Click New Dashboard.
2. Drag the formatted worksheets onto the canvas using a logical layout with the most important information at the top.
3. Add filters such as Region and Year and apply them to all worksheets using the data source.
4. Keep the dashboard simple: consistent colors, clear titles, readable fonts and minimal gridlines.



Dashboard built from the formatted worksheets.

Result

Successfully designed a visually appealing and interactive Tableau dashboard by applying formatting, color schemes, tooltips, Viz in Tooltip and readability principles.

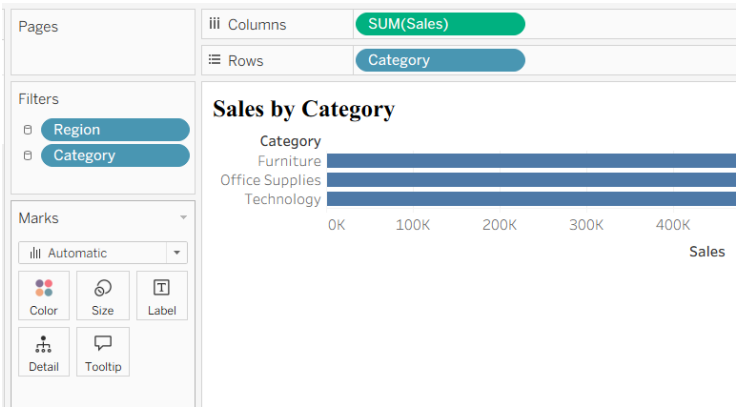
Practical 7: Creating Interactive Dashboards and Data Stories

Aim: To create interactive Tableau dashboards using worksheets, filters and interactive actions (highlight, filter, URL and set actions).

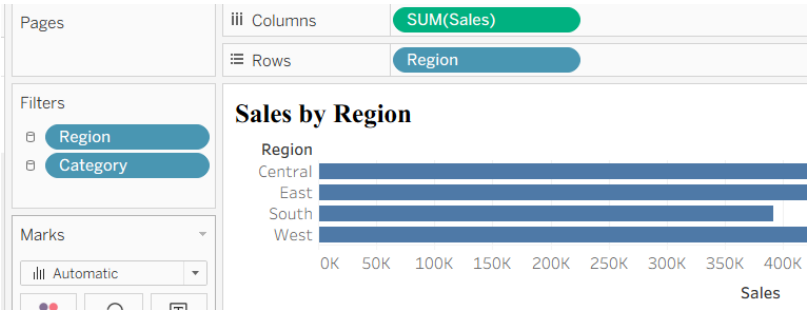
Procedure

a) Create Worksheets for the Dashboard

1. Connect to Sample - Superstore.
2. Create the worksheets Sales by Category, Sales by Region, Monthly Sales Trend, Profit by Category and Sales by State (drag State to the view and Sales to Color).



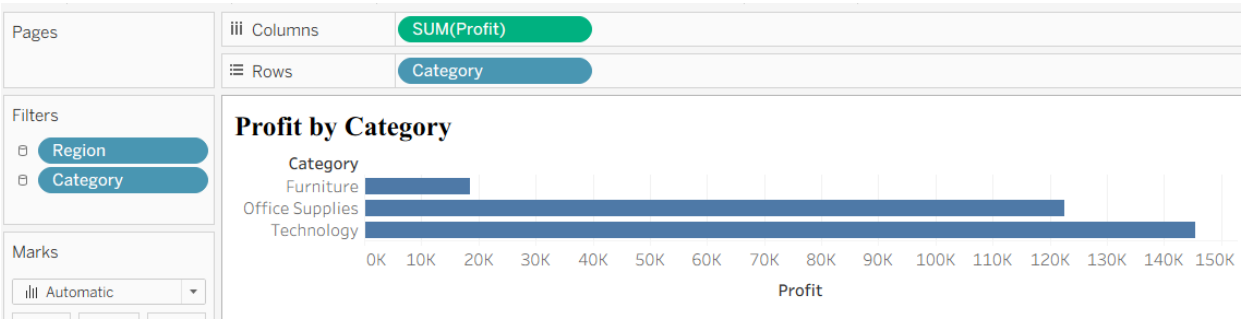
Sales by Category.



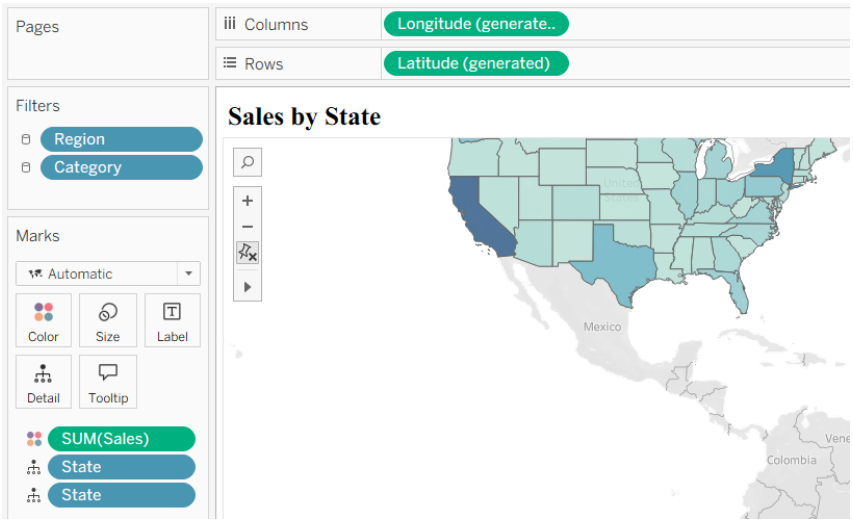
Sales by Region.



Monthly Sales Trend.



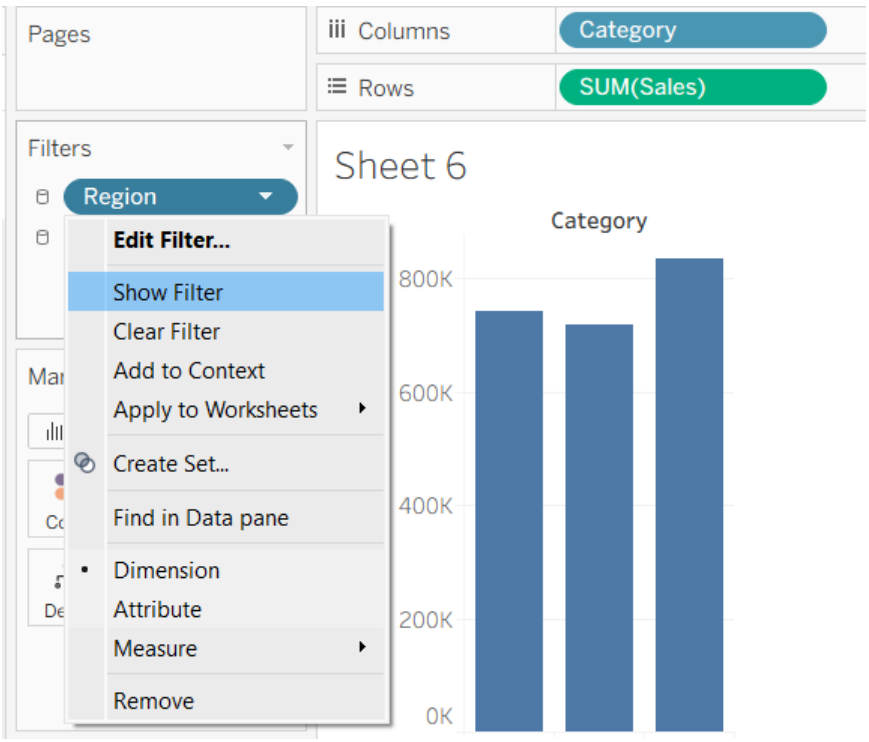
Profit by Category.



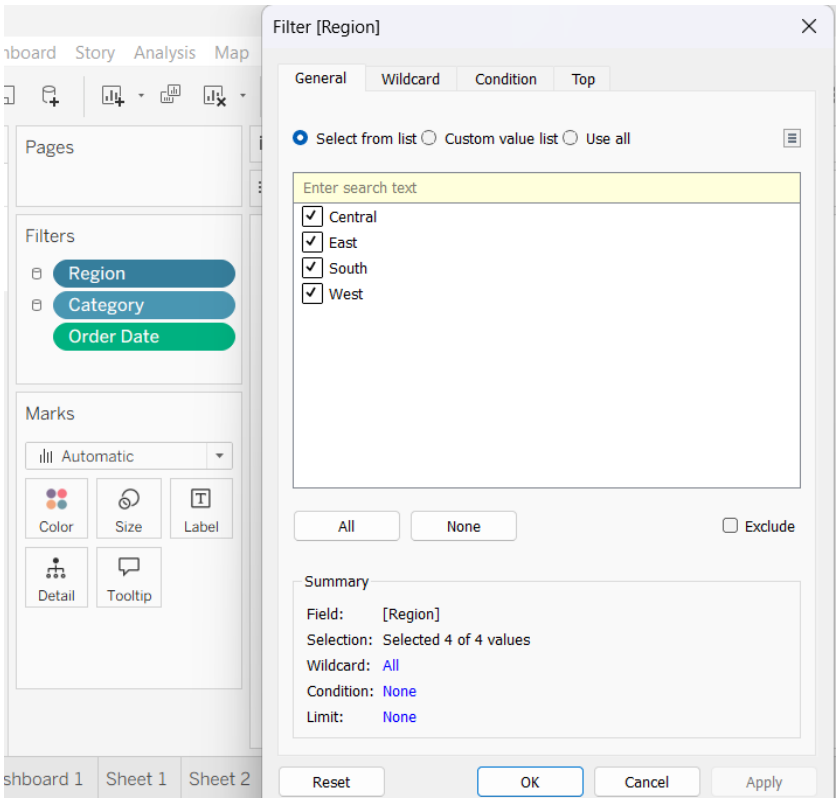
Sales by State (map).

b) Add Filters and Interactive Controls

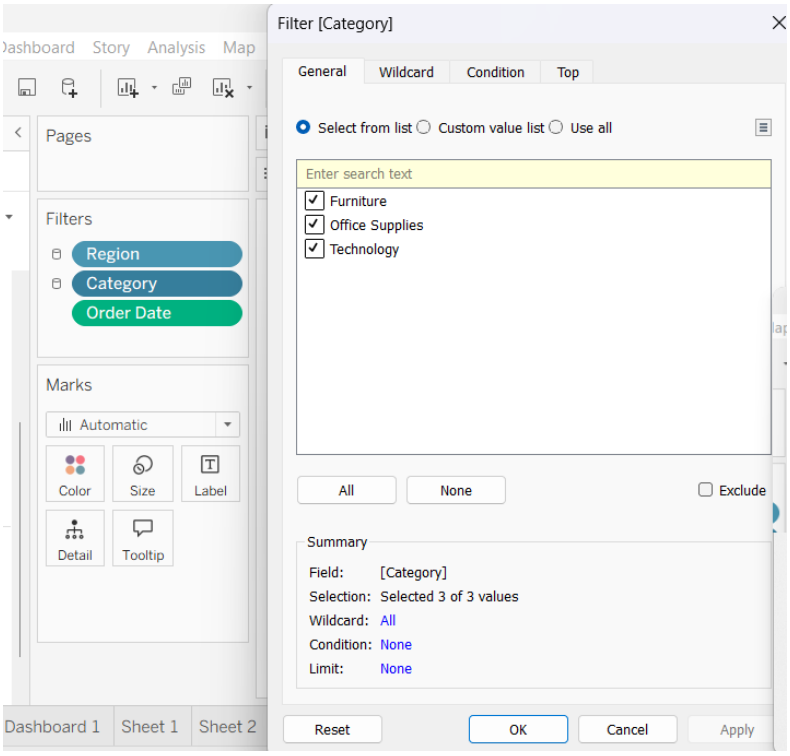
1. Open a worksheet and drag Region to the Filters shelf.
2. Right-click Region and select Show Filter.
3. Repeat for Category.
4. Drag Order Date to Filters, select Years, click OK and select Show Filter.
5. Open the filter menu and select Apply to Worksheets → All Using This Data Source.



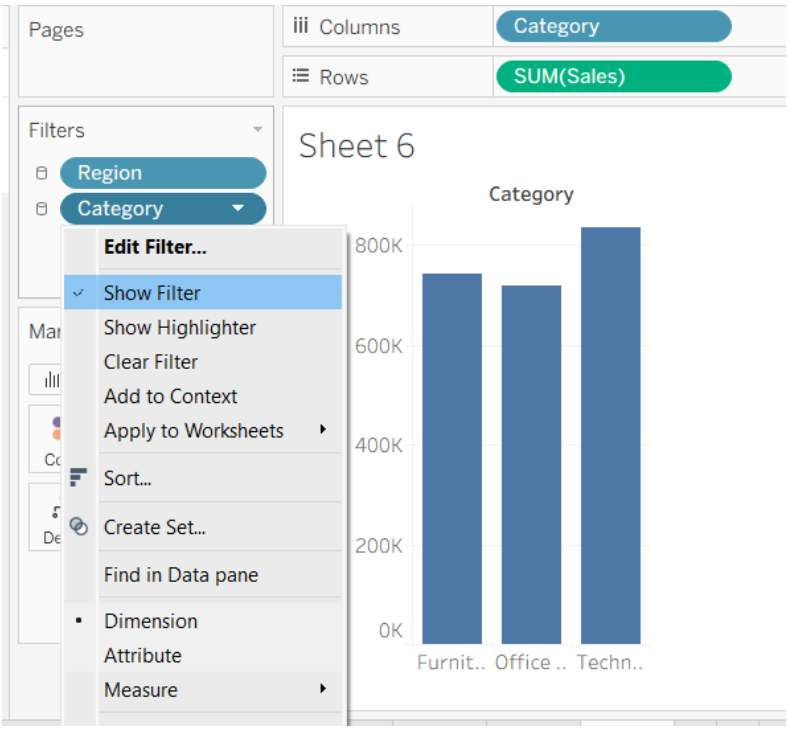
Show Filter option on the filter menu.



Filter [Region] dialog.



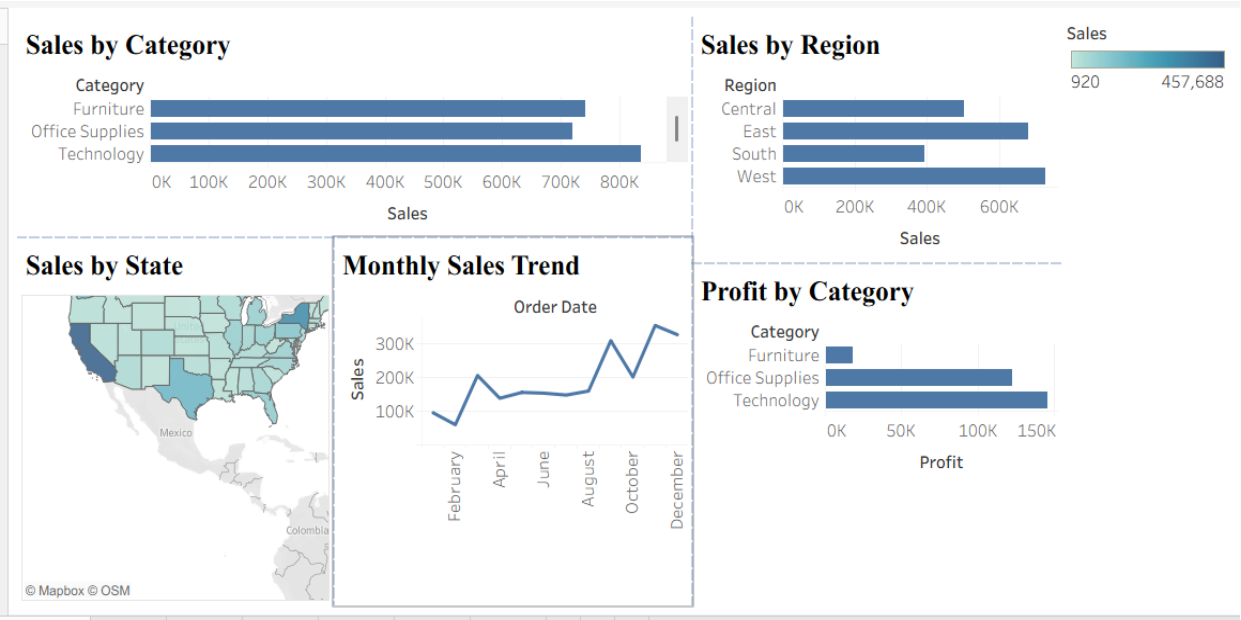
Filter [Category] dialog.



Show Filter enabled for Category.

c) Design the Dashboard Layout

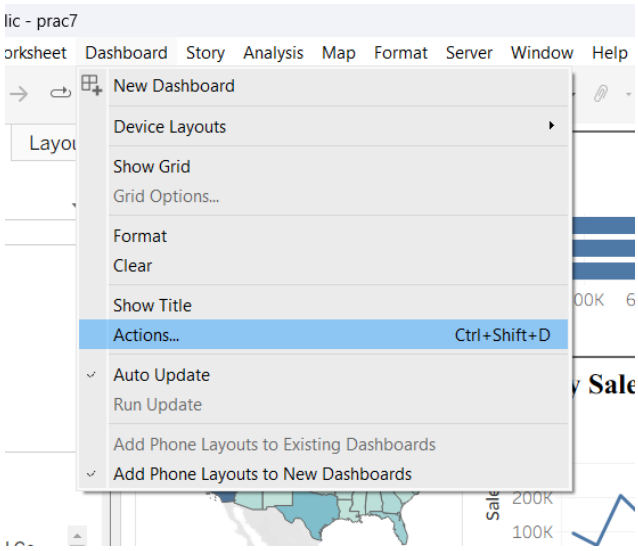
1. Click New Dashboard.
2. Set the Size (for example Desktop Browser).
3. Drag Monthly Sales Trend, Sales by Category, Sales by Region, Profit by Category and Sales by State onto the dashboard.



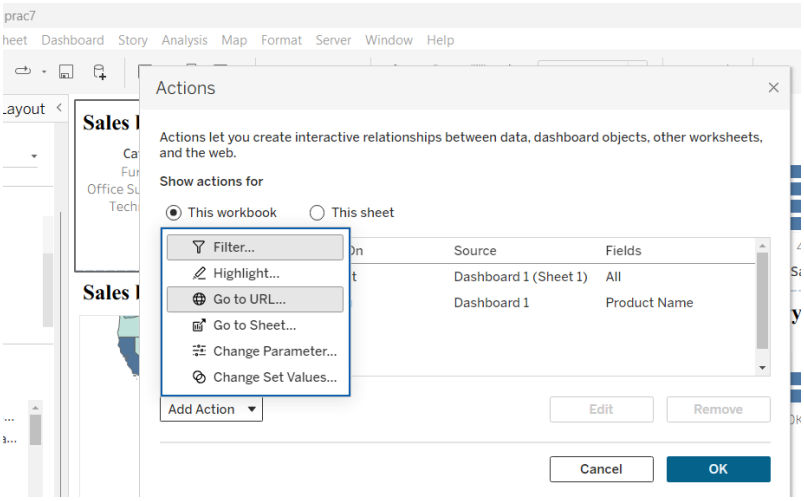
Dashboard containing the five worksheets.

d) Implement Highlight and Filter Actions

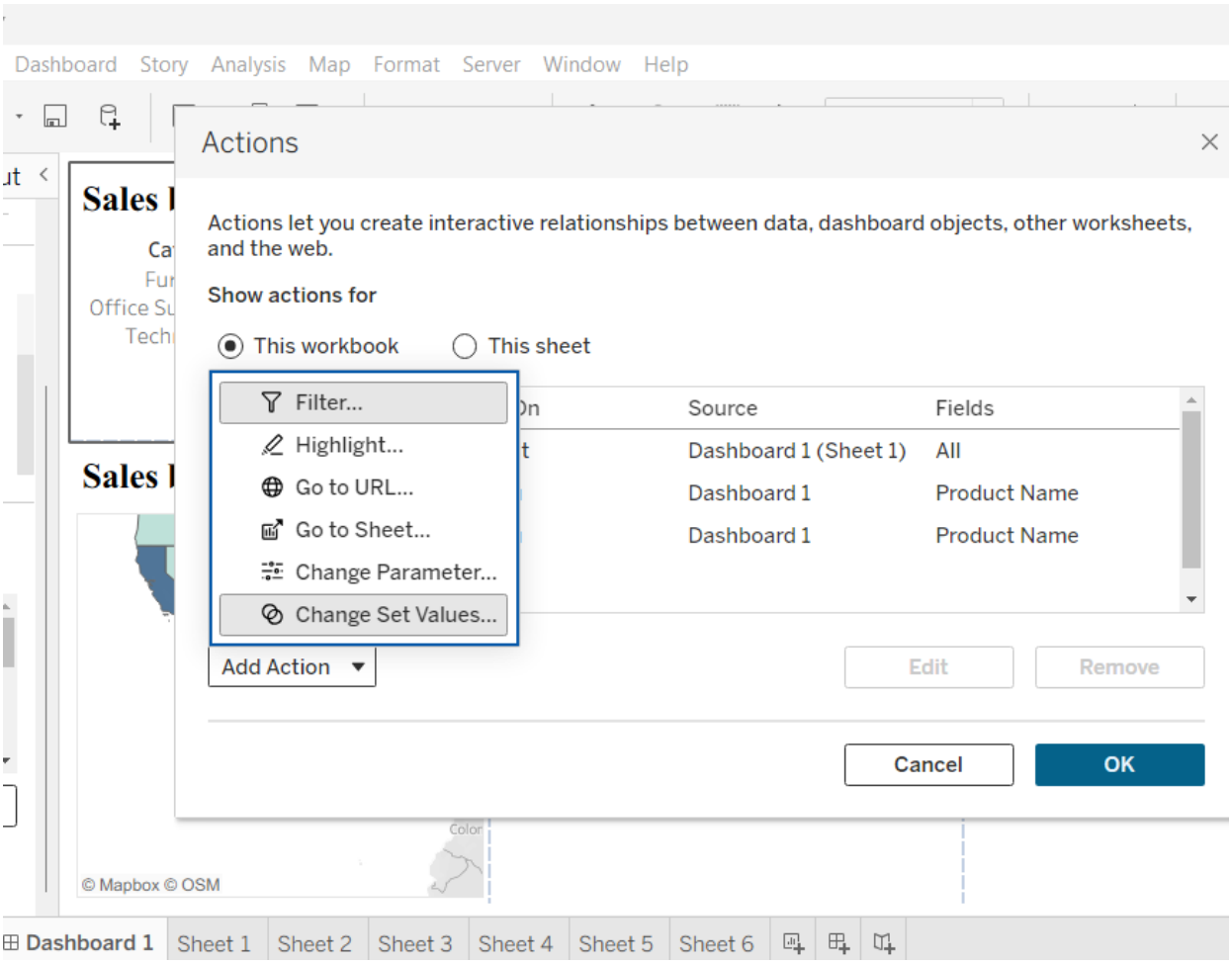
1. Open the dashboard and select Dashboard → Actions.
2. Click Add Action → Highlight (or Filter).
3. Select the source sheet, for example Sales by Category (Highlight) or Sales by Region (Filter).
4. Select the target sheets and the field (Category for Highlight).
5. For a Filter action choose Run action on Select.
6. Click OK.



Dashboard menu with the Actions option.



Actions dialog with the Add Action menu.

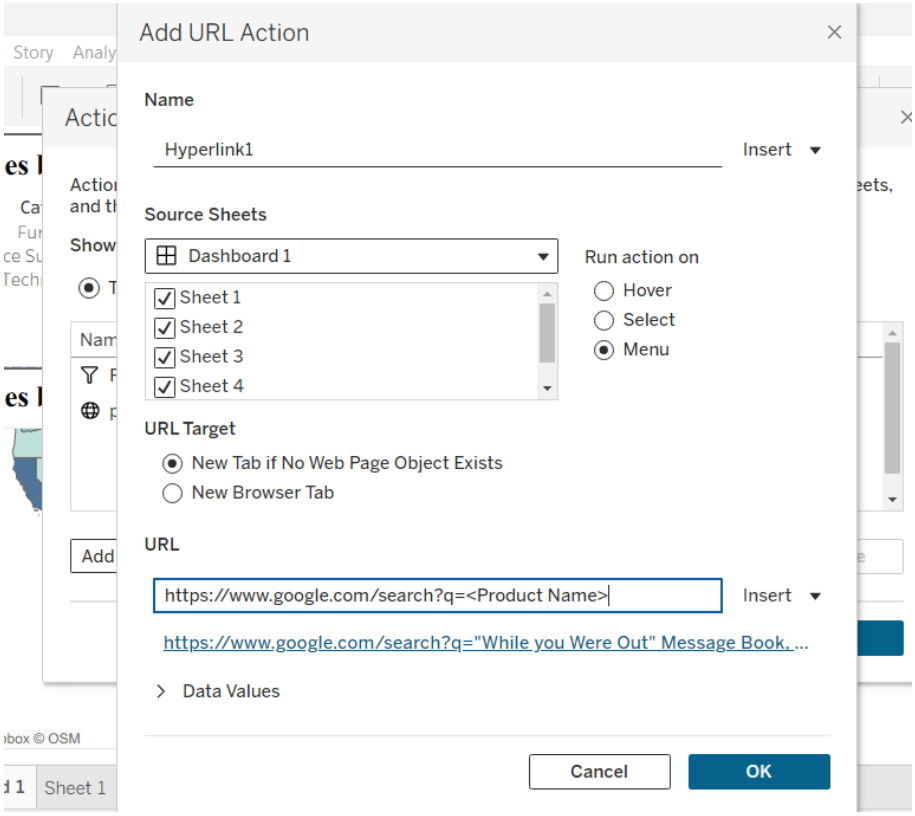


Actions dialog listing the filter, highlight and URL actions.

e) Use URL Actions and Set Actions

URL Action

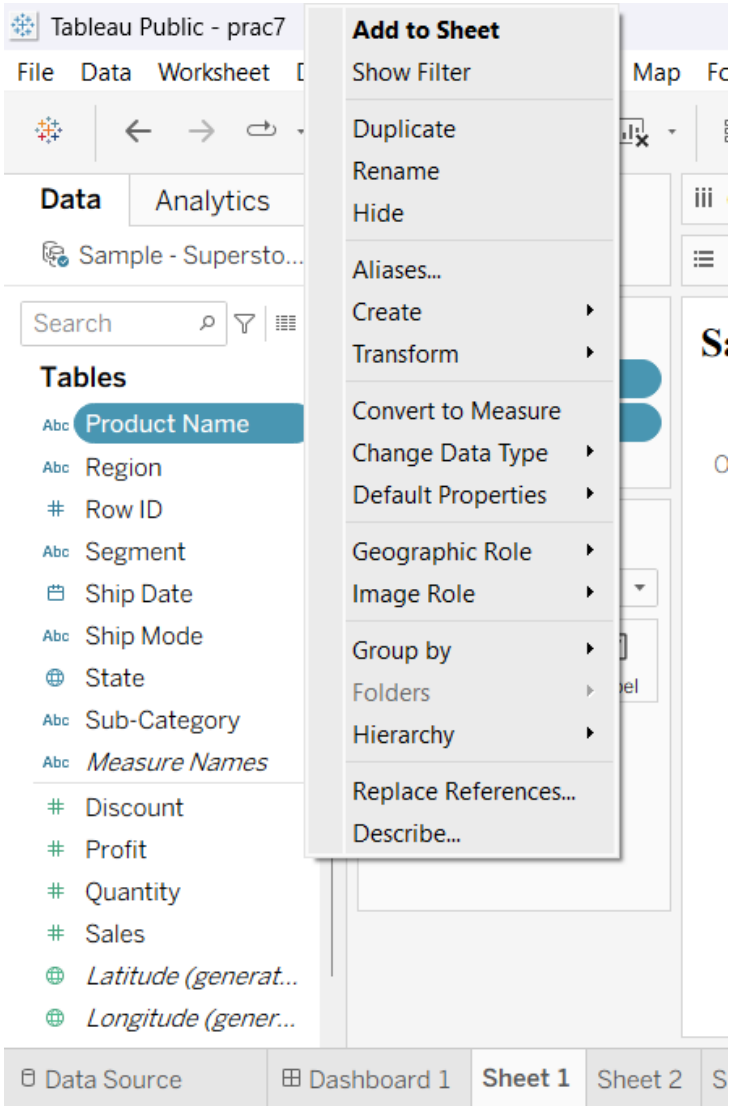
1. Open the dashboard and select Dashboard → Actions.
2. Click Add Action → Go to URL.
3. Enter a URL, for example https://www.google.com/search?q=<Product Name>.
4. Select the worksheet that will trigger the action.
5. Click OK.



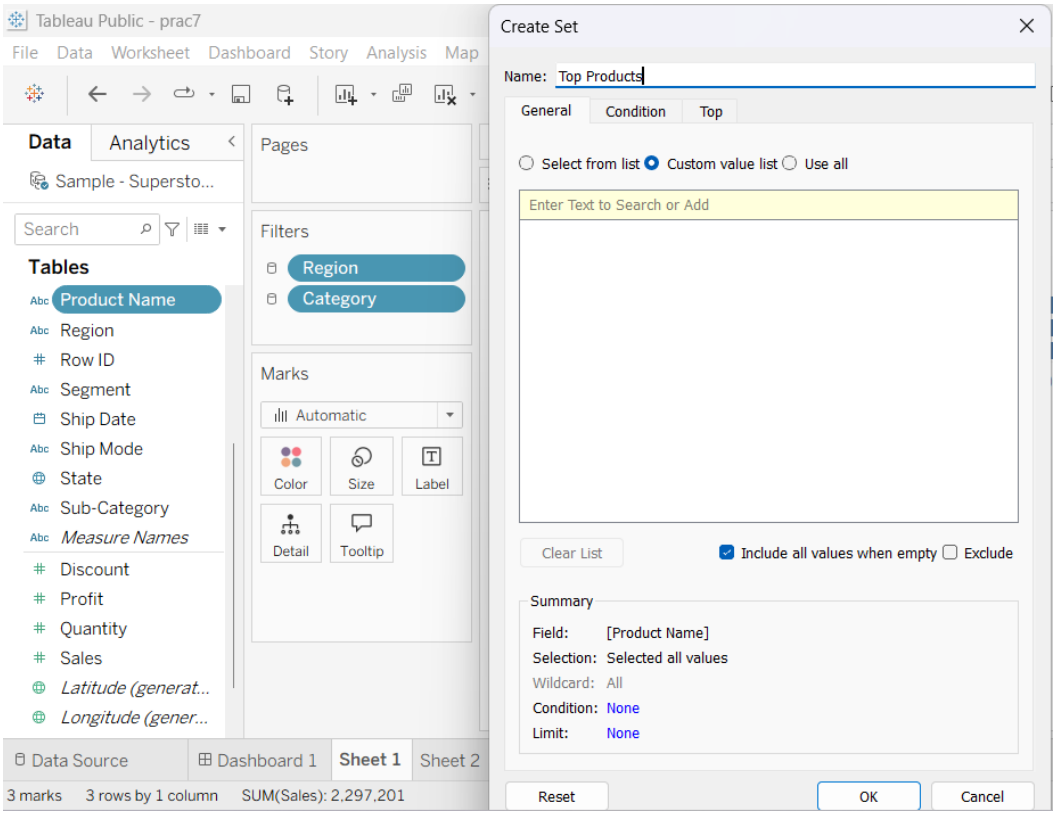
Add URL Action dialog.

Set Action

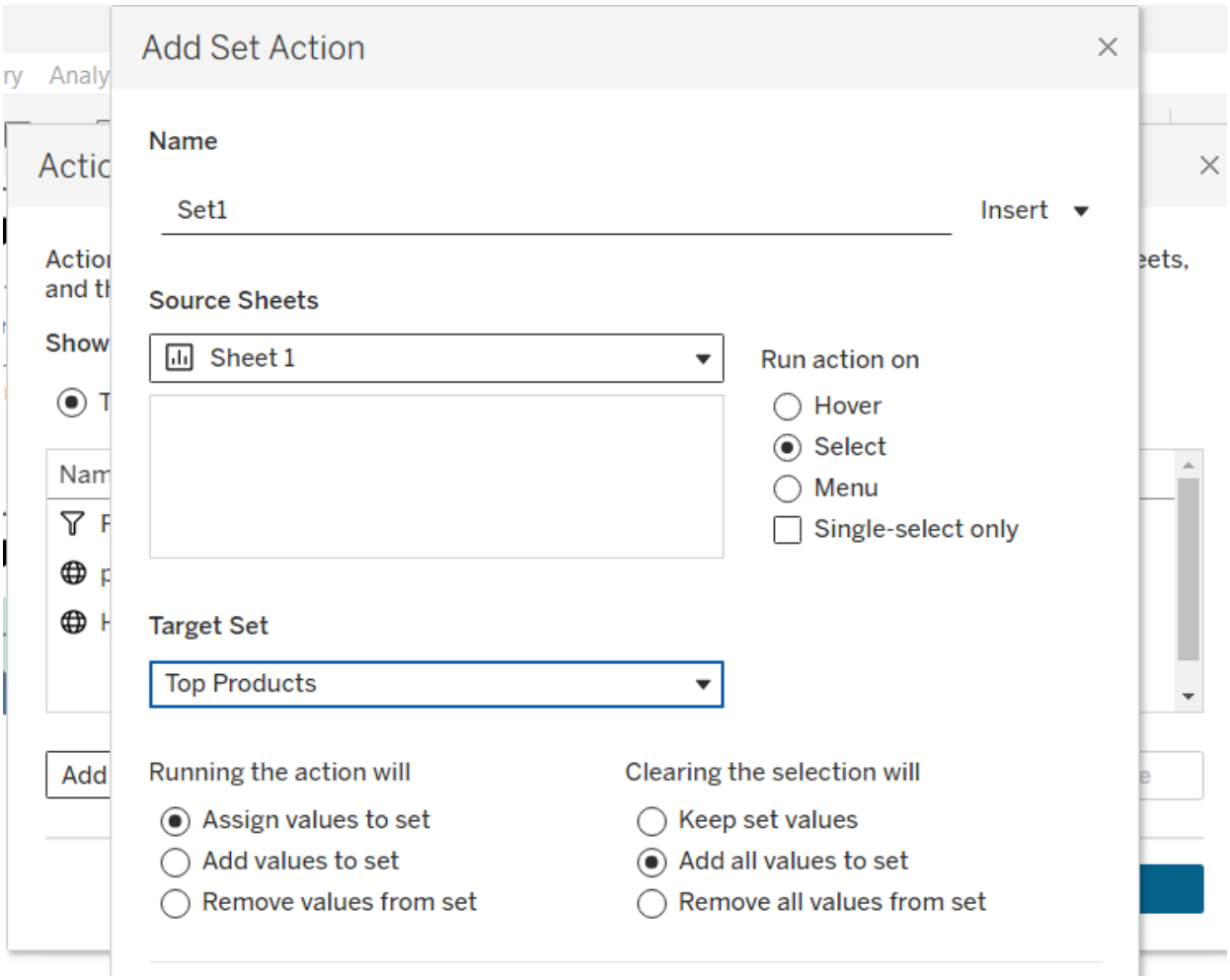
1. Right-click Product Name and select Create → Set.
2. Name it Top Products, select a few products and click OK.
3. Open Dashboard → Actions and select Add Action → Change Set Values.
4. Select the source sheet and the set Top Products.
5. Choose Assign values to set and click OK.



Right-click menu on Product Name used to create a set.

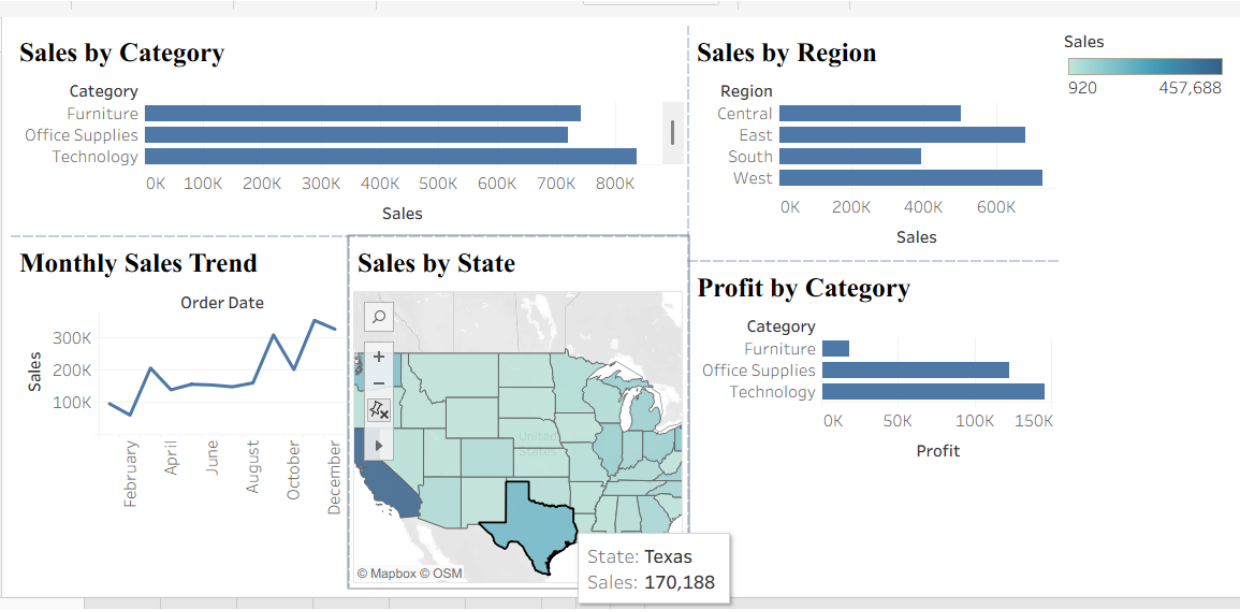


Create Set dialog for the set named Top Products.



Add Set Action dialog.

Output



Final interactive dashboard.

Result

Successfully created an interactive Tableau dashboard using multiple worksheets, filters, highlight and filter actions, URL actions and set actions.

Practical 8: Performing Advanced Analytics using Trends, Clustering, and Forecasting

Aim: To perform advanced analytics in Tableau using trend lines, clustering, distribution analysis and forecasting, to compare different statistical models, and to export and interpret the statistical results.

Software Required

- Tableau Desktop / Tableau Public
- Dataset: Sample - Superstore

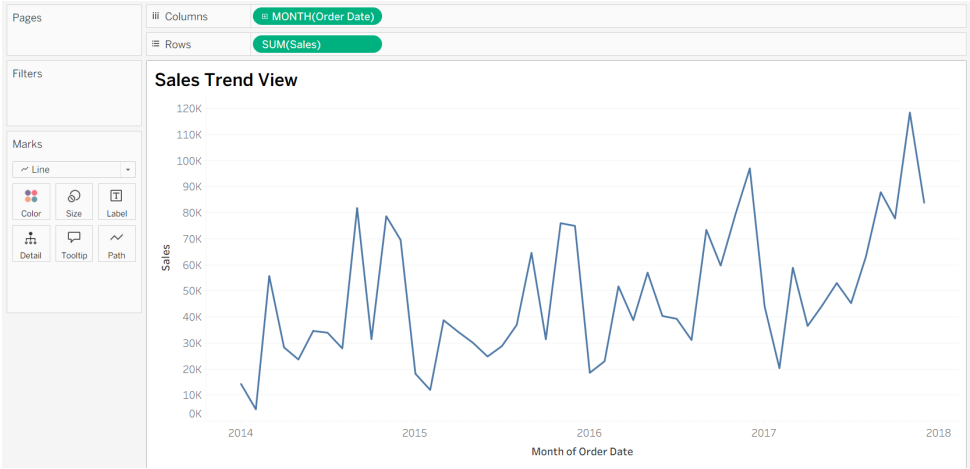
Procedure

a) Add and Interpret Trend Lines (Linear and Exponential)

A trend line is a statistical line added to a view to show the general direction of the data or the relationship between two variables. Tableau supports Linear, Logarithmic, Exponential, Polynomial and Power trend models.

Sales Trend View

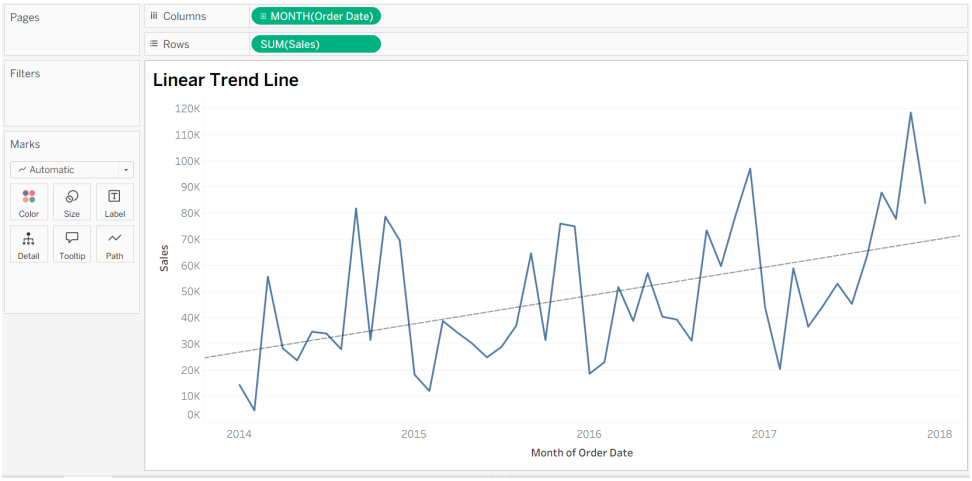
1. Open Tableau and connect to Sample - Superstore.
2. Open a new worksheet.
3. Drag Order Date to Columns and select Month.
4. Drag Sales to Rows.
5. Set the Marks type to Line and rename the sheet title as Sales Trend View.



Sales Trend View: monthly Sales over time.

Linear Trend Line

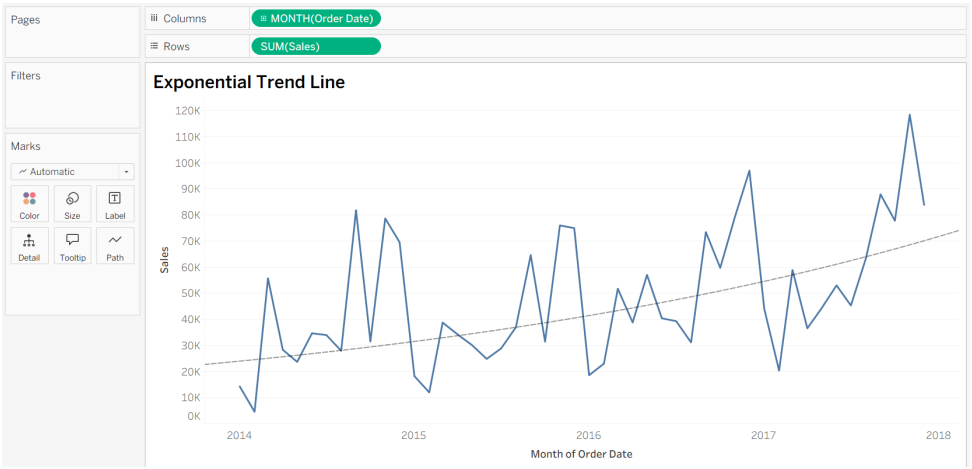
1. Open the Analytics pane.
2. Drag Trend Line onto the view and drop it on Linear. Alternatively select Analysis → Trend Lines → Show Trend Lines.



Linear trend line added to the monthly Sales view.

Exponential Trend Line

1. Select Analysis → Trend Lines → Edit Trend Lines.
2. Select Exponential and click OK.



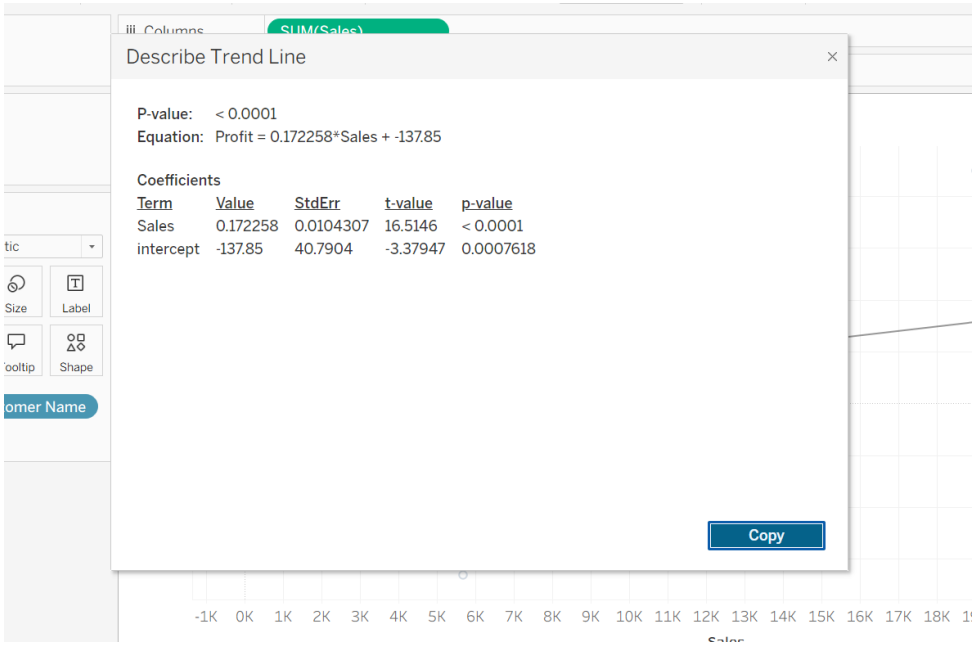
Exponential trend line added to the monthly Sales view.

Observation

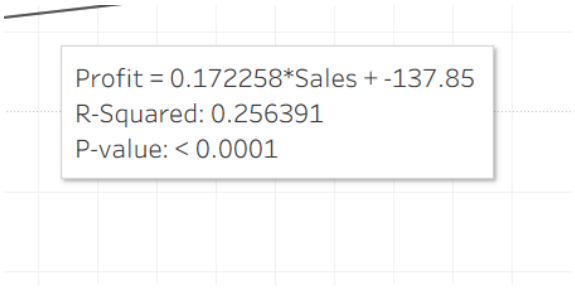
- An upward sloping trend line shows that Sales are generally increasing over time; a downward slope shows a decrease and a flat line shows little overall change.
- The linear trend line slopes upward, so monthly Sales show a general increase over the period.
- The exponential trend line curves upward gradually, showing growth at an increasing rate.

b) Customize and Analyze Trend Models

1. Create a new worksheet.
2. Drag Sales to Columns and Profit to Rows to create a scatter plot.
3. Drag Order ID to Detail so that each order is shown as a mark.
4. Select Analysis → Trend Lines → Show Trend Lines.
5. Right-click the trend line and select Describe Trend Line.
6. Hover over the trend line to view the tooltip.



Describe Trend Line dialog showing the equation, coefficients and P-value.



Trend line tooltip showing the equation, R-Squared and P-value.

Observation

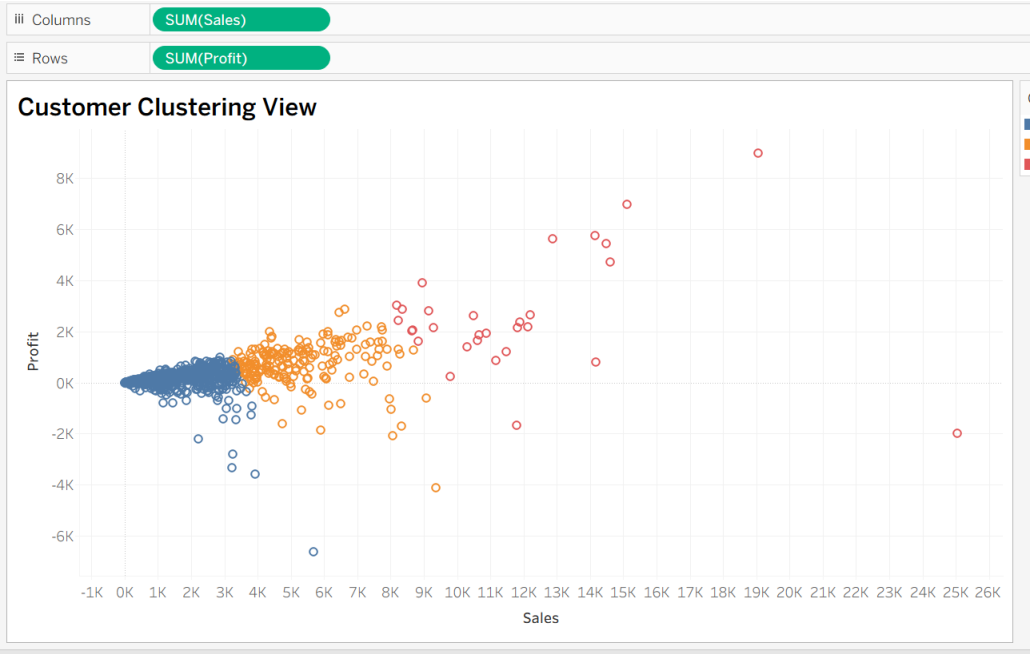
The Describe Trend Line option provides the following statistical information:

- Equation of the model
- R-squared value
- P-value
- Coefficients, standard error and t-value of each term

c) Perform Clustering on Datasets

Clustering divides data into groups of similar observations, for example customers grouped by their Sales and Profit.

1. Create a new worksheet.
2. Drag Sales to Columns and Profit to Rows.
3. Drag Customer Name to Detail.
4. Open the Analytics pane and drag Cluster onto the view.
5. Double-click the Cluster field on the Marks card (or the Cluster indicator) to open the Clusters dialog box.
6. Set the Number of Clusters to 3 and close the dialog box.



Customer Clustering View: customers grouped into three clusters by Sales and Profit.

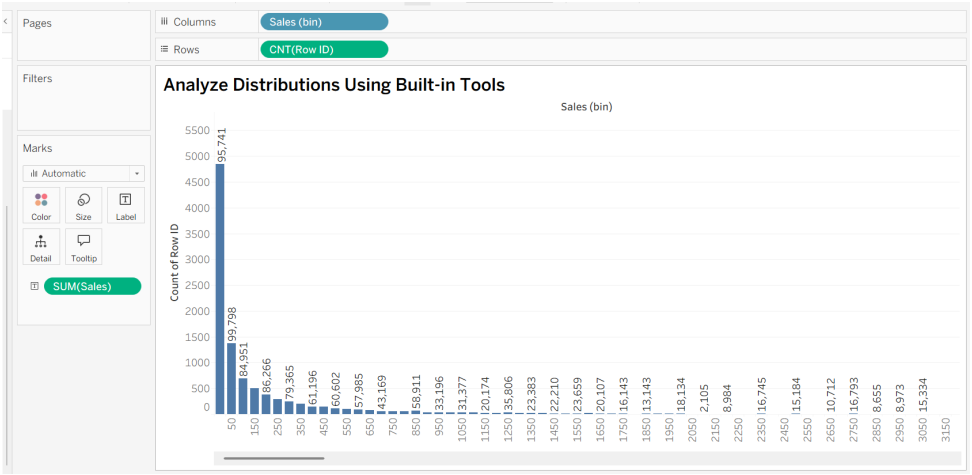
Observation

Tableau divides the customers into three clusters, shown in different colors, based on the similarity of their Sales and Profit values.

d) Analyze Distributions Using Built-in Tools

Histogram

1. Create a new worksheet and drag Sales to the view.
2. Right-click Sales and select Create → Bins.
3. Enter a suitable bin size, for example 500, and click OK.
4. Drag Sales (bin) to Columns.
5. Drag Number of Records (Row ID count) to Rows.



Histogram of Sales (bin) against the count of records.

Box Plot

1. Create a new worksheet.
2. Drag Category to Columns and Profit to Rows.
3. Open the Analytics pane and drag Box Plot onto the view.



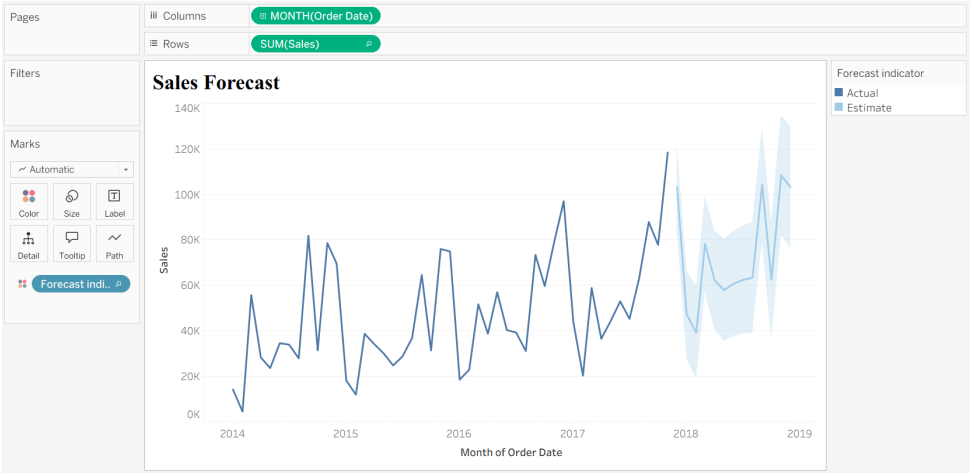
Box plot of Profit for each Category.

Observation

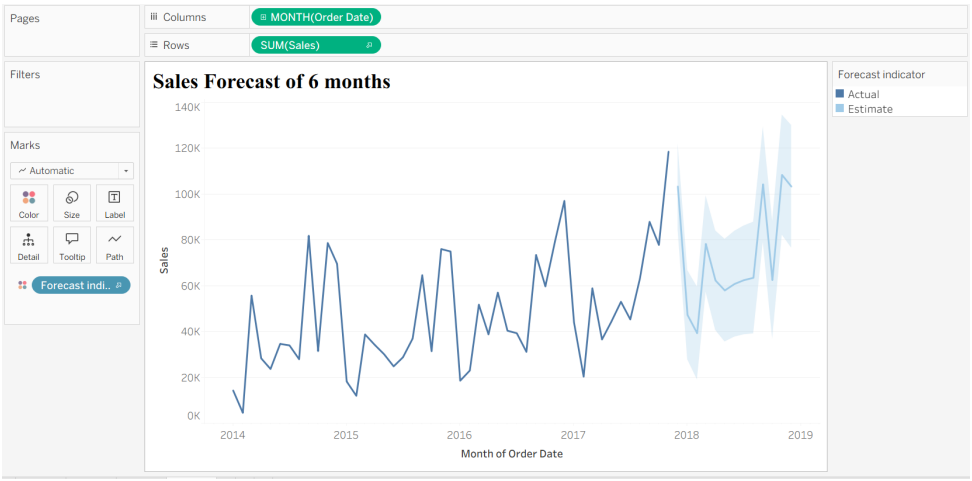
- The histogram shows that most orders fall in the lowest Sales range, while far fewer orders occur at higher Sales values.
- The box plot summarizes the median, quartiles, spread and outliers of Profit for Furniture, Office Supplies and Technology.

e) Apply Forecasting Techniques

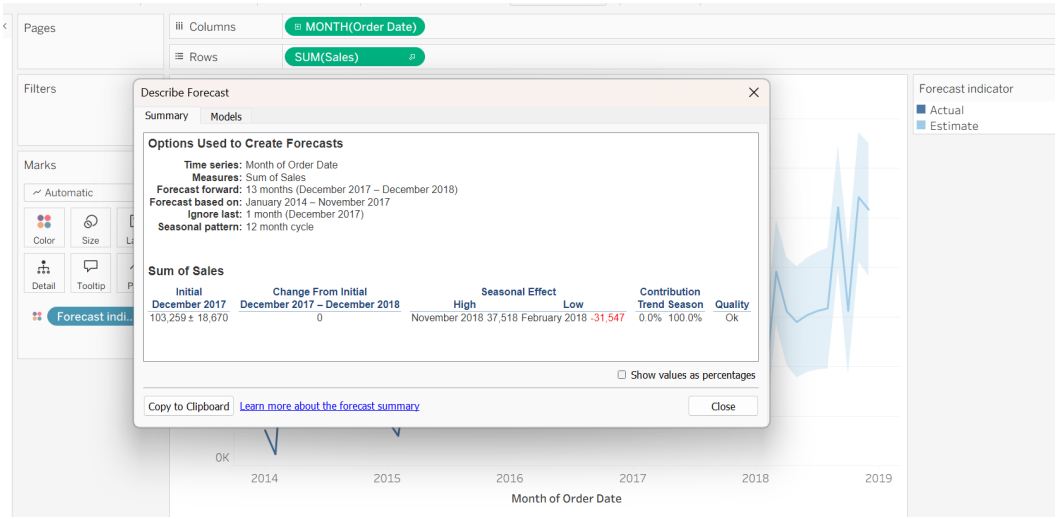
1. Create a new worksheet.
2. Drag Order Date to Columns and select Month.
3. Drag Sales to Rows.
4. Open the Analytics pane and drag Forecast onto the view. Alternatively select Analysis → Forecast → Show Forecast.
5. To change the forecast period, select Analysis → Forecast → Forecast Options.
6. Under Forecast Length select Exactly and enter 6 months, then click OK.
7. Select Analysis → Forecast → Describe Forecast to view the details of the forecasting model.



Sales Forecast with the automatic forecast length.



Sales Forecast of 6 months.



Describe Forecast dialog showing the options used to create the forecast.

Observation

- The forecast extends the Sales line into the future; the shaded band is the prediction interval, which shows the uncertainty of the forecast.
- The Describe Forecast dialog shows the time series, the measure, the forecast length, the period used for forecasting and the seasonal pattern (a 12 month cycle).

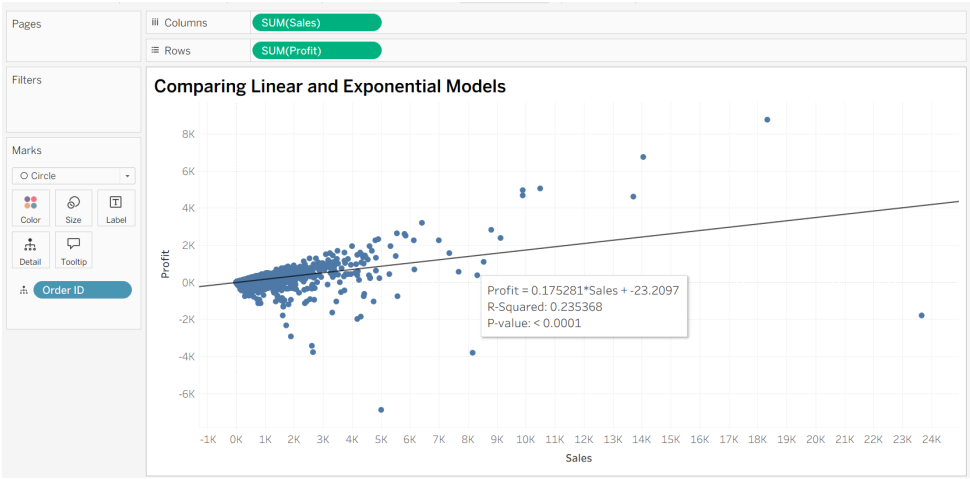
f) Compare Different Statistical Models

Different models suit different patterns of data. The table below lists the models used in this practical.

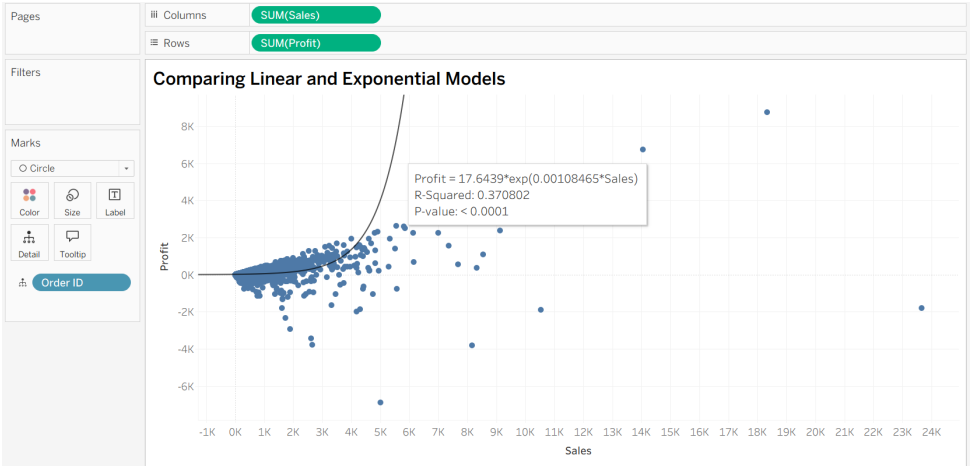
Model	Suitable Pattern
Linear	Straight-line relationship
Exponential	Growth or decay
Logarithmic	Rapid initial change followed by slower change
Polynomial	Curved relationship
Forecasting	Future time-series prediction
Clustering	Grouping similar observations

Compare Linear and Exponential Models

1. Create a scatter plot with Sales on Columns, Profit on Rows and Order ID on Detail.
2. Add a Linear trend line and hover over it (or select Describe Trend Line) to note the R-squared value.
3. Edit the trend line to Exponential and note the R-squared value again.
4. Compare the results of the two models.



Linear trend line on the Sales vs Profit scatter plot (R-Squared: 0.235368).



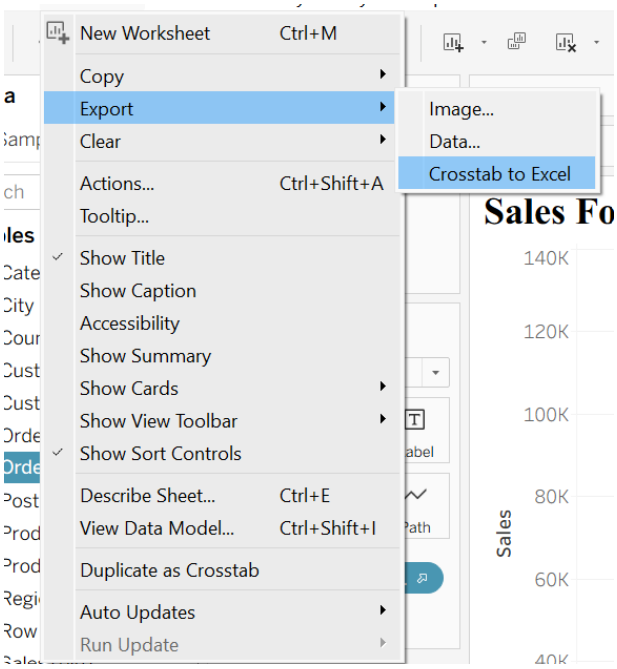
Exponential trend line on the Sales vs Profit scatter plot (R-Squared: 0.370802).

Model	R^2	Interpretation
Linear	0.235368	Explains approximately 23.54% of the variation in Profit.
Exponential	0.370802	Explains approximately 37.08% of the variation in Profit and gives a better fit based on R^2.

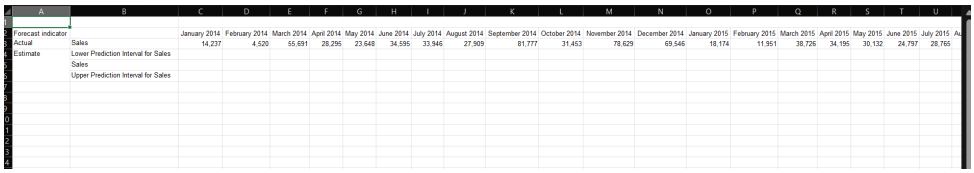
g) Export and Interpret Statistical Results

Export Crosstab

1. Open the required worksheet (for example the Sales forecast).
2. Select Worksheet → Export → Crosstab to Excel.
3. Save the Excel file at the required location.



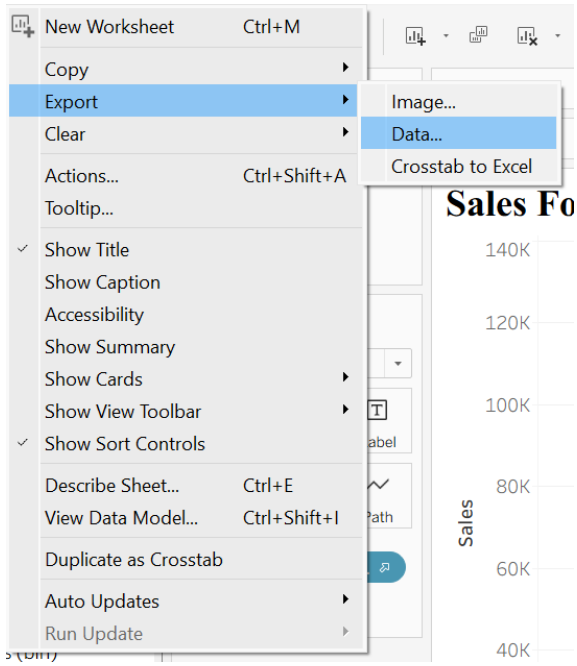
Worksheet → Export → Crosstab to Excel.



Exported crosstab opened in Excel.

Export Data

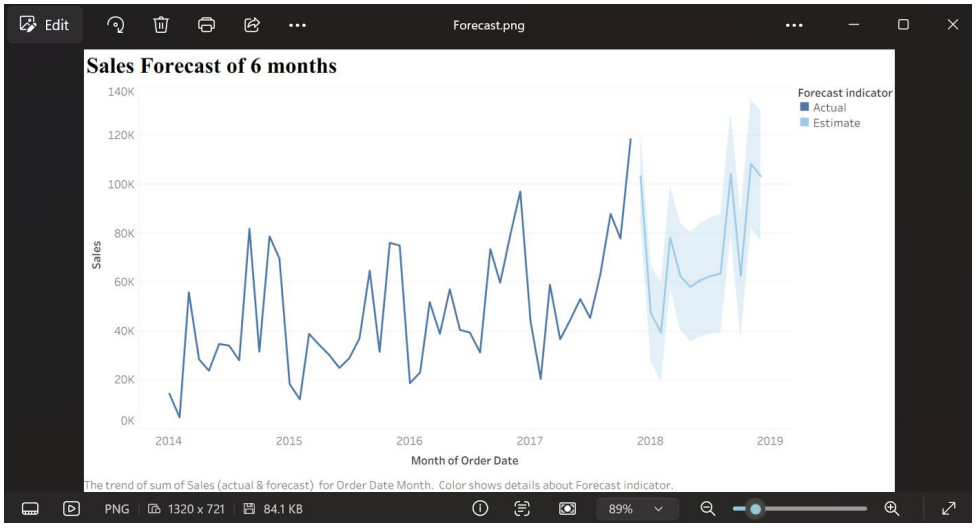
1. Select Worksheet → Export → Data.
2. Review the underlying data and save it if required.



Worksheet → Export → Data.

Export Image

1. Select Worksheet → Export → Image.
2. Choose the required options and save the image.



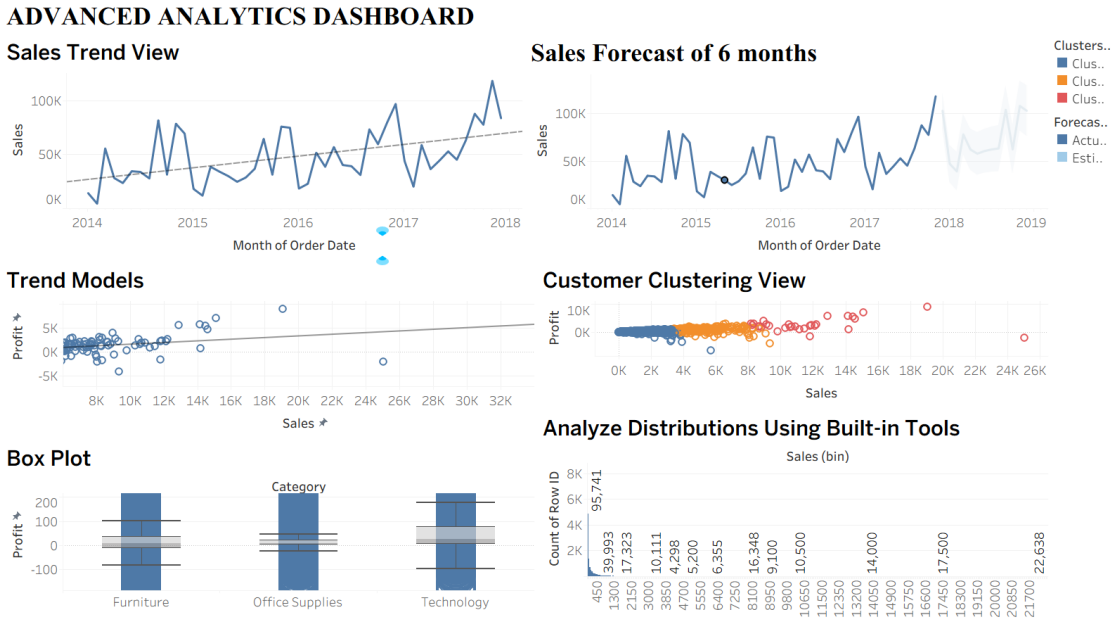
Sales Forecast of 6 months exported as an image (Forecast.png).

Interpretation of Results

- Trend line: a positive slope indicates that Sales generally increased during the selected period, while a negative slope indicates a decrease.
- R-squared (R^2): indicates how well a model fits the data; a higher value means the model explains more of the variation. Here the exponential model (0.370802) fits better than the linear model (0.235368).
- Clustering: groups customers with similar Sales and Profit; the cluster with higher values represents the high-value customers.

- Forecast: estimates future Sales from historical patterns, and the prediction interval shows the uncertainty of the forecast.
- Distribution: the histogram shows how observations are spread across Sales ranges, and the box plot shows the median, spread and outliers of Profit.

Output



Advanced Analytics Dashboard: Sales trend, trend models, customer clusters, forecast, distribution histogram and Profit box plot.

Result

Successfully performed advanced analytics in Tableau by adding and analyzing trend lines, clustering customers, analyzing distributions with a histogram and box plot, forecasting Sales, comparing linear and exponential models using R^2, and exporting and interpreting the statistical results.